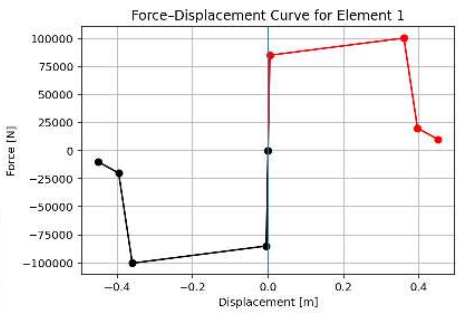
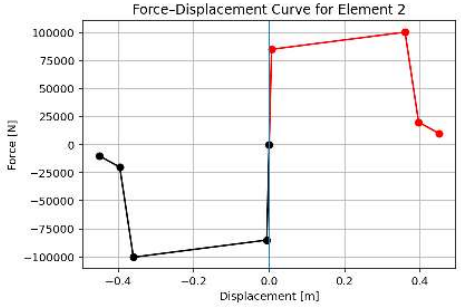
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A MULTI-DEGREE-FREEDOM ELEVATOR AND STRUCTURE SYSTEM: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

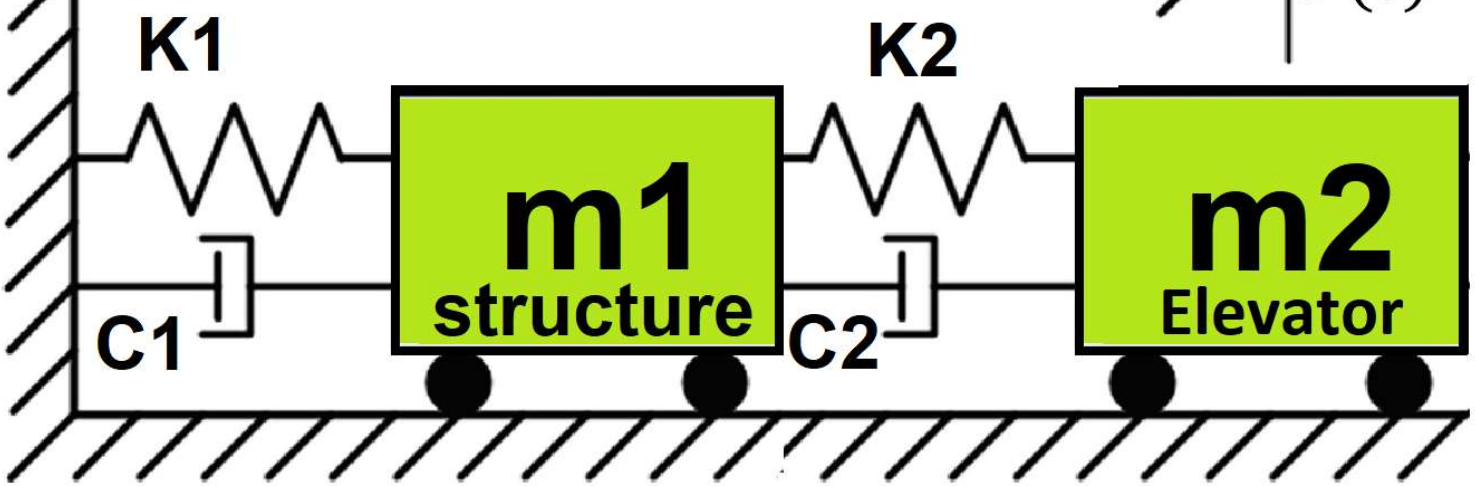


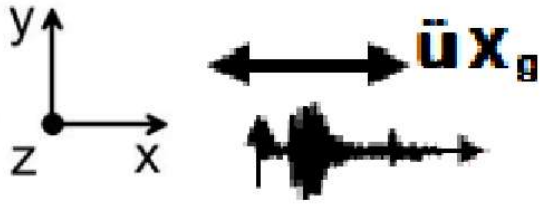


INCREMENTAL DISPLACEMENT

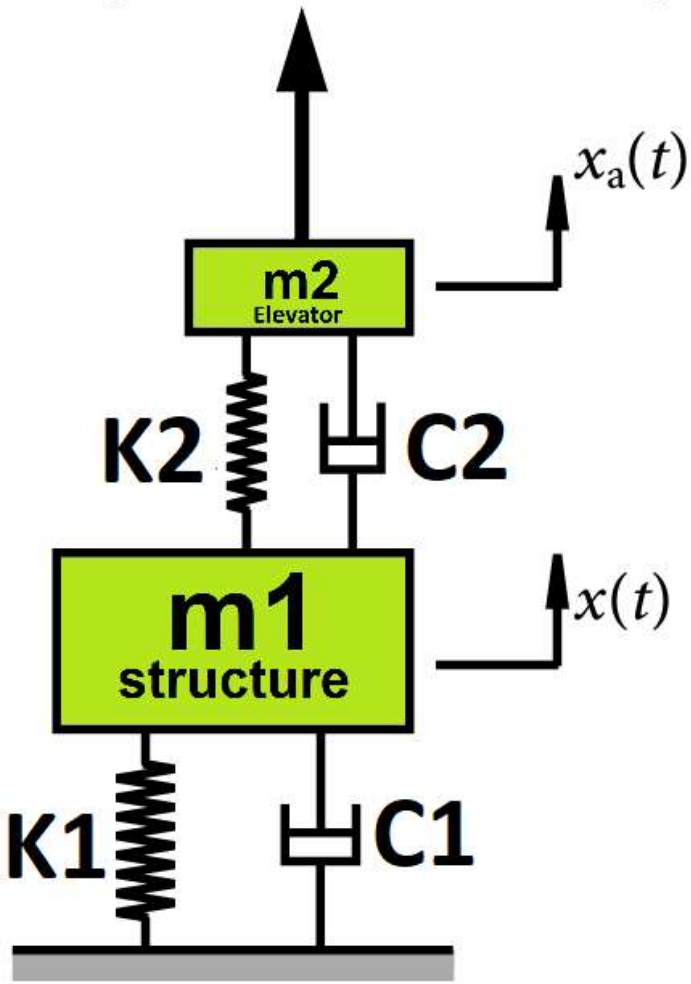
Initial Displacement
Initial Velocity
Initial Acceleration

$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$





$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$



$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

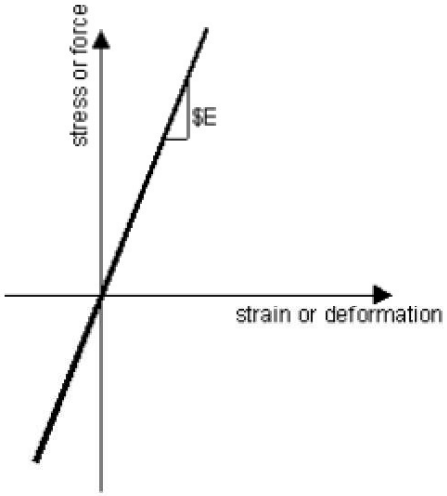
$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

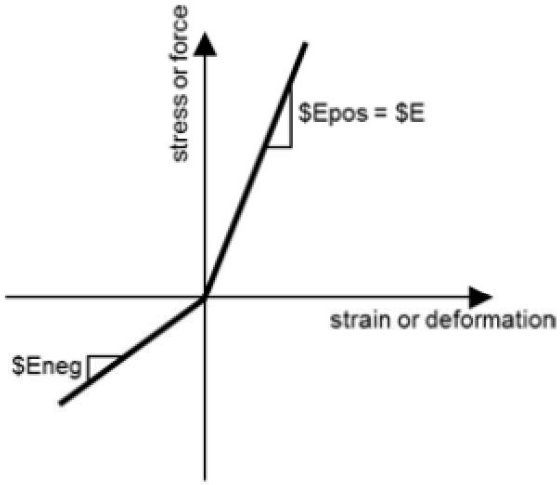
$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

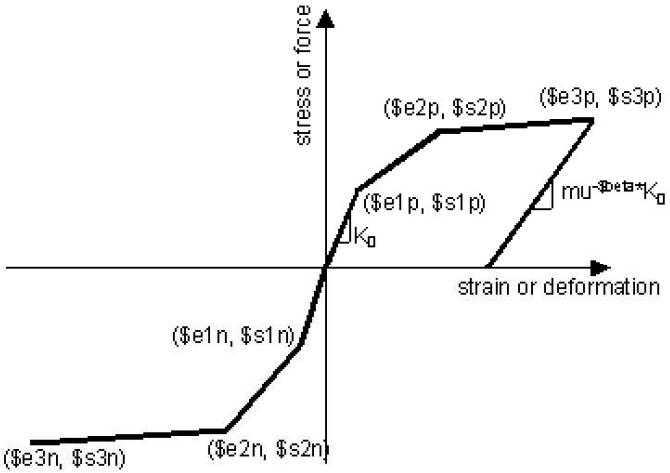
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$



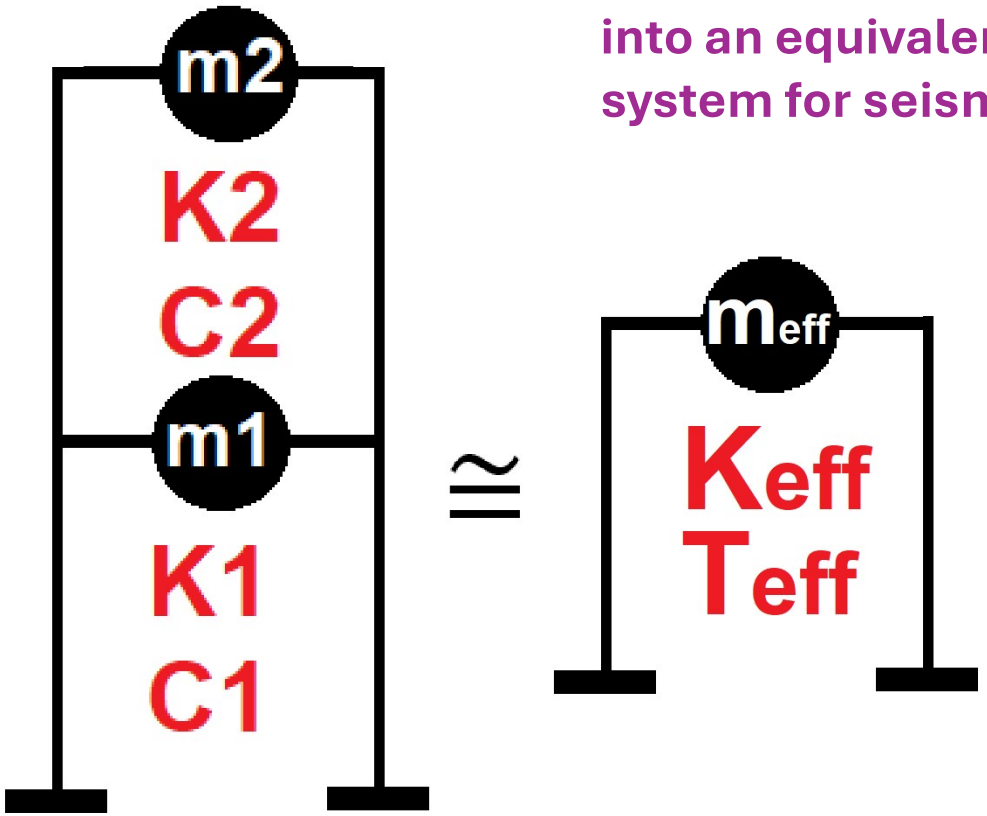


ELASTIC MATERIAL



INELASTIC MATERIAL

Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment



$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i} \qquad m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d} \qquad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

Spyder (Python 3.12)

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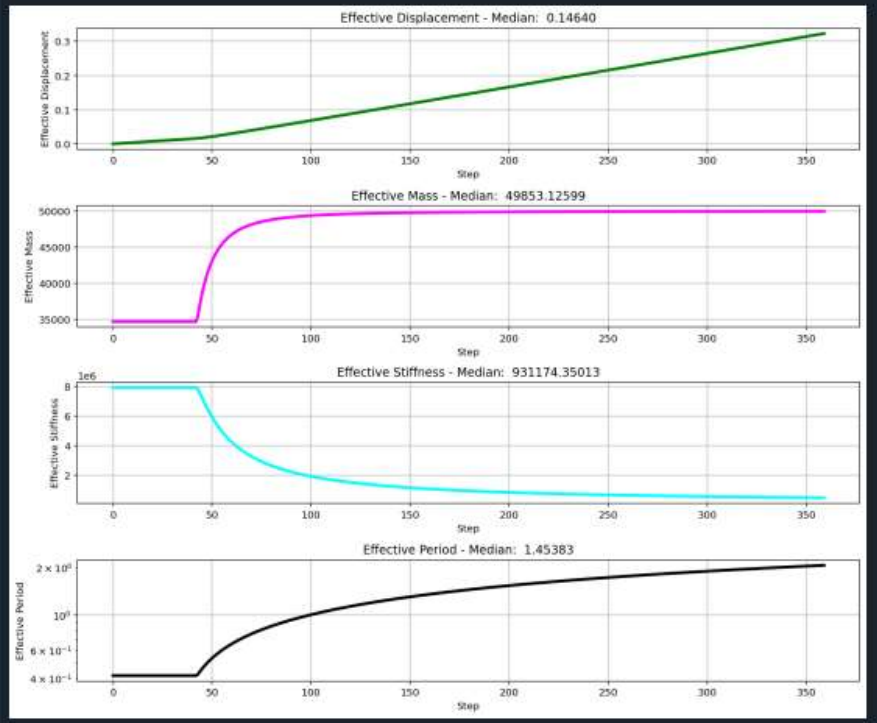
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..._MDOF_8_ANA\P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py

P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py

```
################################################################################
#                  >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST ME
#   COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A MULTI-DEGREE-FREEDOM EL
#               AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADAT
#                        STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANAL
#-------------------------------------------------------------------------------
#EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PR
#-------------------------------------------------------------------------------
#                                              P(t) = P0 sin(wt)
#                                       P(t) = P0 exp(-0.05wt) sin(wt)
#-------------------------------------------------------------------------------
#         ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AN
#                                      OF ENERGY DISSIPATION CAPACITY IND
#-------------------------------------------------------------------------------
#                    THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI
#                               EMAIL: salar.d.ghashghaei@gmail.com
################################################################################
"""
Nonlinear Seismic Performance Assessment of a MDOF Elevator and Structure S
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Stati

This OpenSeesPy script performs rigorous nonlinear static and dynamic analy
 for performance-based earthquake engineering. The 2D model incorporates bo
 (elastic-perfectly plastic or hysteretic steel with strain hardening)
 and geometric nonlinearity (corotational truss formulation) to capture P-d
 and large displacements—essential for collapse assessment.

Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
(2) [STATIC] : Gravity load analysis establishing dead load state
(3) [PUSHOVER] : Displacement-controlled pushover generating full capacity
 and plastic mechanism identification
(4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturin
 pinching behavior, and energy dissipation degradation
```



IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Conda: anaconda3 (Python 3.12.7) LSP: Python Line 170, Col 4 UTF-8 CRLF RW Mem 49%

Spyder (Python 3.12)

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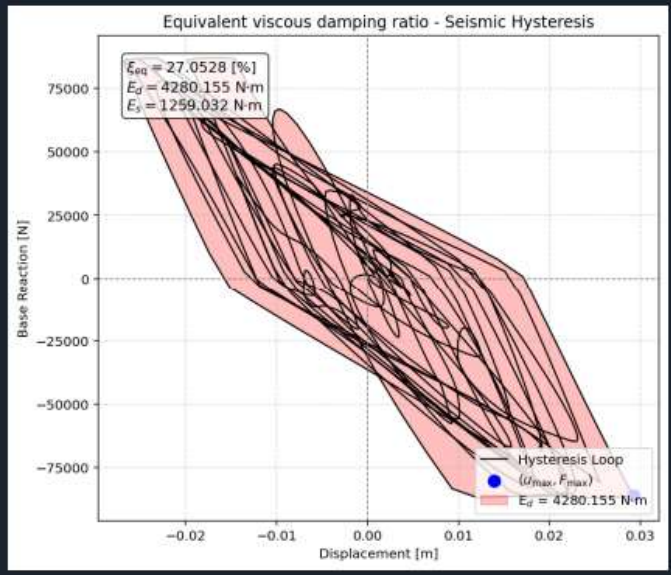
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...E_8_ANA+\EQULIVALENT_VISCOUS_DAMPING_RATIO_FUN.py

P(t)_MDOF_ELEVATOR_STRUCTURE_8_ANA.py | EQULIVALENT_VISCOUS_DAMPING_RATIO_FUN.py

```
#%% EQULIVALENT VISCOUS DAMPING RATIO
"""
[1] The function EQULIVALENT_VISCOUS_DAMPING_RATIO_FUN computes the dissipated energy (E_d) us

[2] Both methods produce the same styled plot: the hysteresis loop in black, the enclosed area

[3] The bottom section generates a synthetic elliptical hysteresis loop (sinusoidal displaceme
THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
"""
def EQULIVALENT_VISCOUS_DAMPING_RATIO_FUN(displacement, base_shear, method, XLABEL, YLABEL, TI
    import numpy as np
    from scipy.spatial import ConvexHull
    import matplotlib.pyplot as plt
    if method == 1:

        """
        Compute the equivalent viscous damping ratio from a hysteresis loop.

        The dissipated energy per cycle (E_d) is the area enclosed by the loop,
        computed using the convex hull (Shoelace formula on the hull). The maximum
        strain energy (E_s) is estimated as 0.5 * F_max * u_max, where u_max is the
        maximum absolute displacement and F_max is the absolute base shear at that
        same displacement.

        Parameters
        ----------
        displacement : array-like
            Displacement values (assumed to form a closed hysteresis loop).
        base_shear : array-like
            Corresponding base shear values.
        XLABEL : str
            Label for the x-axis.
        YLABEL : str
```



IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

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STATIC ANALYSIS

Spyder (Python 3.12)

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...ISSIPATION_CAPACITY_INDEX\54_ELEVATOR_STRUCTURE_MDOF_8_ANA

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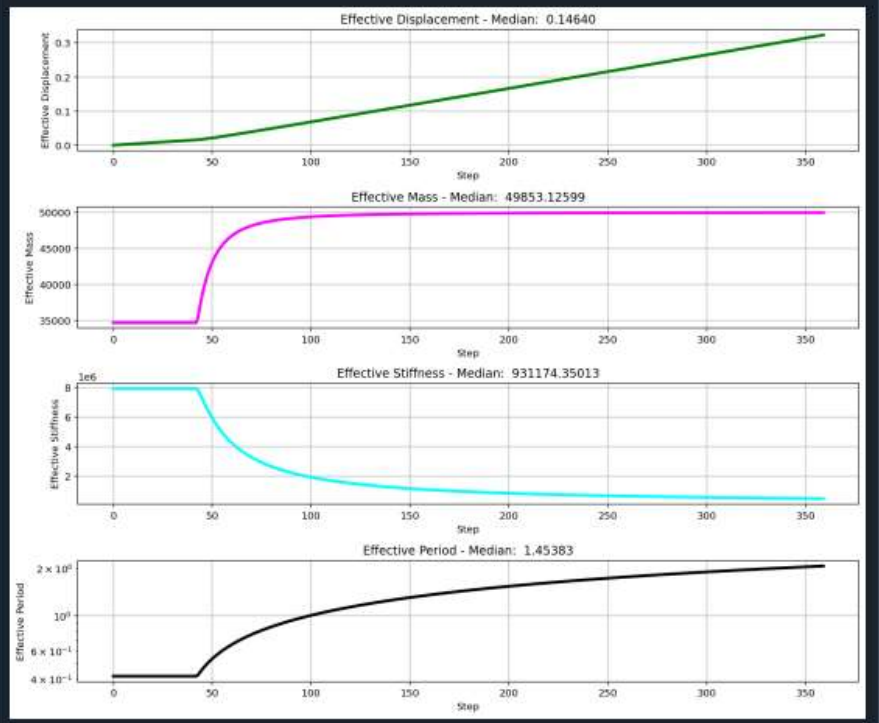
P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py

```
966  #%%-----------------------------------------------------
967  # PERIOD ANALYSIS
968  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
969  ANAL_TYPE = 'PERIOD'
970
971  DATA = ELEVATOR_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
972  (PERIOD_MIN_X, PERIOD_MAX_X) = DATA
973  print('Structure First Period:  ', PERIOD_MIN_X)
974  print('Structure Second Period: ', PERIOD_MAX_X)
975
976
977  #%%-----------------------------------------------------
978  # STATIC ANALYSIS
979  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
980  ANAL_TYPE = 'STATIC'
981
982  DATA = ELEVATOR_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
983  (reaction, disp, DI) = DATA
984
985  #%%-----------------------------------------------------
986  # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
987  MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
988  ANAL_TYPE = 'PUSHOVER'
989
990  DATA = ELEVATOR_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
991  (reaction_PUSH, disp_PUSH, DI_PUSH,
992   ele_force_PUSH, node_displacements_PUSH,
993   PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
994
995
996  XDATA = disp_PUSH
997  YDATA = reaction_PUSH
998  XLABEL = 'Displacement [m]'
999  YLABEL = 'Base Reaction [N]'
```

32 %



IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 170, Col 4 UTF-8 CRLF RW Mem 50%

PUSHOVER ANALYSIS

Spyder (Python 3.12)

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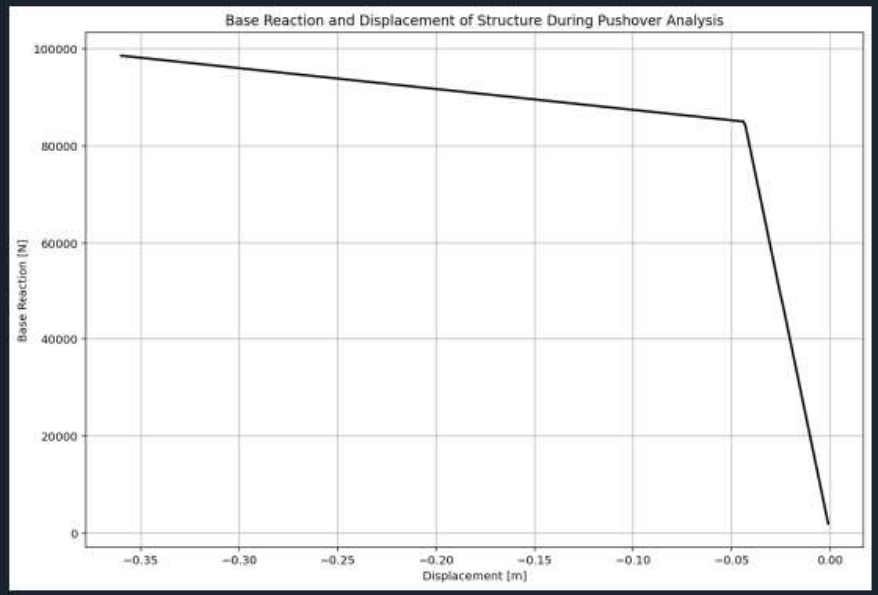
P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py

```
985   #%%-------------------------------------------------
986   # PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
987   MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
988   ANAL_TYPE = 'PUSHOVER'
989
990   DATA = ELEVATOR_STRUCTURE_MDOF(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
991   (reaction_PUSH, disp_PUSH, DI_PUSH,
992    ele_force_PUSH, node_displacements_PUSH,
993    PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
994
995
996   XDATA = disp_PUSH
997   YDATA = reaction_PUSH
998   XLABEL = 'Displacement [m]'
999   YLABEL = 'Base Reaction [N]'
1000  TITLE = 'Base Reaction and Displacement of Structure During Pushover Analys
1001  COLOR = 'black'
1002  SEMILOGY = False
1003  PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1004
1005  DATA = S07.BILNEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NO
1006  (X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_S
1007
1008  # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1009  plt.figure(0, figsize=(12, 8))
1010  plt.plot(disp_PUSH, PERIOD_MIN_PUSH, linewidth=3)
1011  plt.plot(disp_PUSH, PERIOD_MAX_PUSH, linewidth=3)
1012  plt.title('Period of Structure During Pushover Analysis')
1013  plt.ylabel('Structural Period [s]')
1014  plt.xlabel('Displacement [m]')
1015  #plt.semilogy()
1016  plt.grid()
1017  plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_PUSH):.3f} (s) -
1018              f'PERIOD - MAX VALUES:  Min: {np.min(PERIOD_MAX_PUSH):.3f} (s)
```



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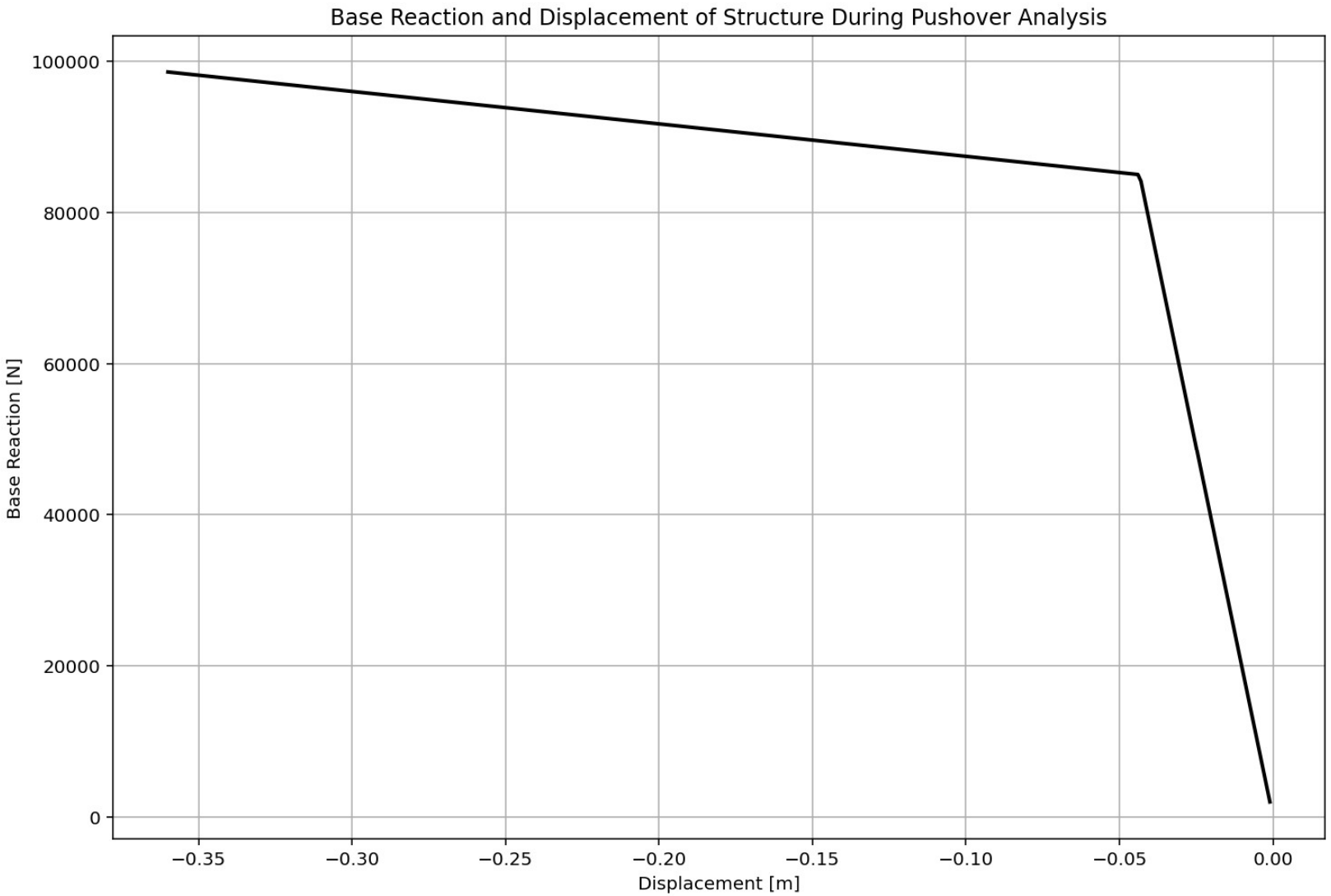
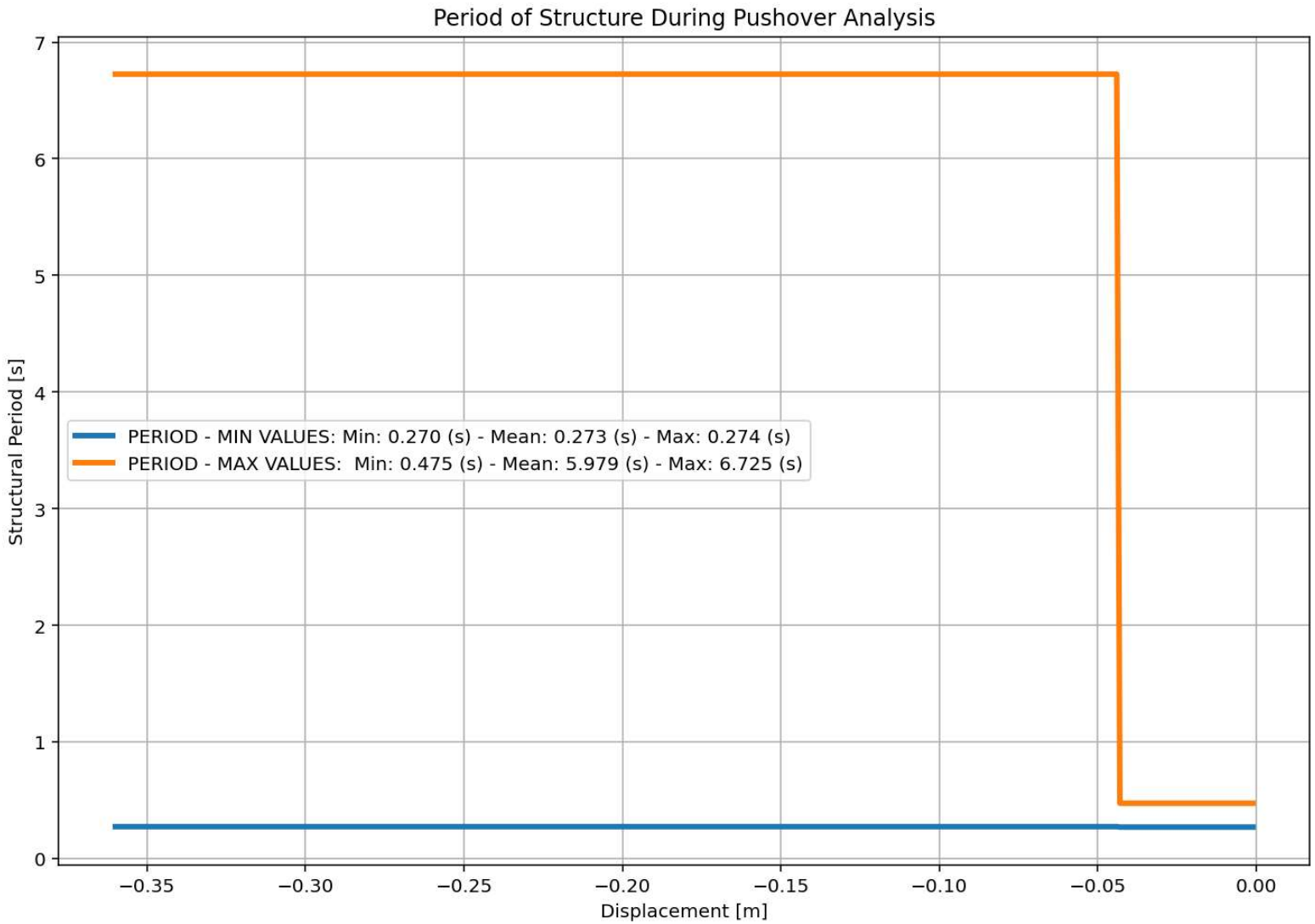
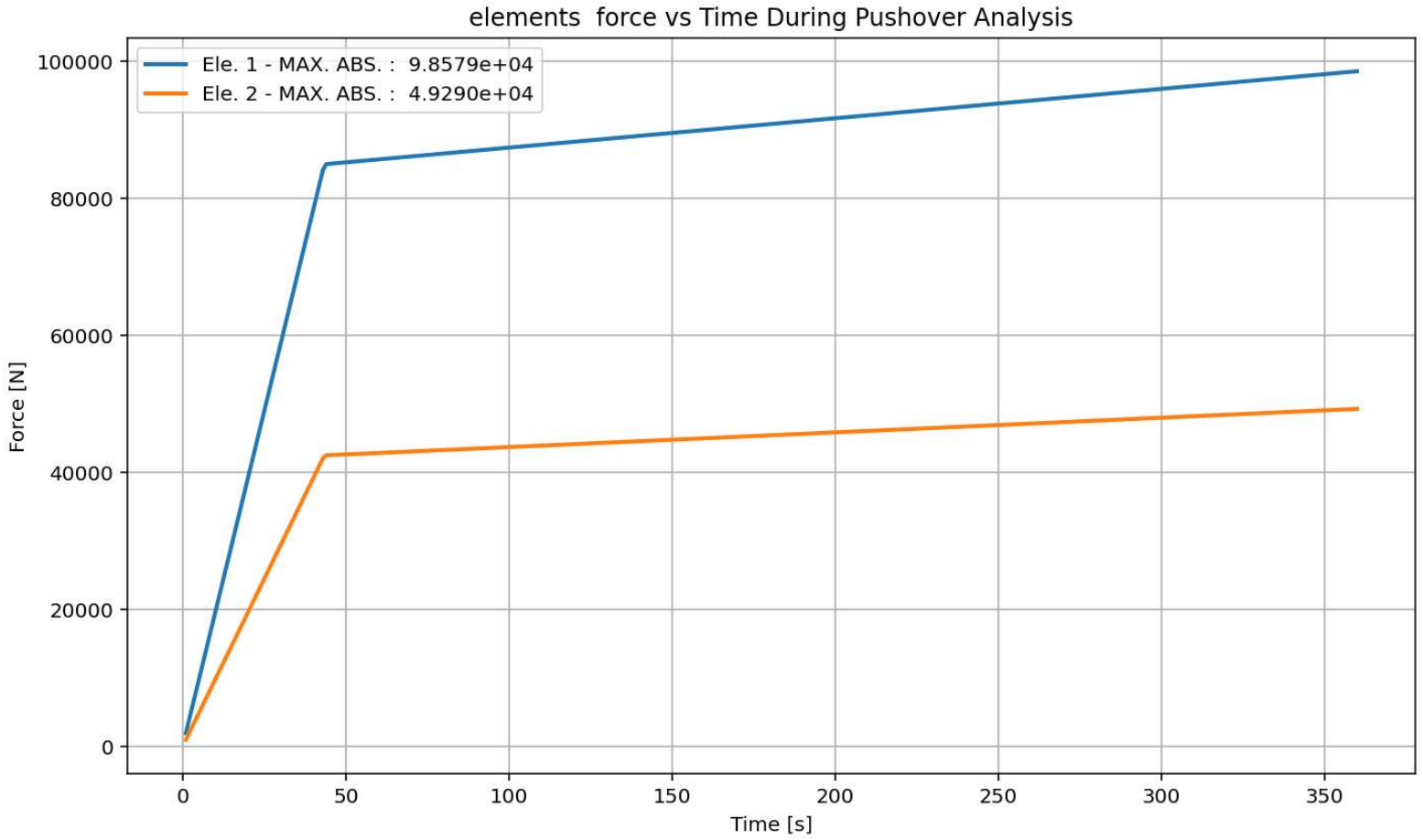
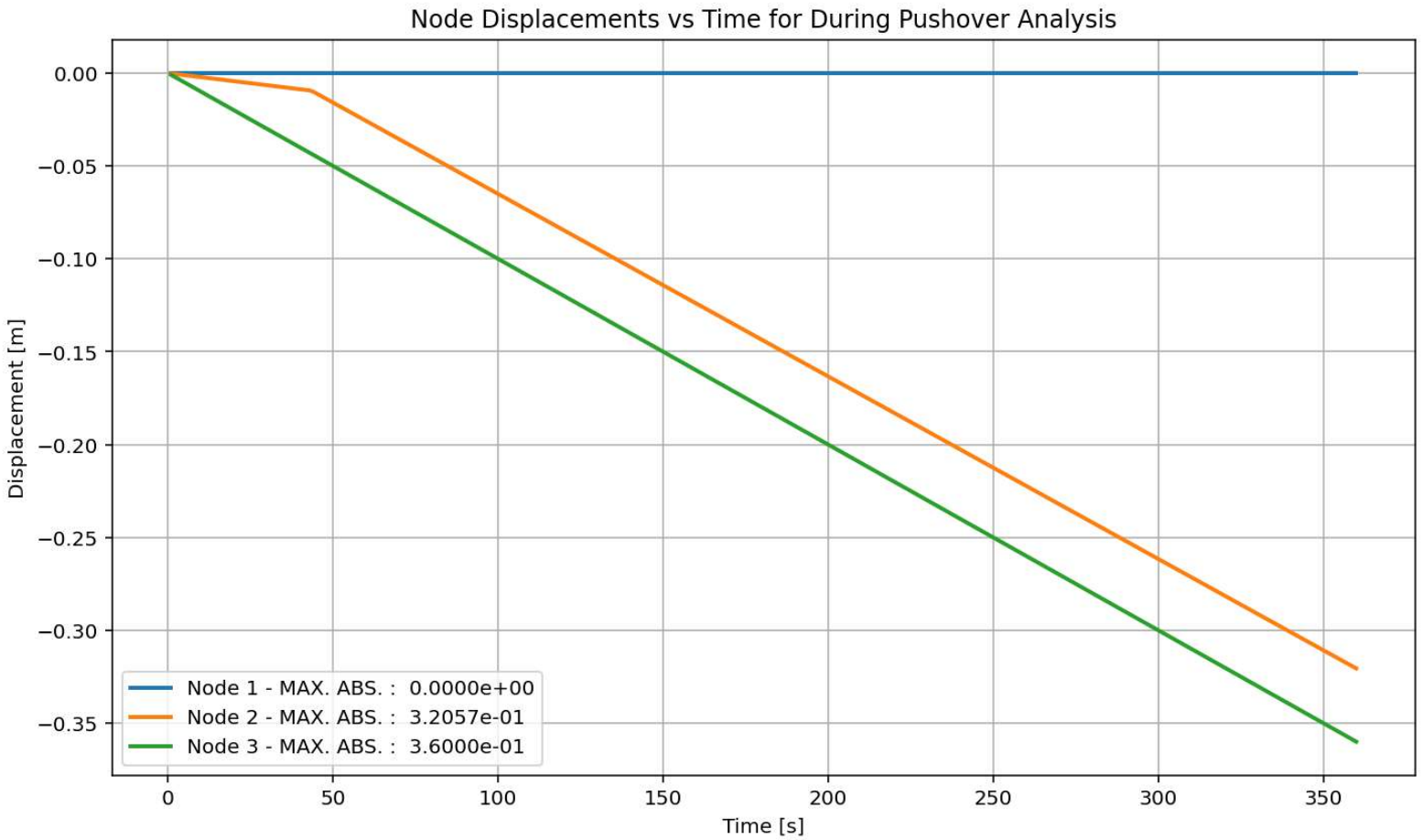
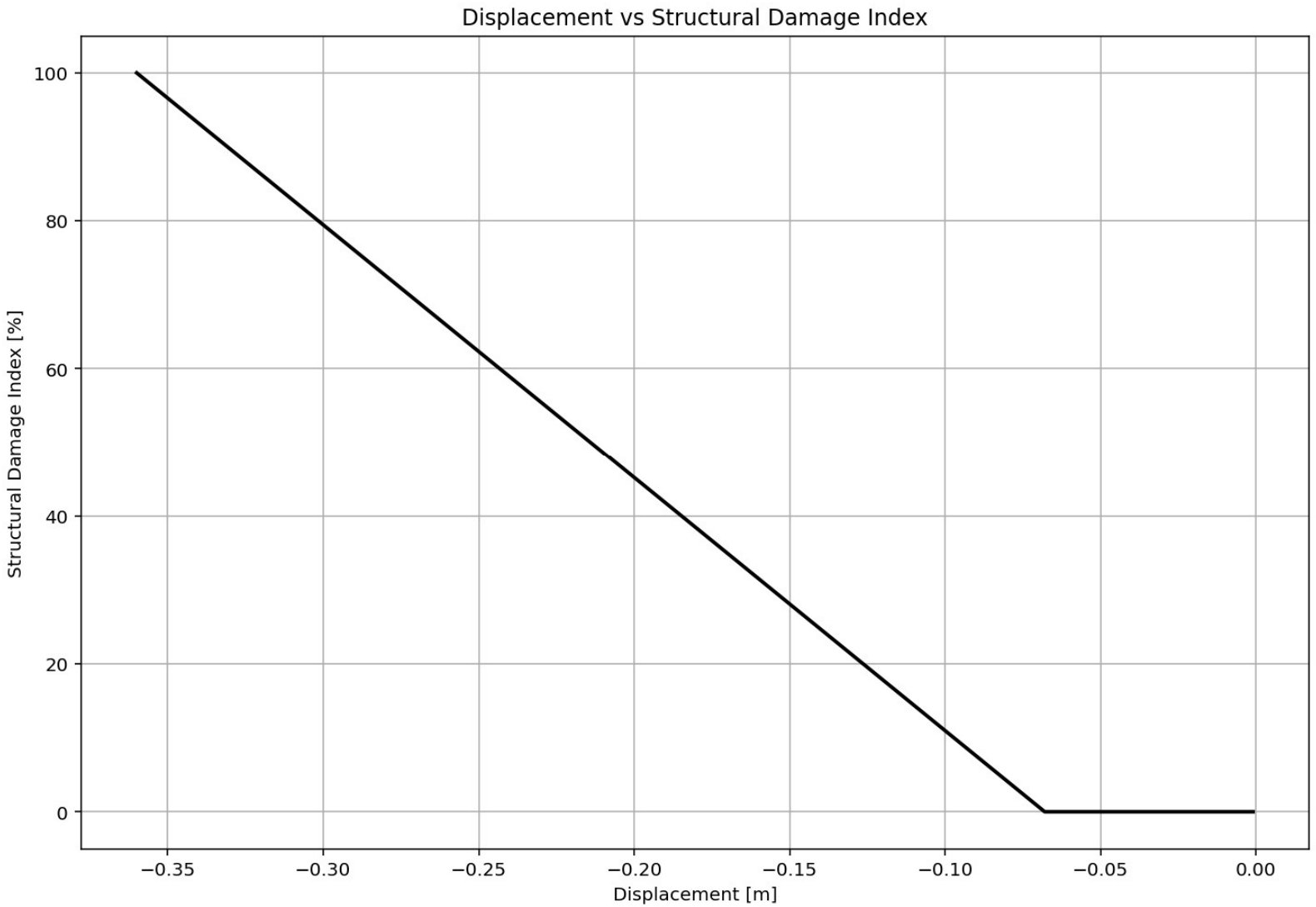

Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
100000
80000
60000
40000
20000
0
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Displacement [m]


Period of Structure During Pushover Analysis
Structural Period [s]
Displacement [m]
PERIOD - MIN VALUES: Min: 0.270 (s) - Mean: 0.273 (s) - Max: 0.274 (s)
PERIOD - MAX VALUES: Min: 0.475 (s) - Mean: 5.979 (s) - Max: 6.725 (s)
0
1
2
3
4
5
6
7
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00


elements force vs Time During Pushover Analysis
Ele. 1 - MAX. ABS. : 9.8579e+04
Ele. 2 - MAX. ABS. : 4.9290e+04
Force [N]
Time [s]
100000
80000
60000
40000
20000
0
0
50
100
150
200
250
300
350


Node Displacements vs Time for During Pushover Analysis
Displacement [m]
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
0
50
100
150
200
250
300
350
Time [s]
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.2057e-01
Node 3 - MAX. ABS. : 3.6000e-01


Displacement vs Structural Damage Index
Structural Damage Index [%]
100
80
60
40
20
0
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Displacement [m]

Spyder (Python 3.12)

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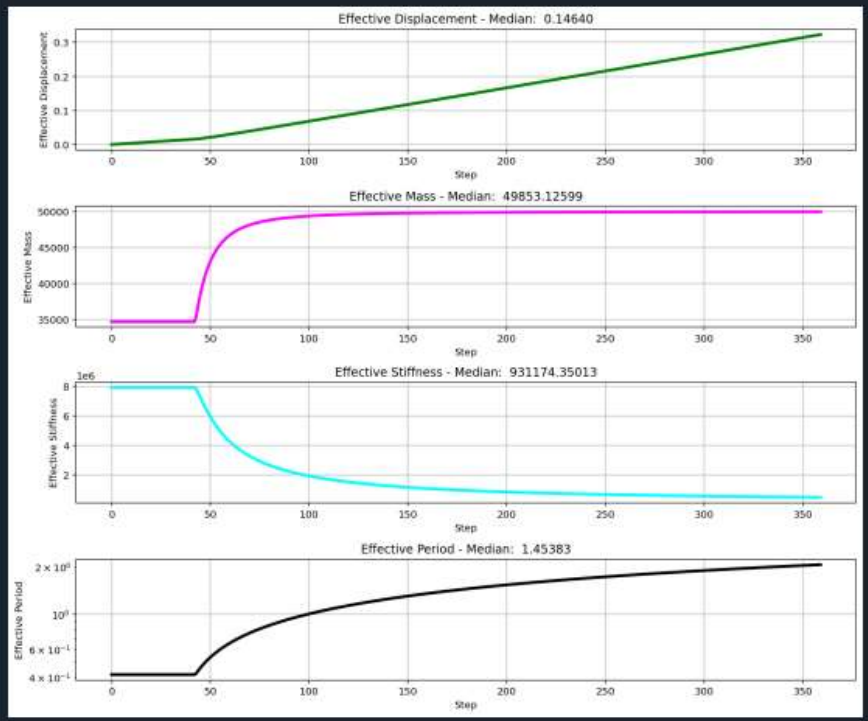
P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py | COMPUTE_EFFECTIVE_PROPERTIES_FUN.py

```
#%% EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN P
# Change MDOF to SDOF System with Displacement Based Design Concept
# THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
"""
This script implements a displacement-based pushover transformation,
 converting a multi-degree-of-freedom (MDOF) system into an equivalent
 single-degree-of-freedom (SDOF) system for seismic assessment.
 It calculates effective modal properties—displacement, mass, and
 stiffness—by weighting element forces and nodal displacements according
 to a presumed deformed shape.
 The derived effective period provides a simplified dynamic characteristic
 for performance-based engineering. The visualizations effectively track the
 evolution of these equivalent parameters throughout the nonlinear analysis s
"""
#-----------------------------------------------------------------------
displacement_X_1 = np.array(list(node_displacements.values())[1])   # DOF 02
displacement_X_2 = np.array(list(node_displacements.values())[2])   # DOF 03


ele_force_01 = np.array(list(ele_force.values())[0])   # ELEMENT 01
ele_force_02 = np.array(list(ele_force.values())[1])   # ELEMENT 02

STIFF_X_1 = np.abs(ele_force_01 / displacement_X_1)
STIFF_X_2 = np.abs(ele_force_02 / displacement_X_2)

MX2 = (MASS[0] * np.square(displacement_X_1) +
       MASS[1] * np.square(displacement_X_2))

MX = (MASS[0] * np.array(displacement_X_1) +
      MASS[1] * np.array(displacement_X_2))
EFFECTIVE_DISP_X = MX2 / np.abs(MX)

EFFECTIVE_MASS_X = np.abs(MX) / EFFECTIVE_DISP_X
```



IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

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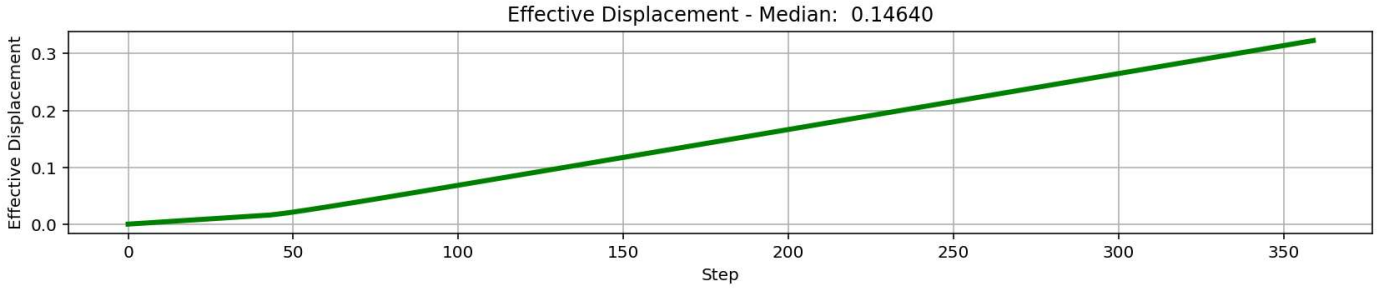
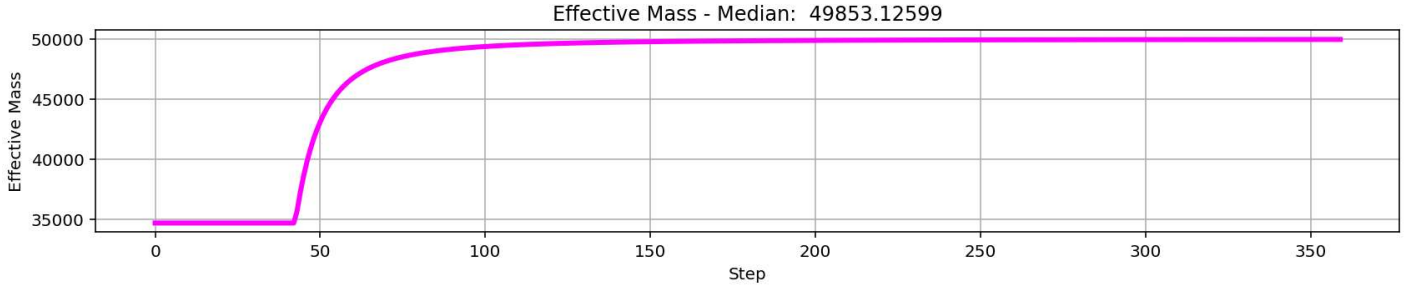
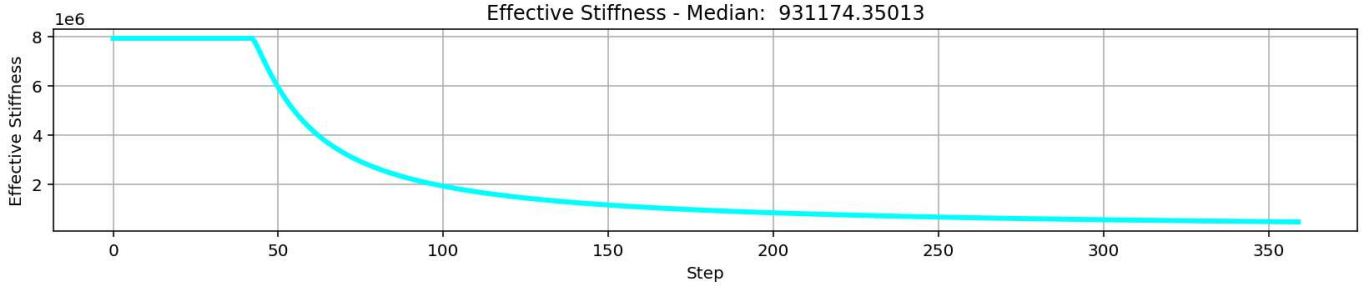


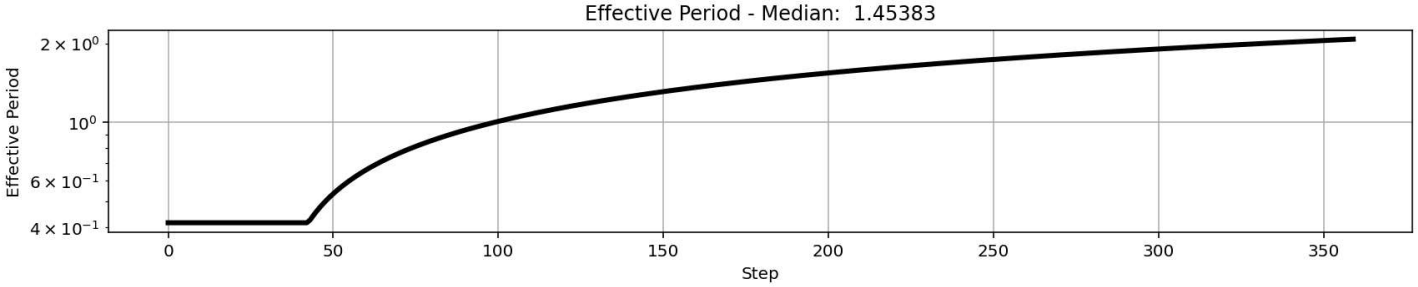
$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i}$$



$$m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$



$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d}$$



$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

CYCLIC DISPLACEMENT ANALYSIS

Spyder (Python 3.12)

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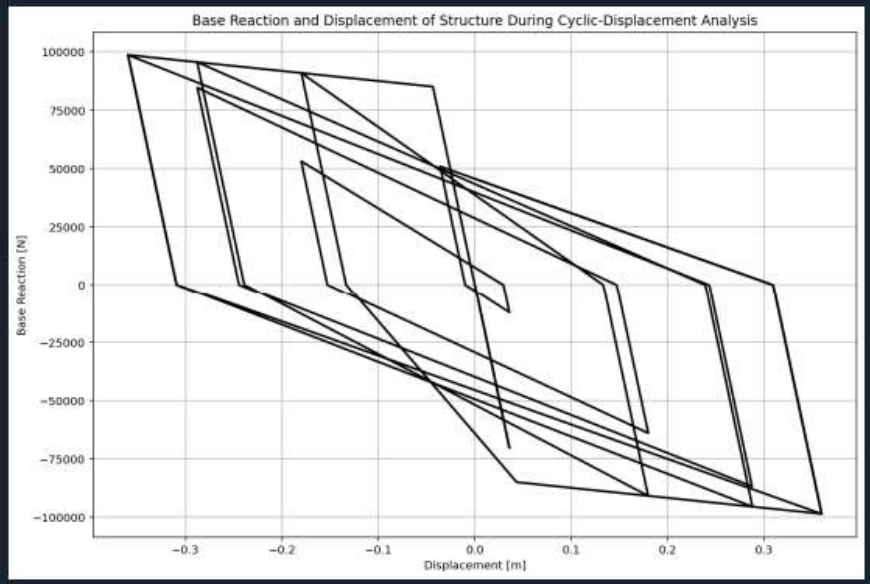
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..._MDOF_8_ANA\P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py

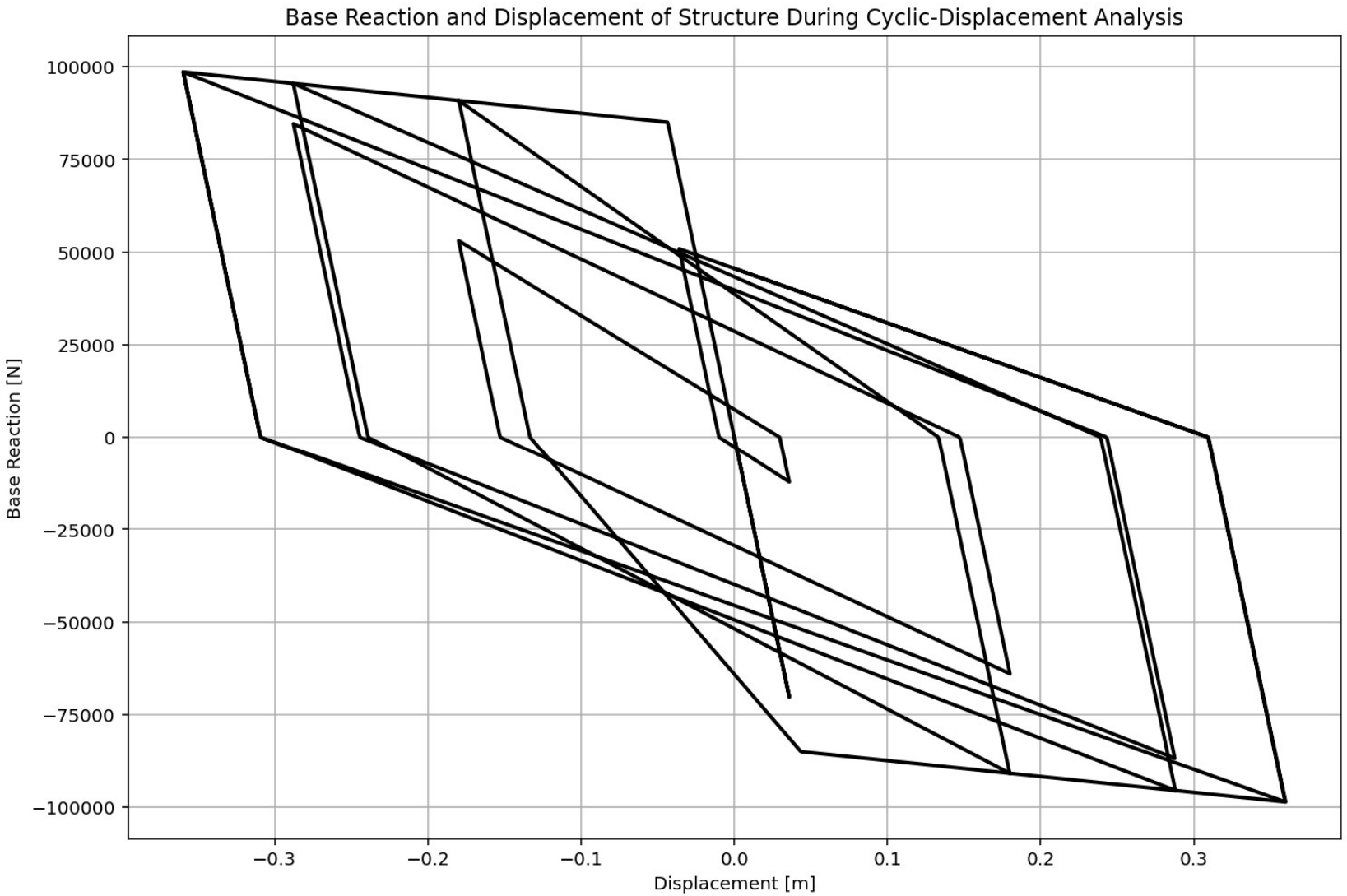
P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py | COMPUTE_EFFECTIVE_PROPERTIES_FUN.py

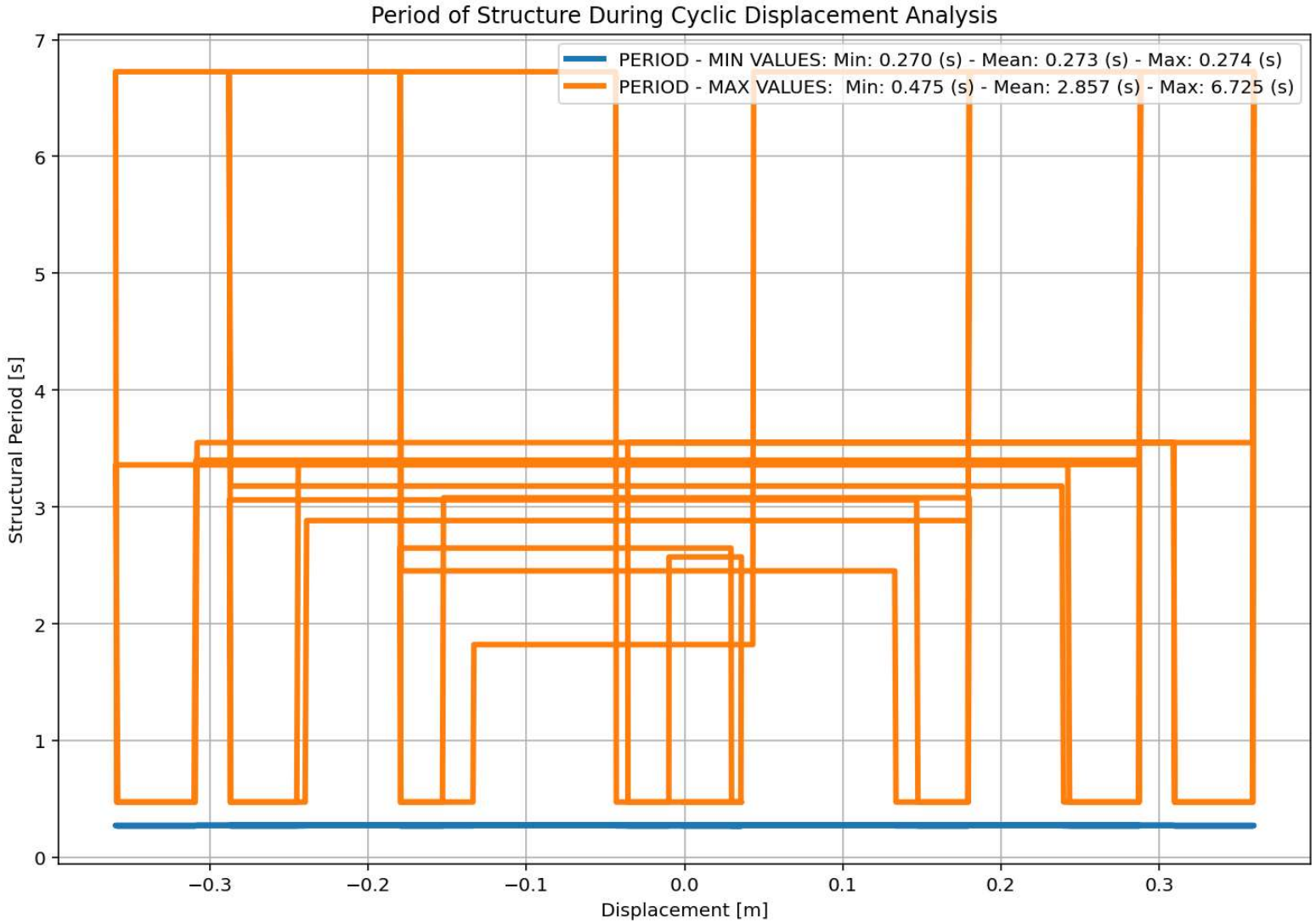
```
366     if ANAL_TYPE == 'CYCLIC_DISPLACEMENT': # STATIC TIME-HISTORY ANALYSIS
367         IDctrlDOF = 1   # 1: Horizental Dispalcement - 2: Vertical Dispalce
368         DMAX = -DSU     # [m] Max. Displacement
369         n_points = 10000
370         # 4. CYCLIC DISPALCEMENT PROTOCOL
371         # Key strain points (same logic as your protocol)
372         key_disp = np.array([
373               0.0,
374               0.1*DMAX,   -0.1*DMAX,
375               0.5*DMAX,   -0.5*DMAX,
376               0.8*DMAX,   -0.8*DMAX,
377               DMAX,       -DMAX,
378               0.1*DMAX,   -0.1*DMAX,
379               0.5*DMAX,   -0.5*DMAX,
380               0.8*DMAX,   -0.8*DMAX,
381               DMAX,       -DMAX,
382               0.0
383         ])
384
385         # Generate 1000-point displacement protocol
386         disp_protocol = np.interp(
387             np.linspace(0, len(key_disp) - 1, n_points),
388             np.arange(len(key_disp)),
389             key_disp
390         )
391
392         ops.analysis('Static')
393         STEP = 0.0
394         for target_disp in disp_protocol:
395             current_disp = ops.nodeDisp(center_node, 1) # DISPALCEMENT APPL
396             dU = target_disp - current_disp
397             ops.integrator('DisplacementControl', center_node, IDctrlDOF, d
398             OK = ops.analyze(1)
399             S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
```

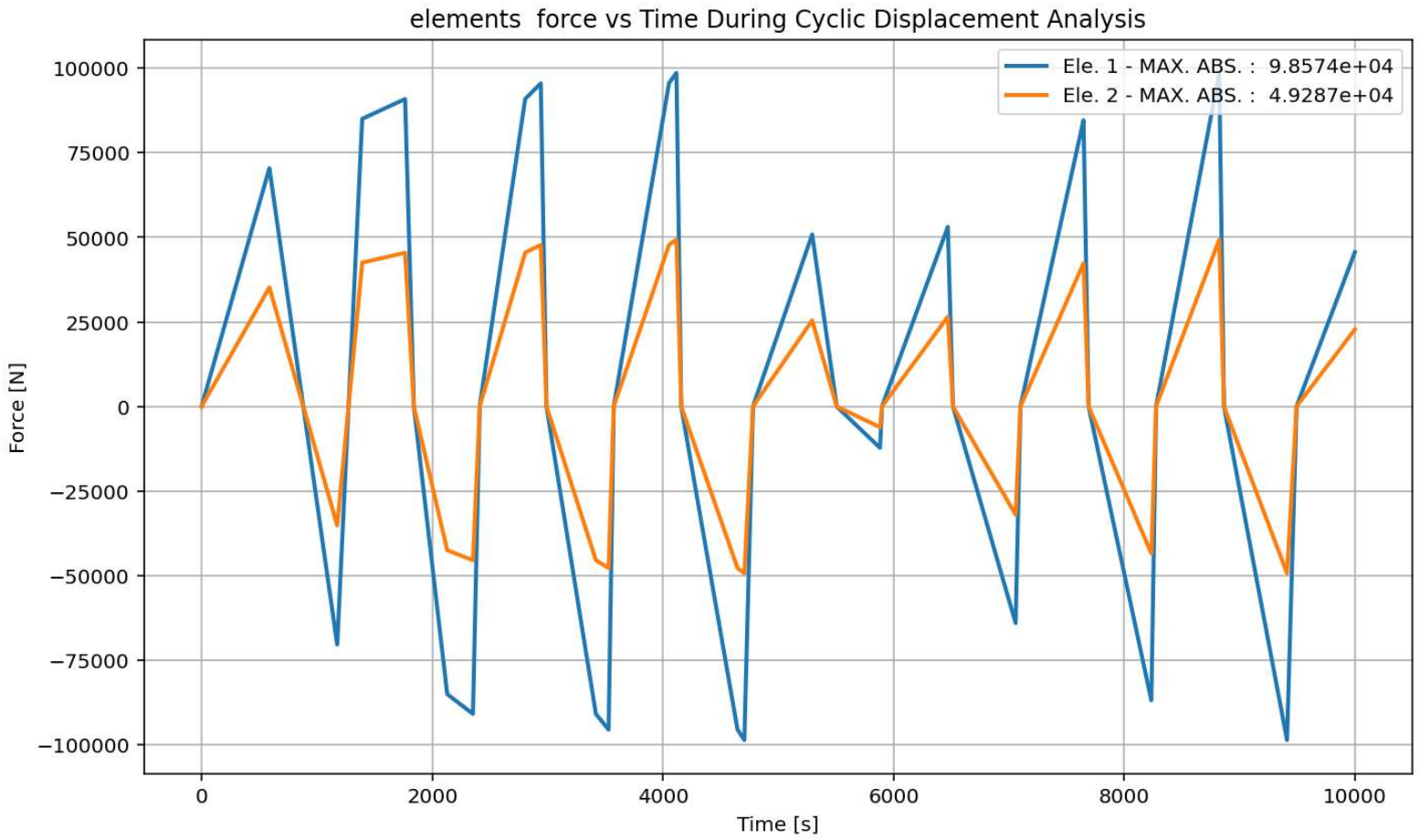


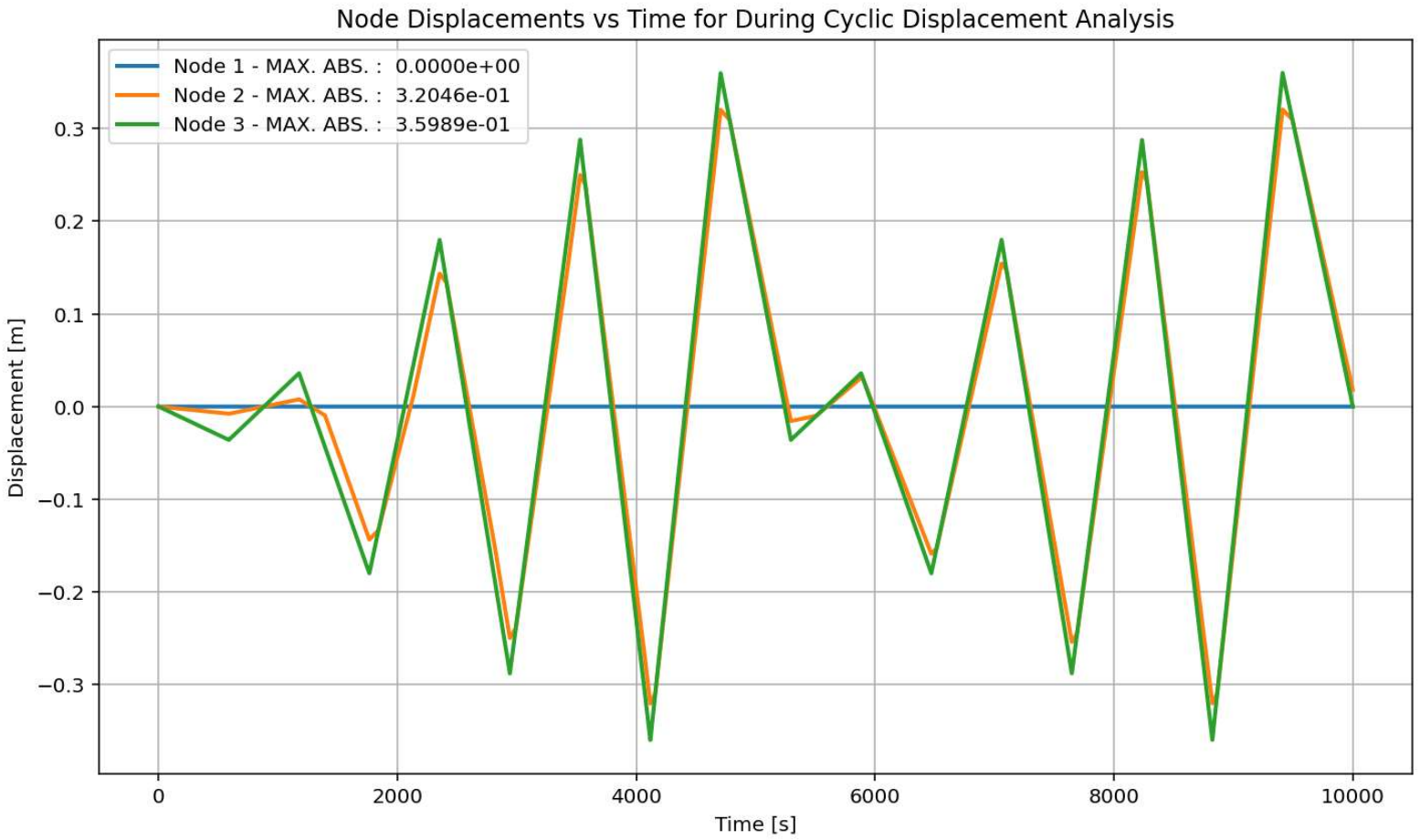
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

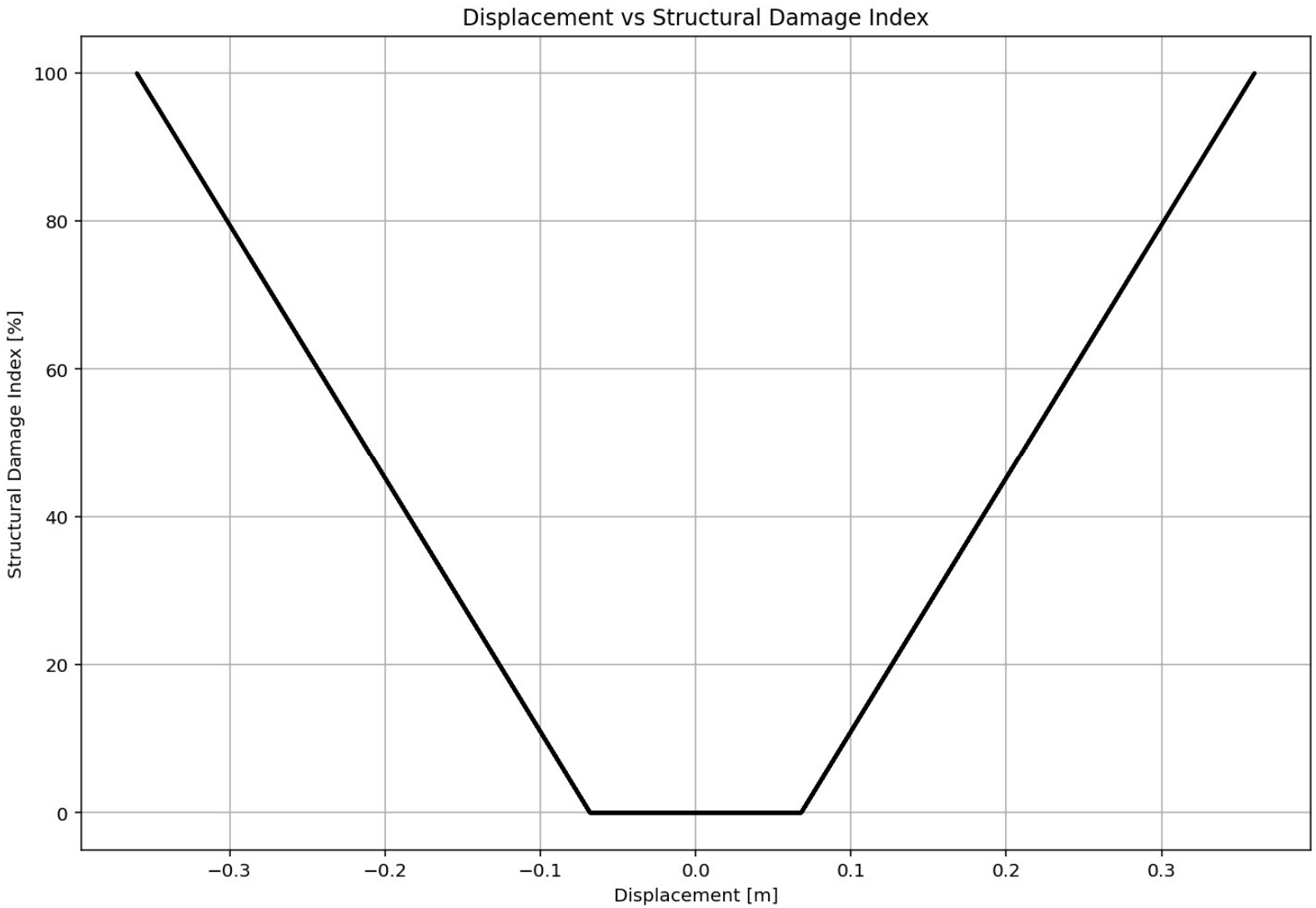
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 362, Col 44 UTF-8 CRLF RW Mem 51%

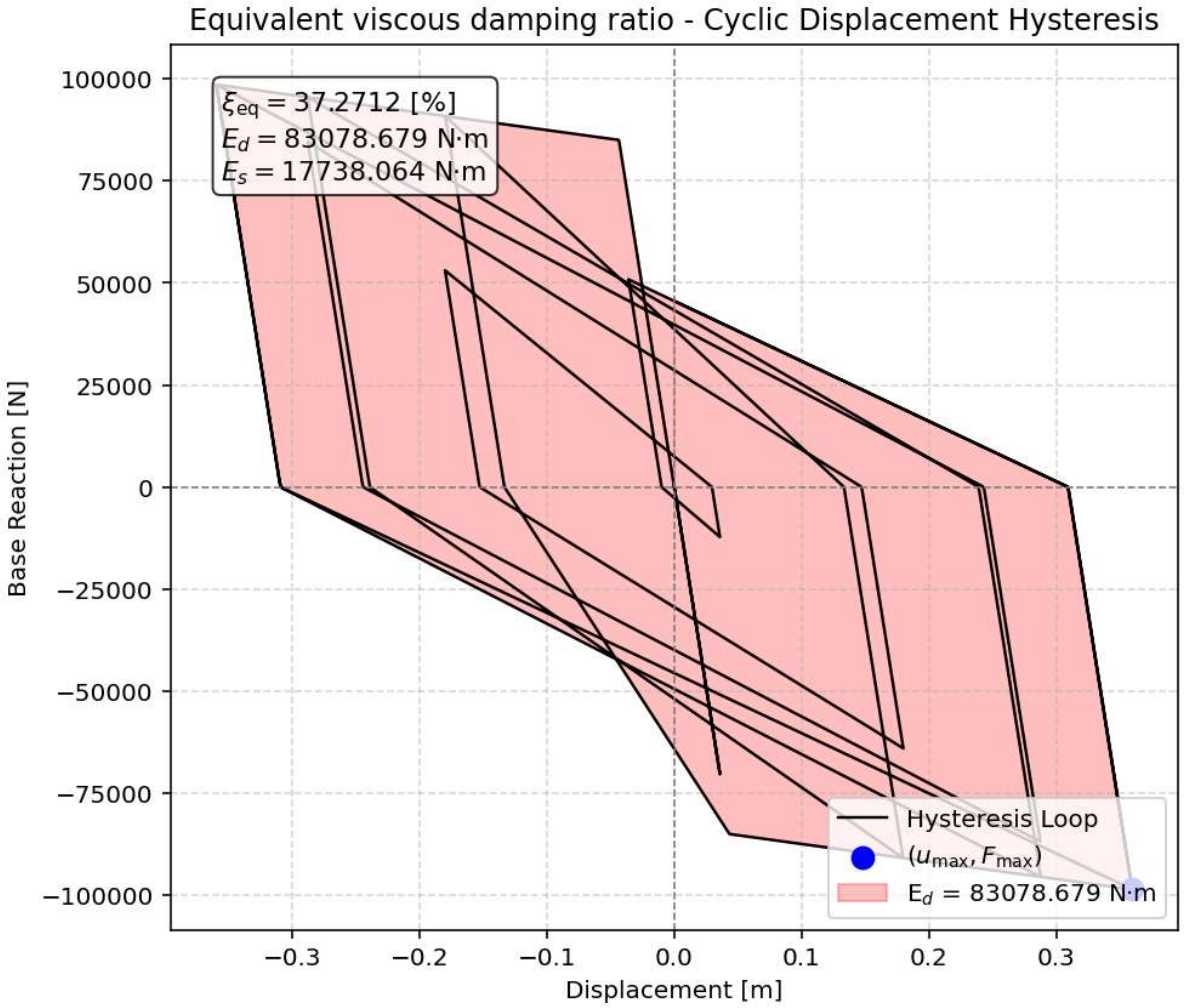

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
Base Reaction [N]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [m]


Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: 0.270 (s) - Mean: 0.273 (s) - Max: 0.274 (s)
PERIOD - MAX VALUES: Min: 0.475 (s) - Mean: 2.857 (s) - Max: 6.725 (s)
Structural Period [s]
Displacement [m]
0
1
2
3
4
5
6
7
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3


elements force vs Time During Cyclic Displacement Analysis
Ele. 1 - MAX. ABS. : 9.8574e+04
Ele. 2 - MAX. ABS. : 4.9287e+04
Force [N]
Time [s]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
0
2000
4000
6000
8000
10000


Node Displacements vs Time for During Cyclic Displacement Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.2046e-01
Node 3 - MAX. ABS. : 3.5989e-01
Displacement [m]
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
2000
4000
6000
8000
10000
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
100
80
60
40
20
0
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [m]


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
$\xi_{eq} = 37.2712$ [%]
$E_d = 83078.679$ N·m
$E_s = 17738.064$ N·m
Hysteresis Loop
(u_{max}, F_{max})
E_d = 83078.679 N·m
Base Reaction [N]
Displacement [m]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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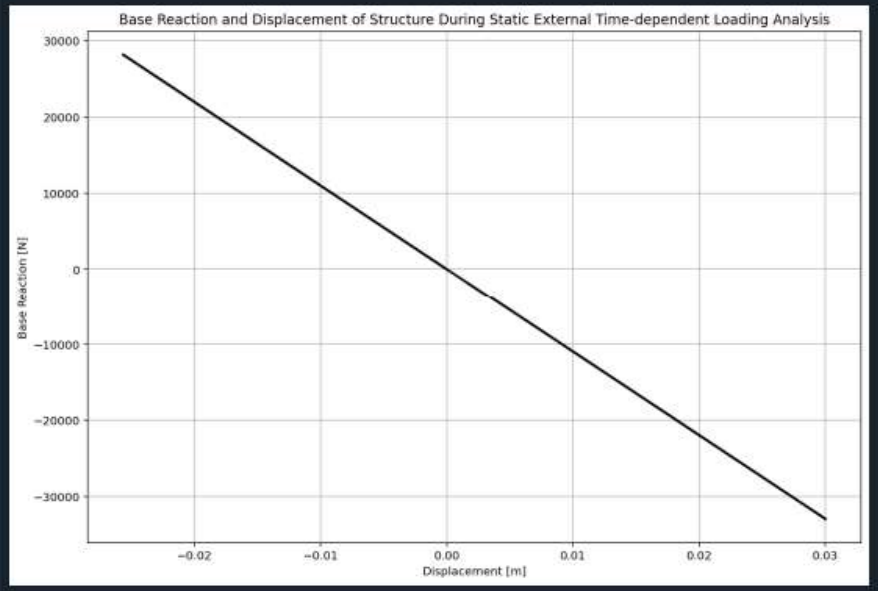
...ISSIPATION_CAPACITY_INDEX\54_ELEVATOR_STRUCTURE_MDOF_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..._MDOF_8_ANA\P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py

P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py | COMPUTE_EFFECTIVE_PROPERTIES_FUN.py

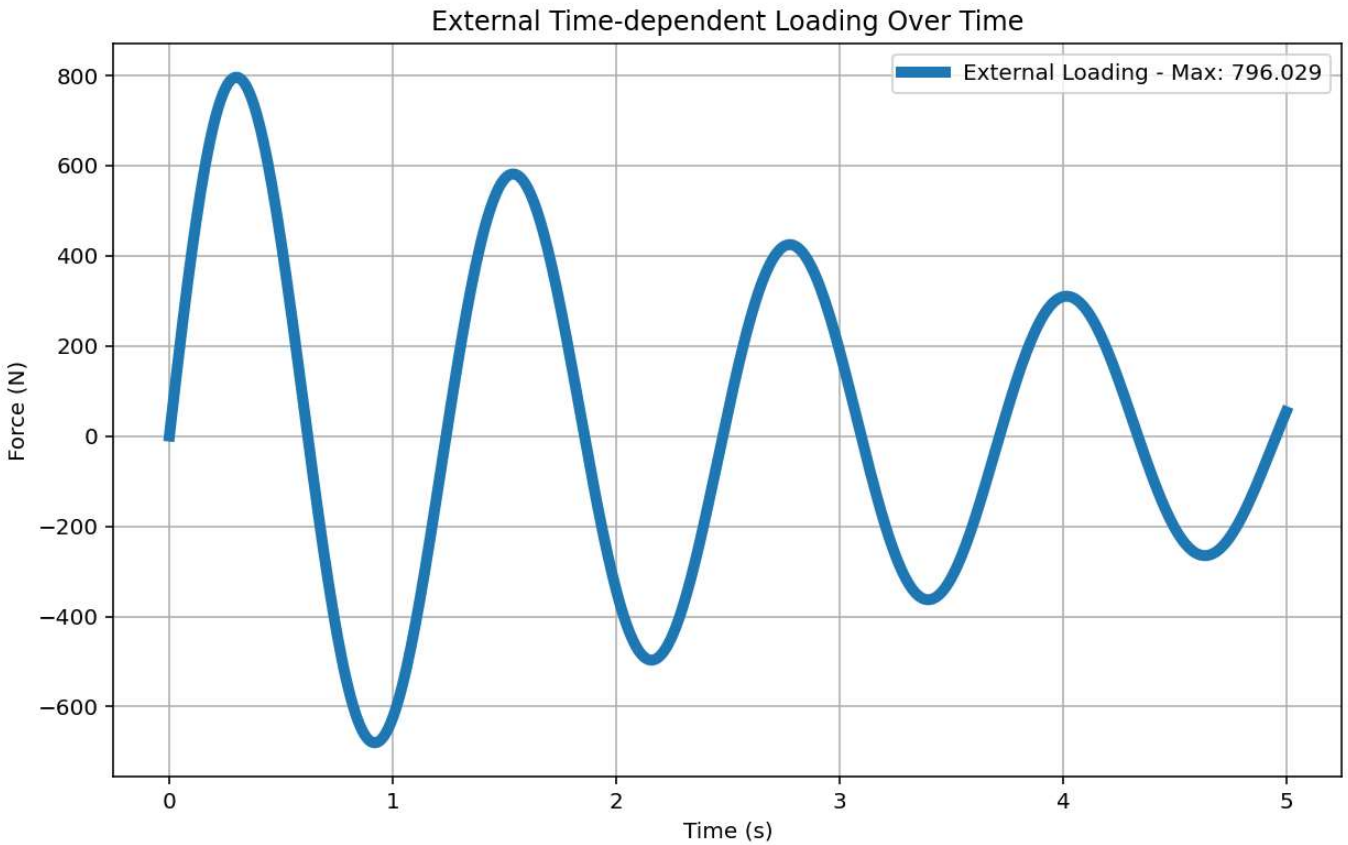
```
431     if ANAL_TYPE == 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING': # STATIC TIME
432         #%% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
433         # IN HERE ANALYSIS TIME AND DURATION ARE LOAD STEPS
434         duration = 20.0             # [s] Analysis duration
435         dt = 0.01                   # [s] Time step
436         DT = dt                     # [s] Time step
437         DT_time = 5.0               # [s] Total external Load Analysis Dura
438         W_kg = MASS[1]#2500         # [kg] Total weight (cabin + capacity +
439         v_mps = 2.0                 # [m/s] Impact velocity
440         S_m = 0.45                  # [m] Buffer stroke
441         g = 9.81                    # [m/s²] standard acceleration due to g
442         W = W_kg * g                # Weight in Newtons
443         # Maximum impact force using buffer formula
444         R_max = W * (1 + (v_mps**2) / (2 * g * S_m))
445         force_amplitude = R_max     # [N] Amplitude Force
446         omega_DT = 5.0715           # [rad/s] Natural angular frequency
447         time_steps = int(duration/dt)
448
449         # Check function
450         def CHECK_FUN(DT_time ,duration):
451             if DT_time > duration:
452                 print('\n\nAnalysis Duration Must be greater than External
453                 exit()
454             return -1
455
456         CHECK_FUN(DT_time ,duration)
457         def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time):
458             import numpy as np
459             import matplotlib.pyplot as plt
460             # External Load Durations
461             num_steps = int(DT_time / DT)
462             load_time = np.linspace(0, DT_time, num_steps)
463             target_frequency = 1.0 * omega_DT  # Target excitation frequenc
464             DT_load = force_amplitude * np.sin(target_frequency * load_time
```

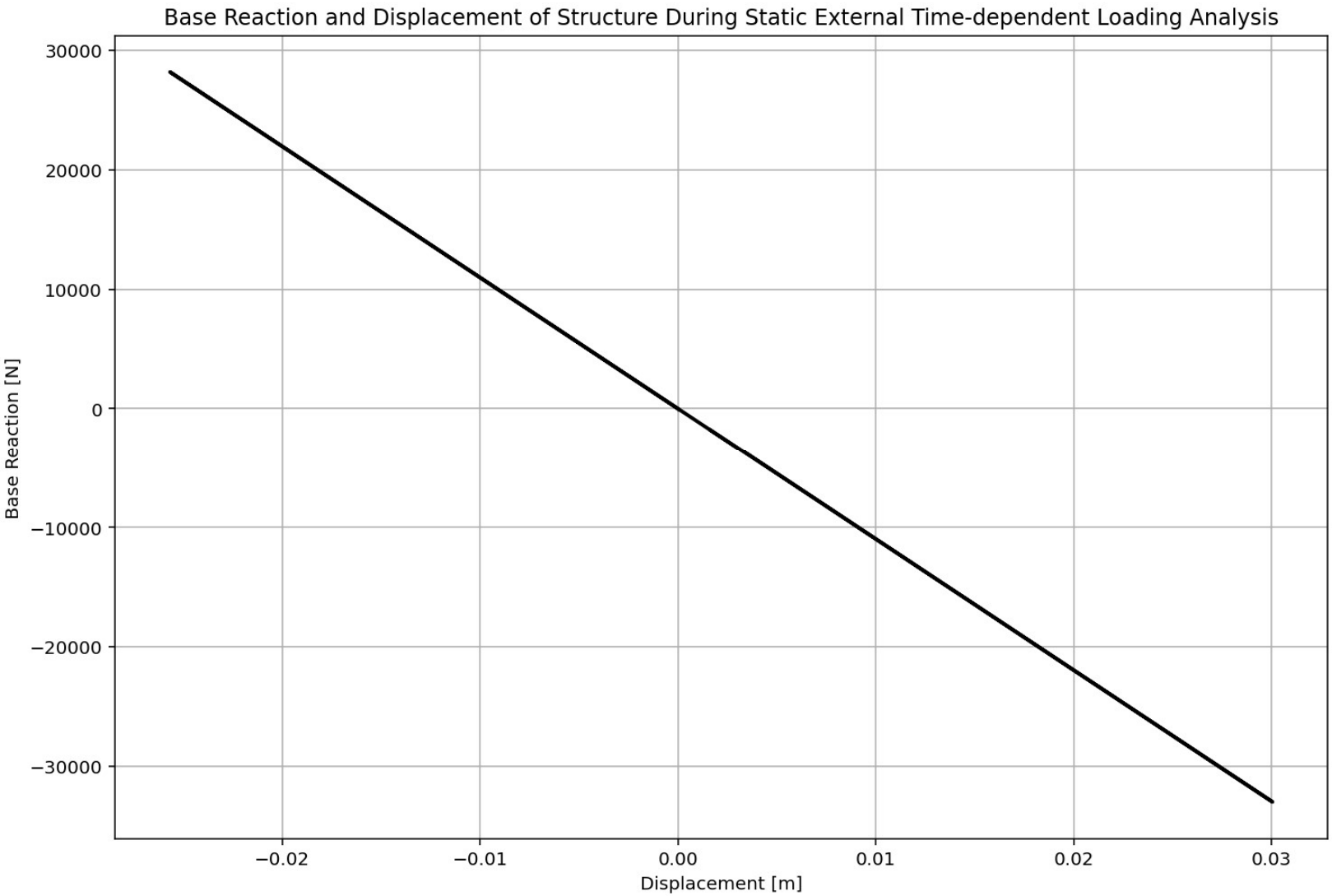


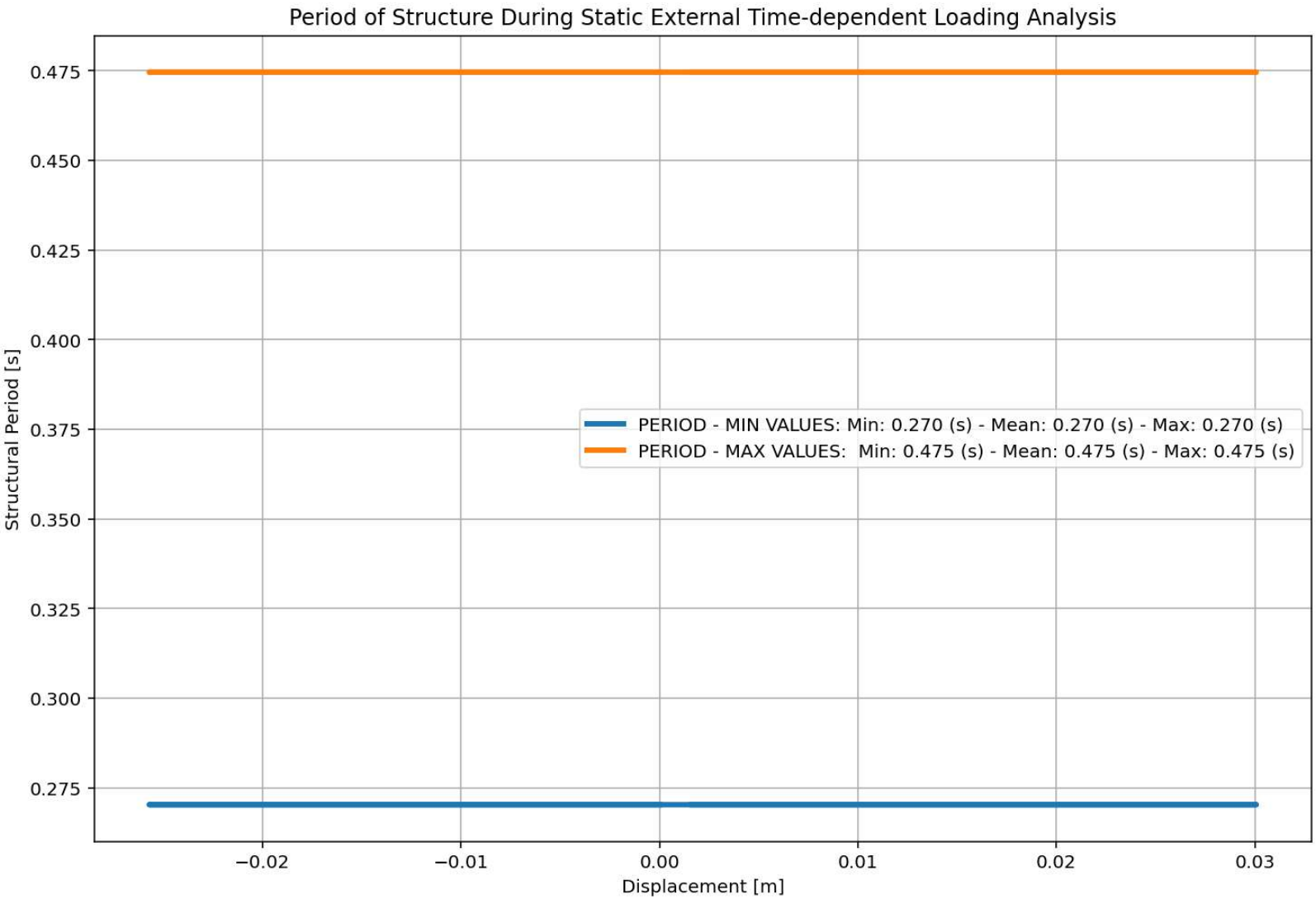
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

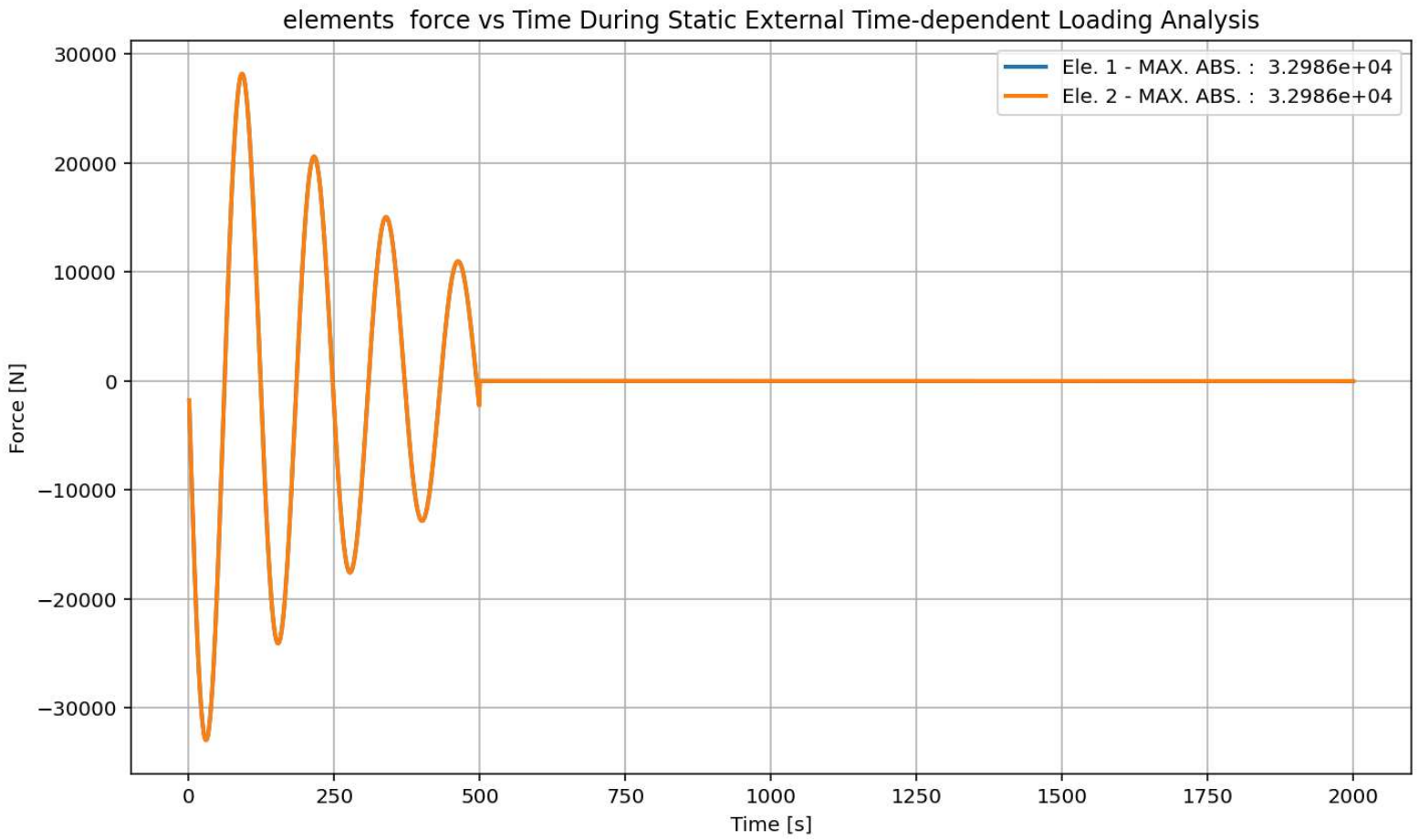
Browse a working directory

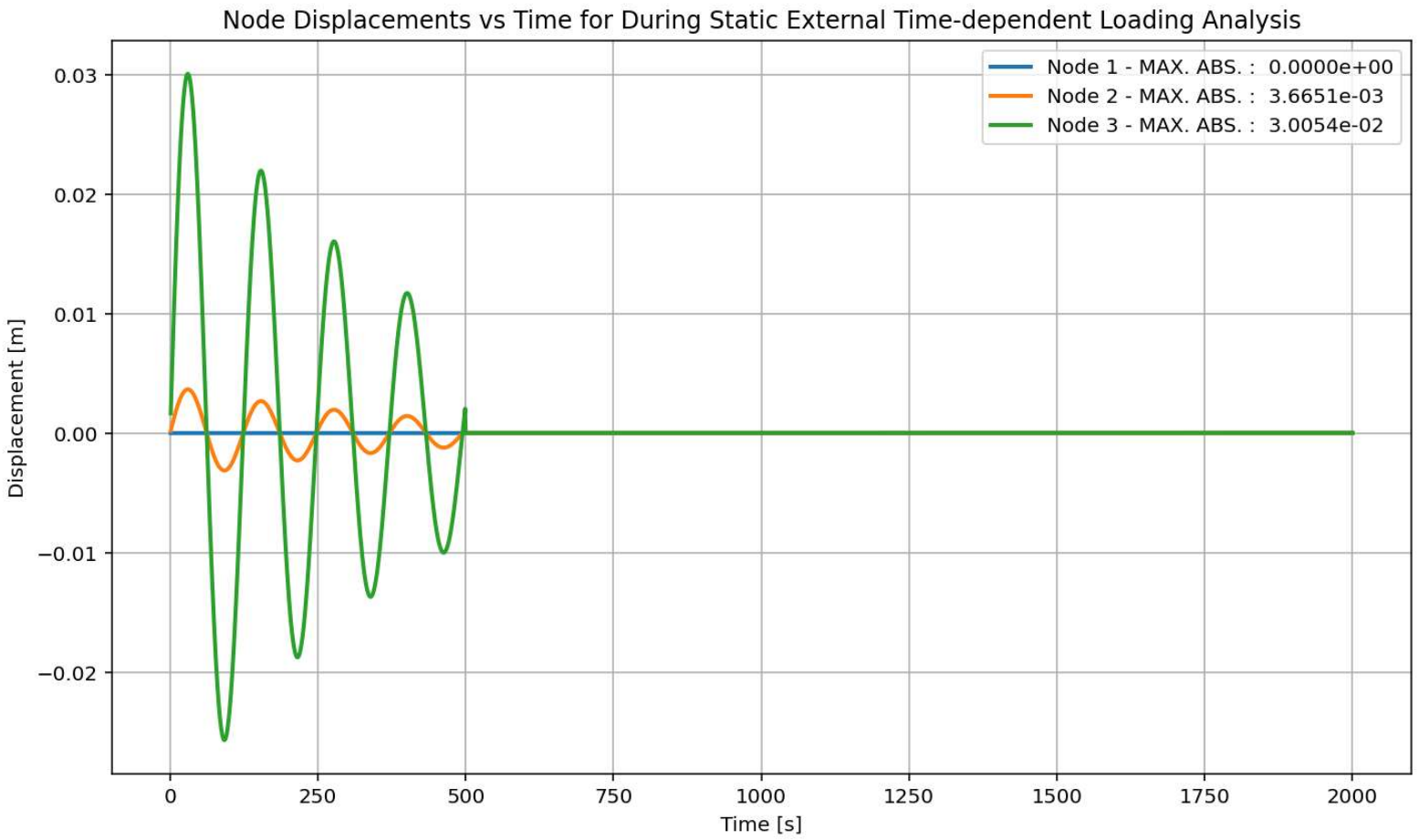
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 362, Col 44 | UTF-8 | CRLF | RW | Mem 51%

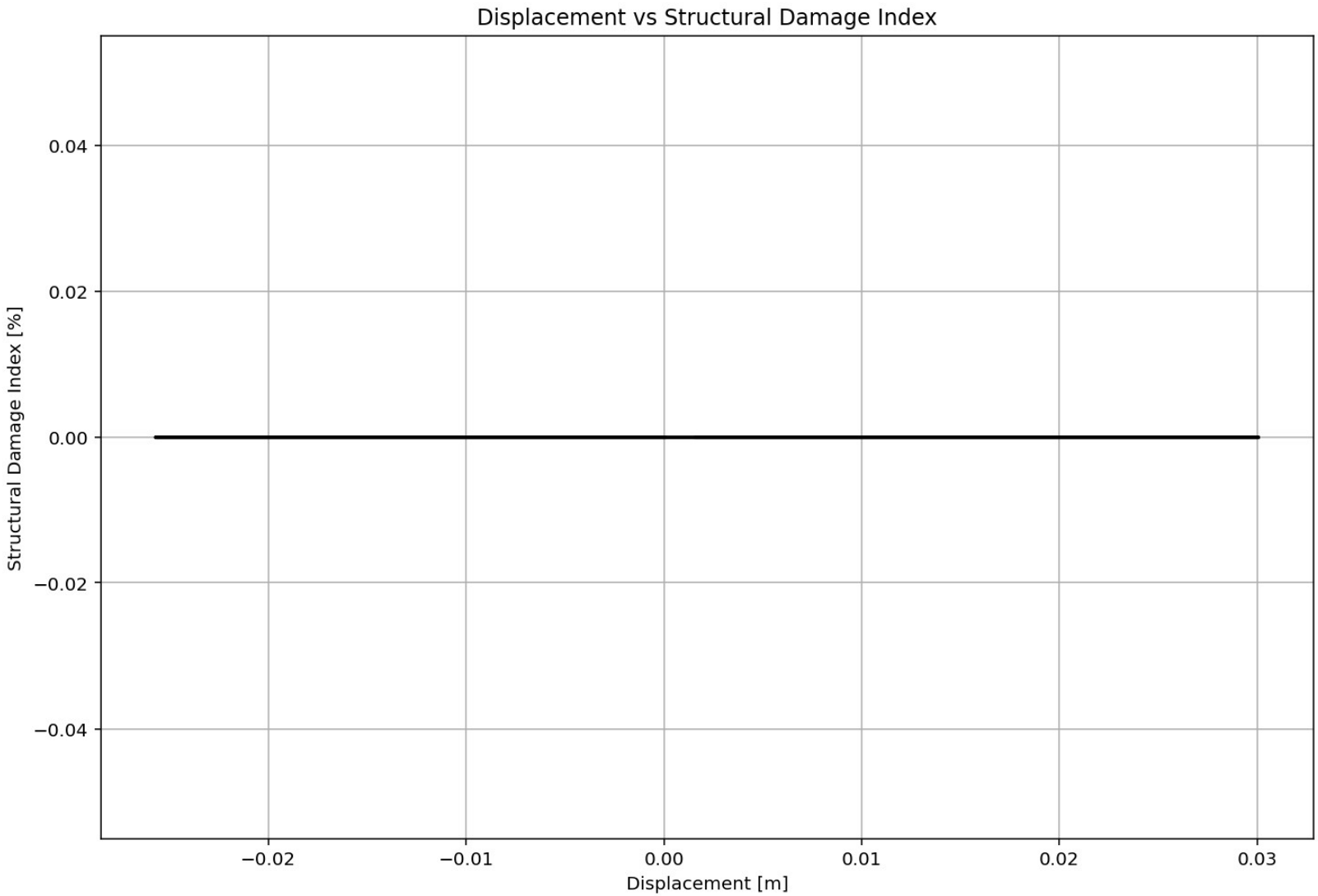

External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5
Time (s)


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
30000
20000
10000
0
−10000
−20000
−30000
−0.02
−0.01
0.00
0.01
0.02
0.03
Displacement [m]


Period of Structure During Static External Time-dependent Loading Analysis
Structural Period [s]
0.475
0.450
0.425
0.400
0.375
0.350
0.325
0.300
0.275
PERIOD - MIN VALUES: Min: 0.270 (s) - Mean: 0.270 (s) - Max: 0.270 (s)
PERIOD - MAX VALUES: Min: 0.475 (s) - Mean: 0.475 (s) - Max: 0.475 (s)
−0.02
−0.01
0.00
0.01
0.02
0.03
Displacement [m]


elements force vs Time During Static External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 3.2986e+04
Ele. 2 - MAX. ABS. : 3.2986e+04
Force [N]
30000
20000
10000
0
−10000
−20000
−30000
0
250
500
750
1000
1250
1500
1750
2000
Time [s]


Node Displacements vs Time for During Static External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.6651e-03
Node 3 - MAX. ABS. : 3.0054e-02
Displacement [m]
0.03
0.02
0.01
0.00
−0.01
−0.02
0
250
500
750
1000
1250
1500
1750
2000
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.02
−0.01
0.00
0.01
0.02
0.03
Displacement [m]

DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

Spyder (Python 3.12)

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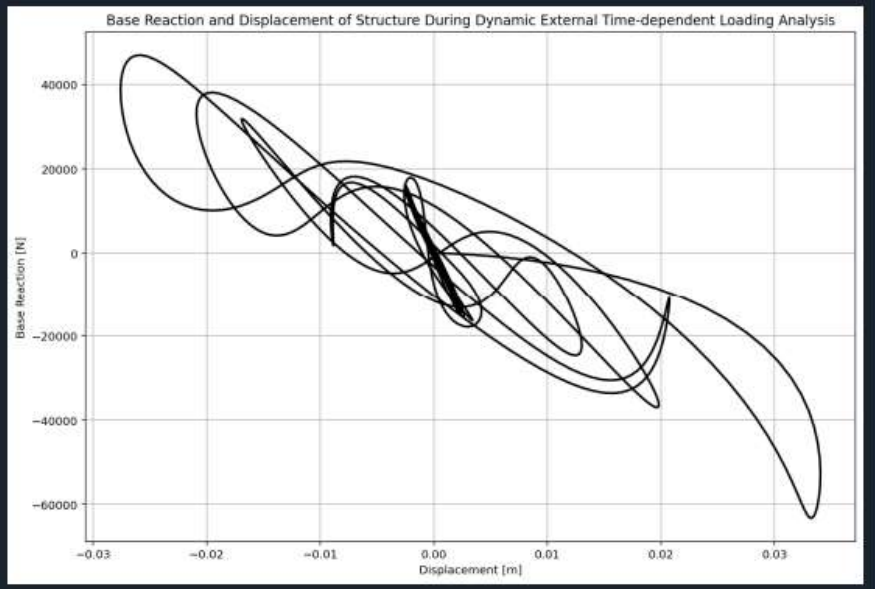
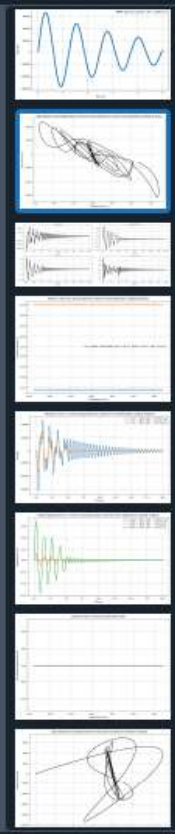
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..._MDOF_8_ANA\P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py

P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py | COMPUTE_EFFECTIVE_PROPERTIES_FUN.py

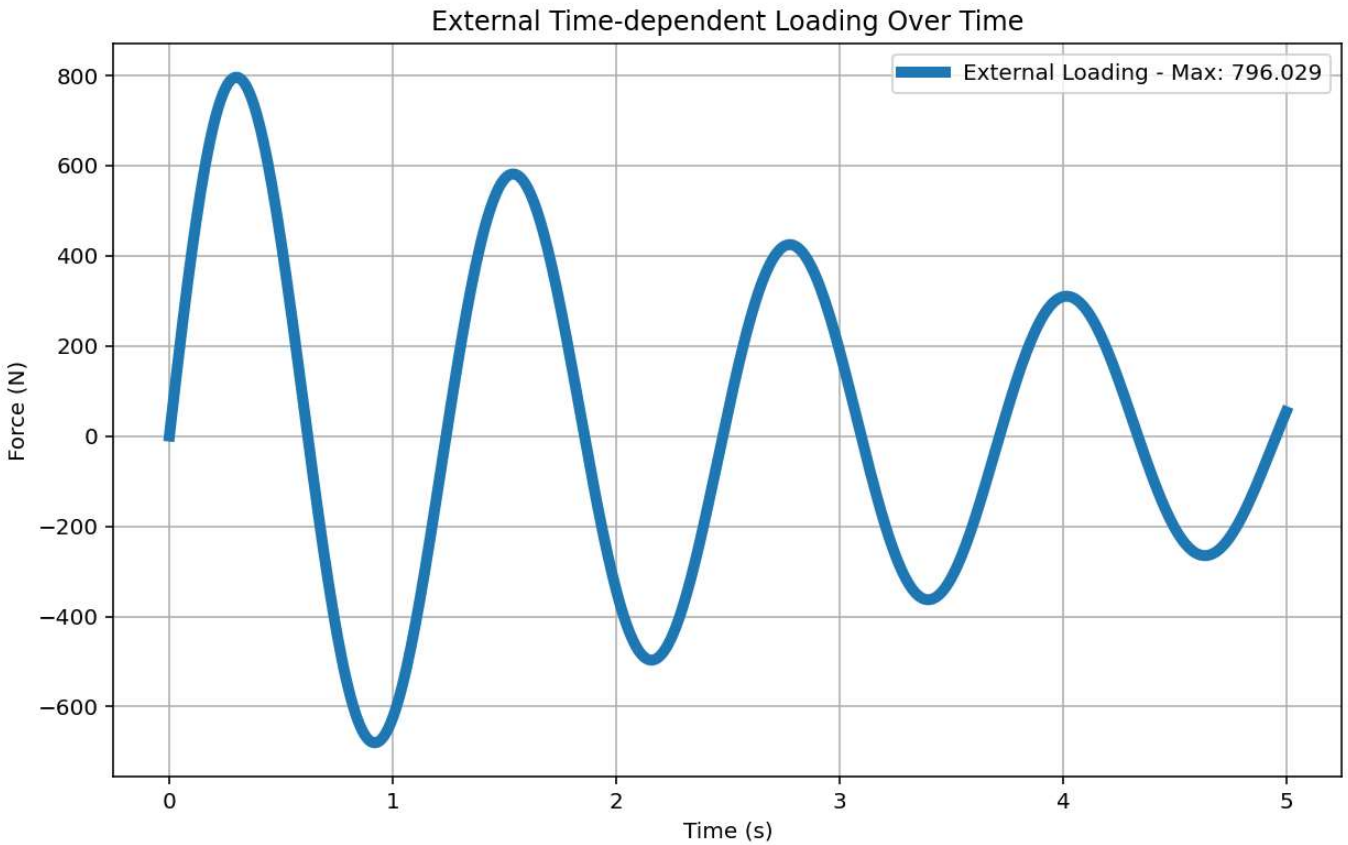
```
551     if ANAL_TYPE == 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING': # DYNAMIC TI
552         #%% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
553         # IN HERE ANALYSIS TIME AND DURATION ARE LOAD STEPS
554         duration = 20.0             # [s] Analysis duration
555         dt = 0.01                   # [s] Time step
556         DT = dt                     # [s] Time step
557         DT_time = 5.0               # [s] Total external Load Analysis Dura
558         W_kg = MASS[1]#2500         # [kg] Total weight (cabin + capacity +
559         v_mps = 2.0                 # [m/s] Impact velocity
560         S_m = 0.45                  # [m] Buffer stroke
561         g = 9.81                    # [m/s²] standard acceleration due to g
562         W = W_kg * g                # Weight in Newtons
563         # Maximum impact force using buffer formula
564         R_max = W * (1 + (v_mps**2) / (2 * g * S_m))
565         force_amplitude = R_max     # [N] Amplitude Force
566         omega_DT = 5.0715           # [rad/s] Natural angular frequency
567         time_steps = int(duration/dt)
568
569         # Check function
570         def CHECK_FUN(DT_time ,duration):
571             if DT_time > duration:
572                 print('\n\nAnalysis Duration Must be greater than External
573                 exit()
574             return -1
575
576         CHECK_FUN(DT_time ,duration)
577         def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time):
578             import numpy as np
579             import matplotlib.pyplot as plt
580             # External Load Durations
581             num_steps = int(DT_time / DT)
582             load_time = np.linspace(0, DT_time, num_steps)
583             target_frequency = 1.0 * omega_DT  # Target excitation frequenc
584             DT_load = force_amplitude * np.sin(target_frequency * load_time
```

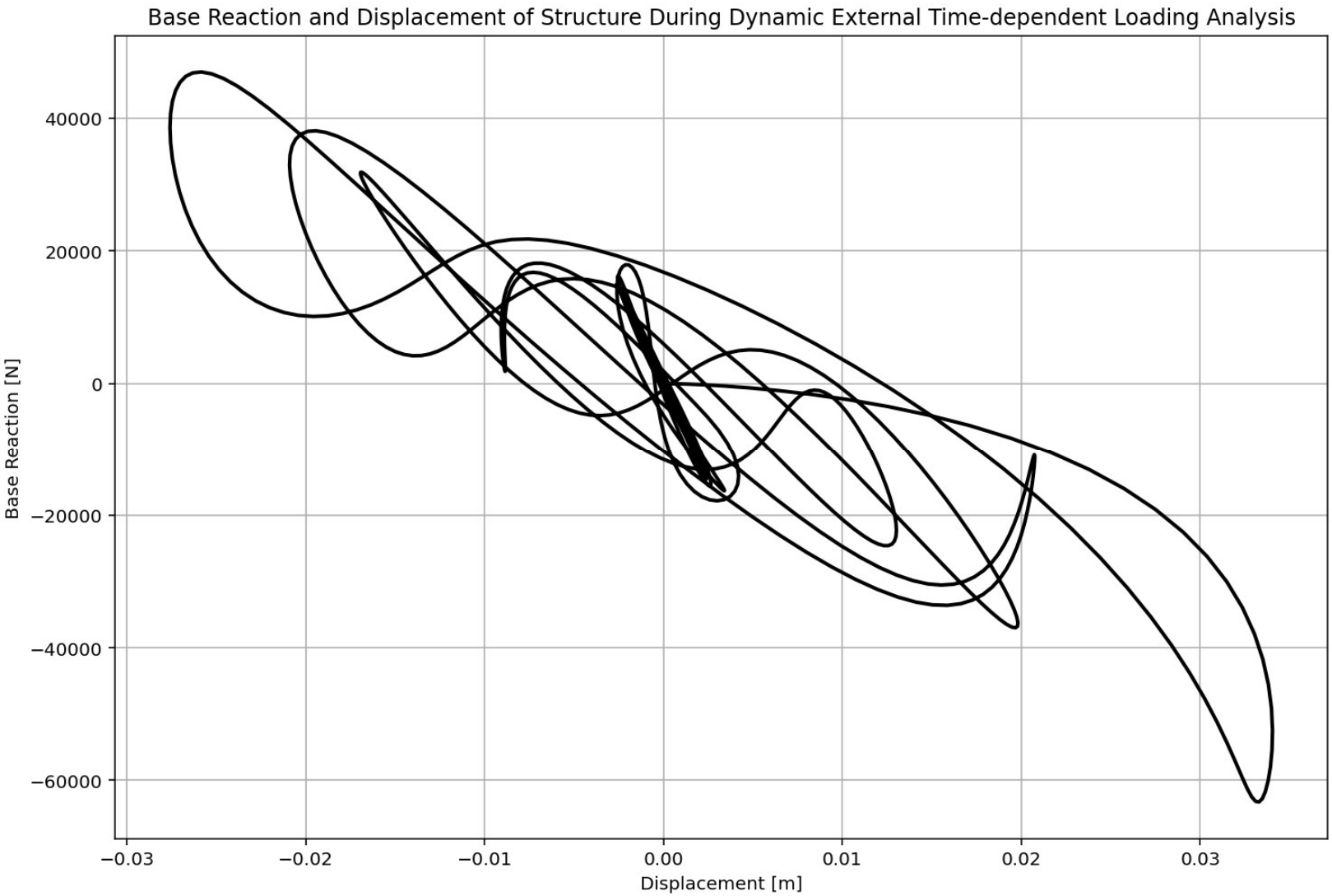


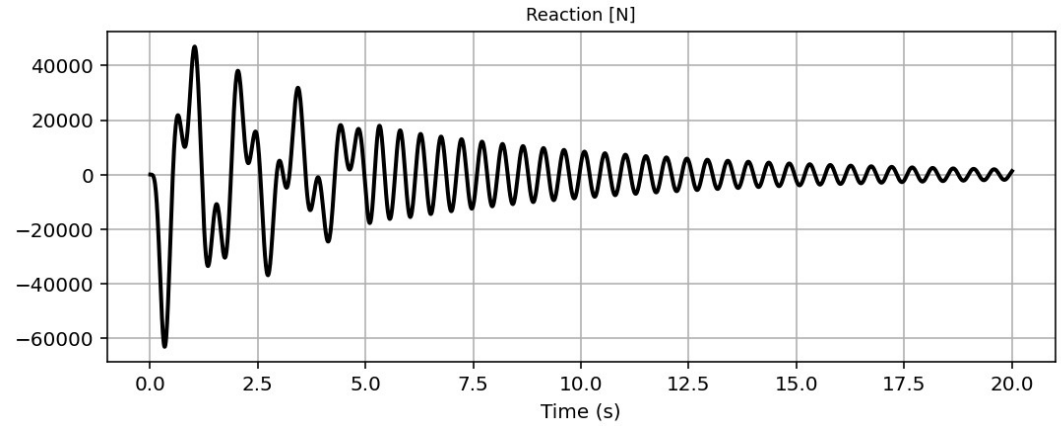


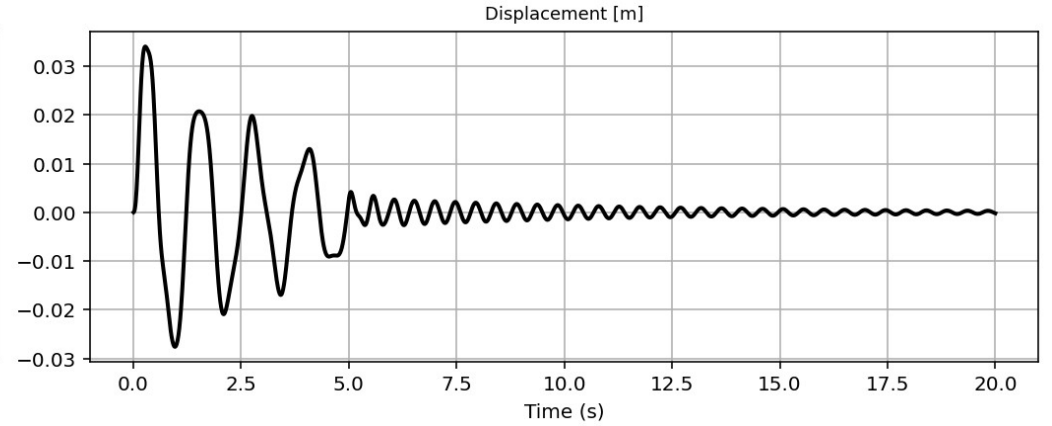
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

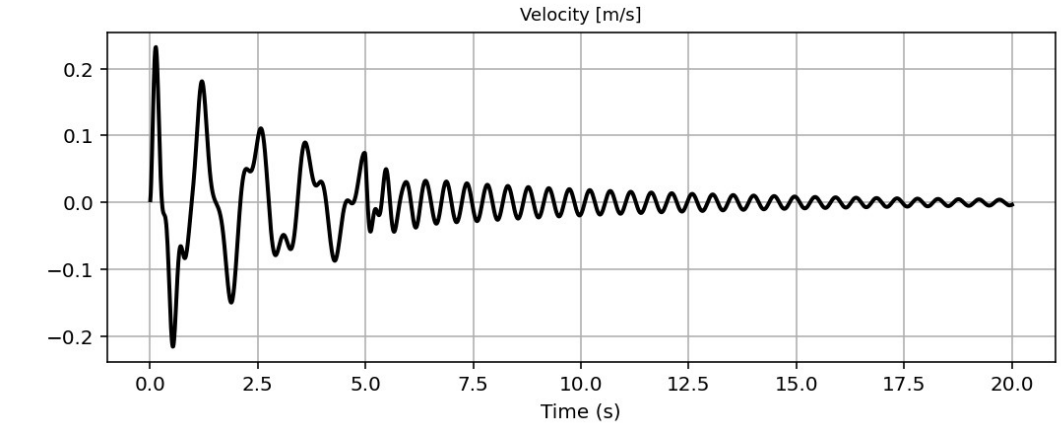
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 362, Col 44 UTF-8 CRLF RW Mem 52%

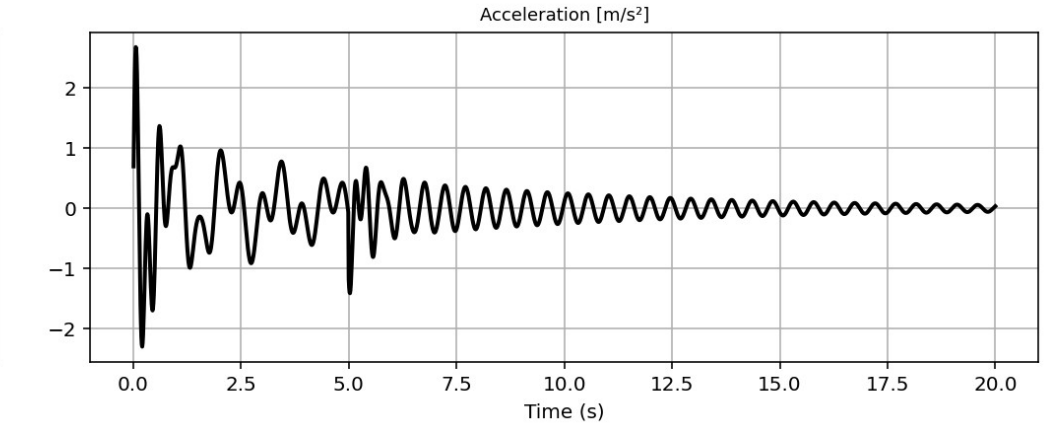

External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5
Time (s)

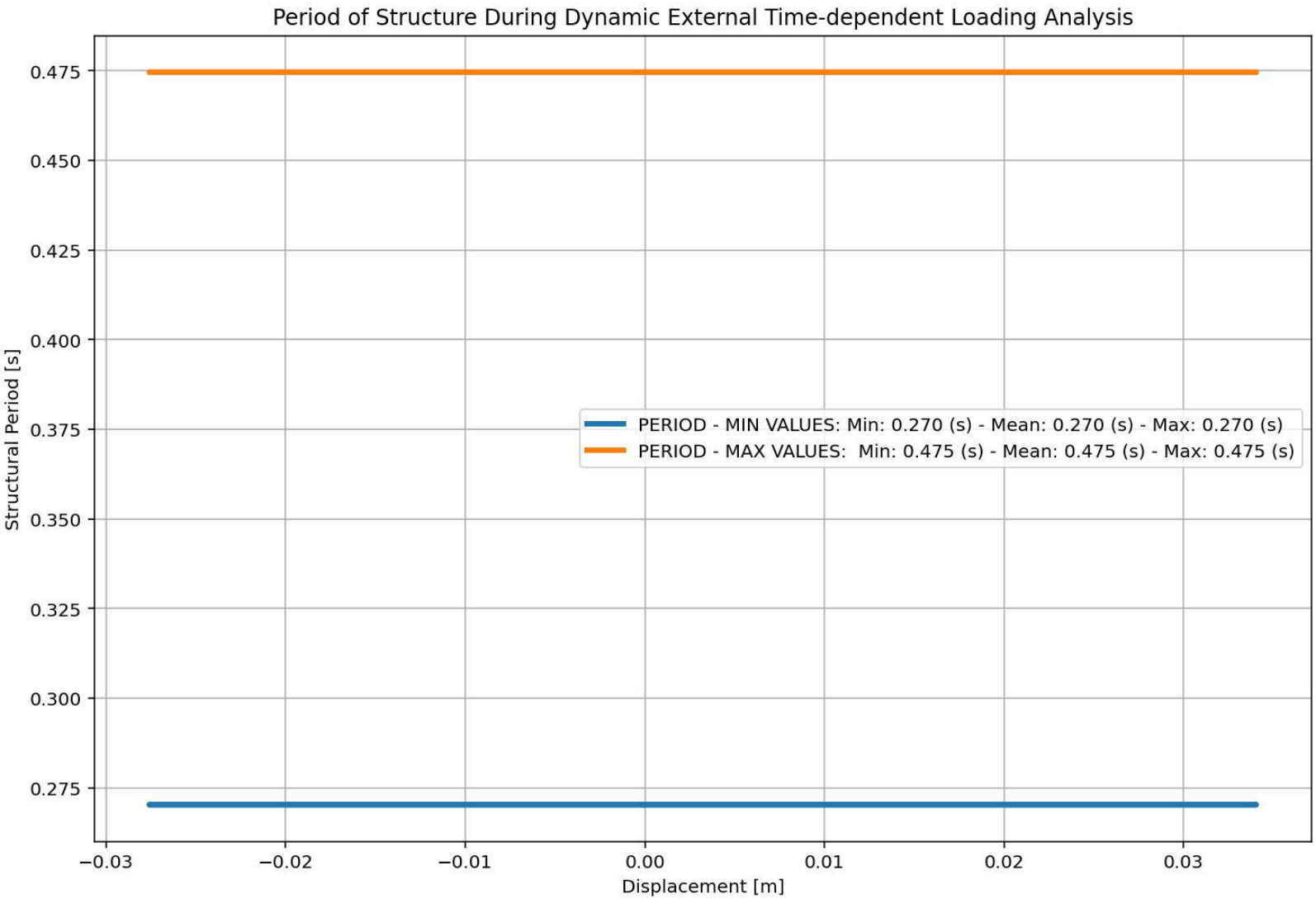

Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
40000
20000
0
−20000
−40000
−60000
−0.03
−0.02
−0.01
0.00
0.01
0.02
0.03
Displacement [m]

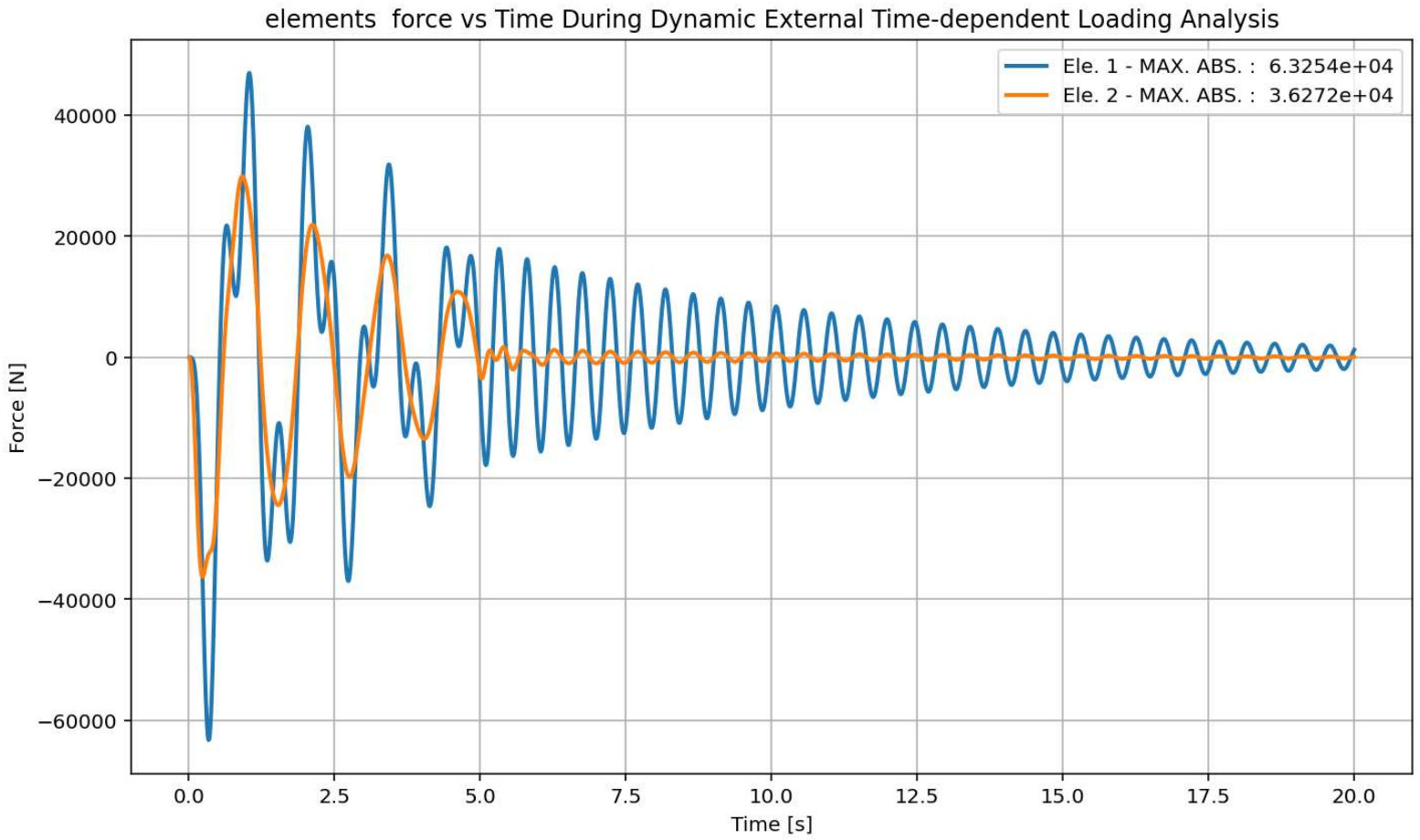

Reaction [N]
40000
20000
0
−20000
−40000
−60000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

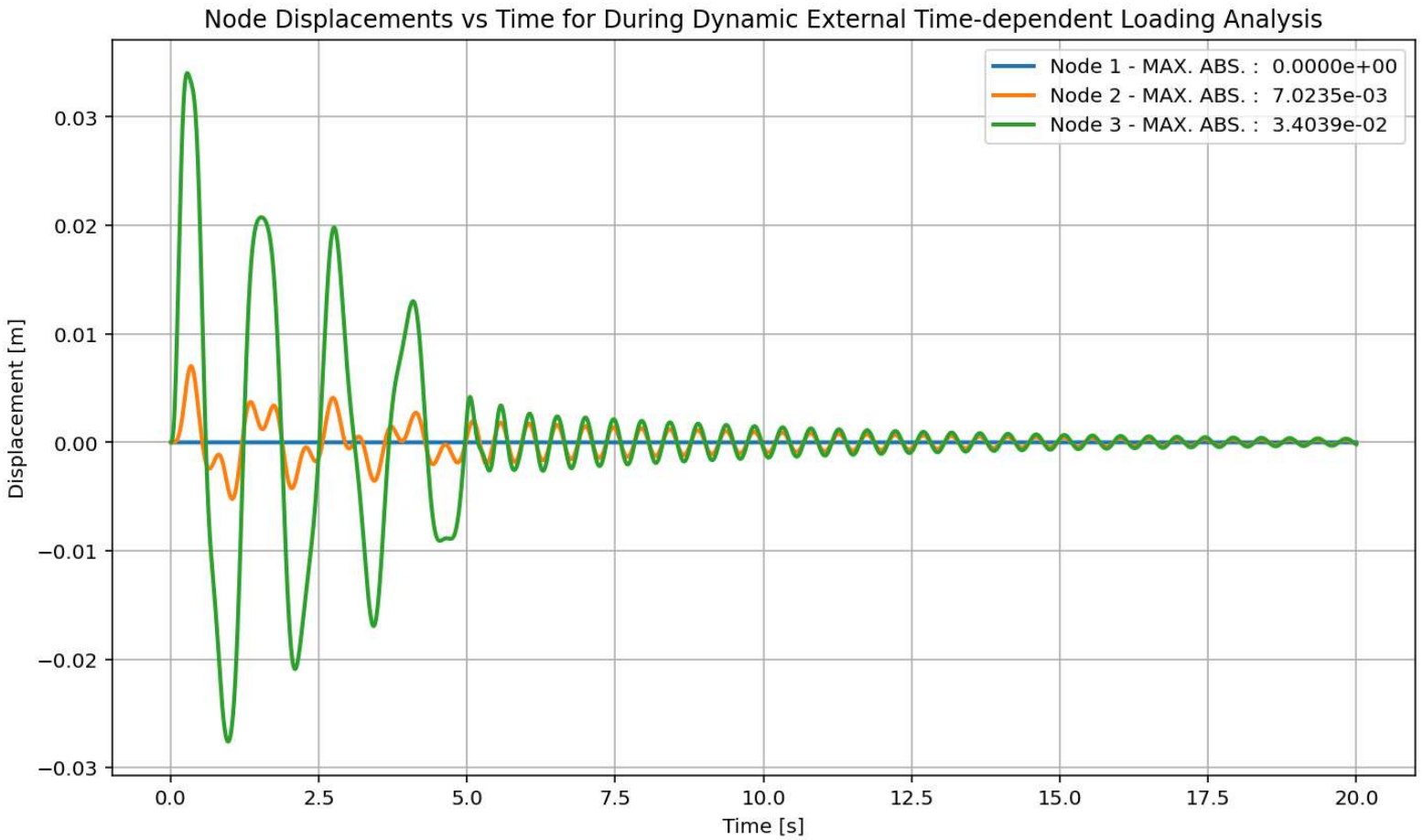

Displacement [m]
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

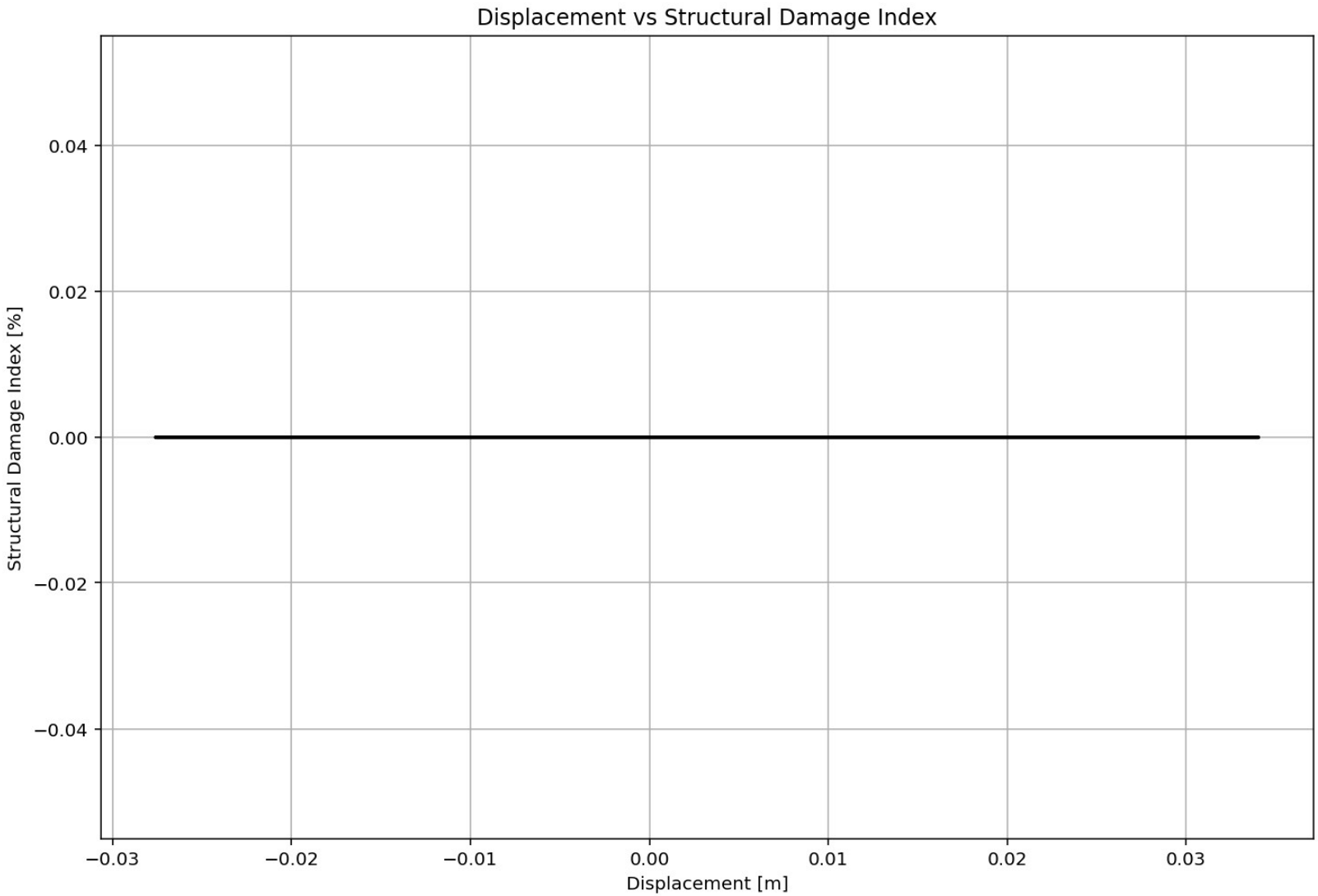

Velocity [m/s]
0.2
0.1
0.0
−0.1
−0.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


Acceleration [m/s²]
2
1
0
−1
−2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


Period of Structure During Dynamic External Time-dependent Loading Analysis
Structural Period [s]
0.475
0.450
0.425
0.400
0.375
0.350
0.325
0.300
0.275
−0.03
−0.02
−0.01
0.00
0.01
0.02
0.03
Displacement [m]
PERIOD - MIN VALUES: Min: 0.270 (s) - Mean: 0.270 (s) - Max: 0.270 (s)
PERIOD - MAX VALUES: Min: 0.475 (s) - Mean: 0.475 (s) - Max: 0.475 (s)


elements force vs Time During Dynamic External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 6.3254e+04
Ele. 2 - MAX. ABS. : 3.6272e+04
Force [N]
40000
20000
0
−20000
−40000
−60000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Node Displacements vs Time for During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 7.0235e-03
Node 3 - MAX. ABS. : 3.4039e-02
Displacement [m]
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.03
−0.02
−0.01
0.00
0.01
0.02
0.03
Displacement [m]

FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

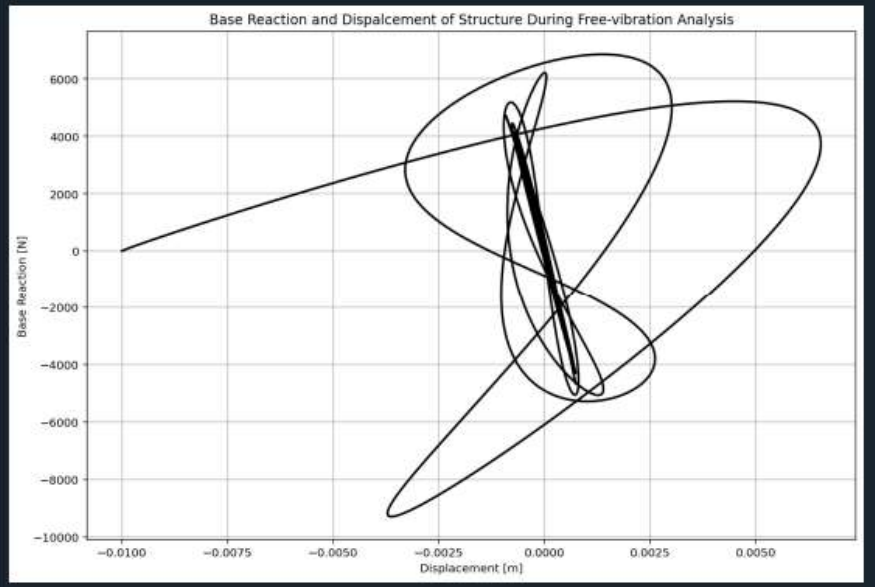
File Edit Search Source Run Debug Consoles Projects Tools View Help

...ISSIPATION_CAPACITY_INDEX\54_ELEVATOR_STRUCTURE_MDOF_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..._MDOF_8_ANA\P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py

P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py | COMPUTE_EFFECTIVE_PROPERTIES_FUN.py

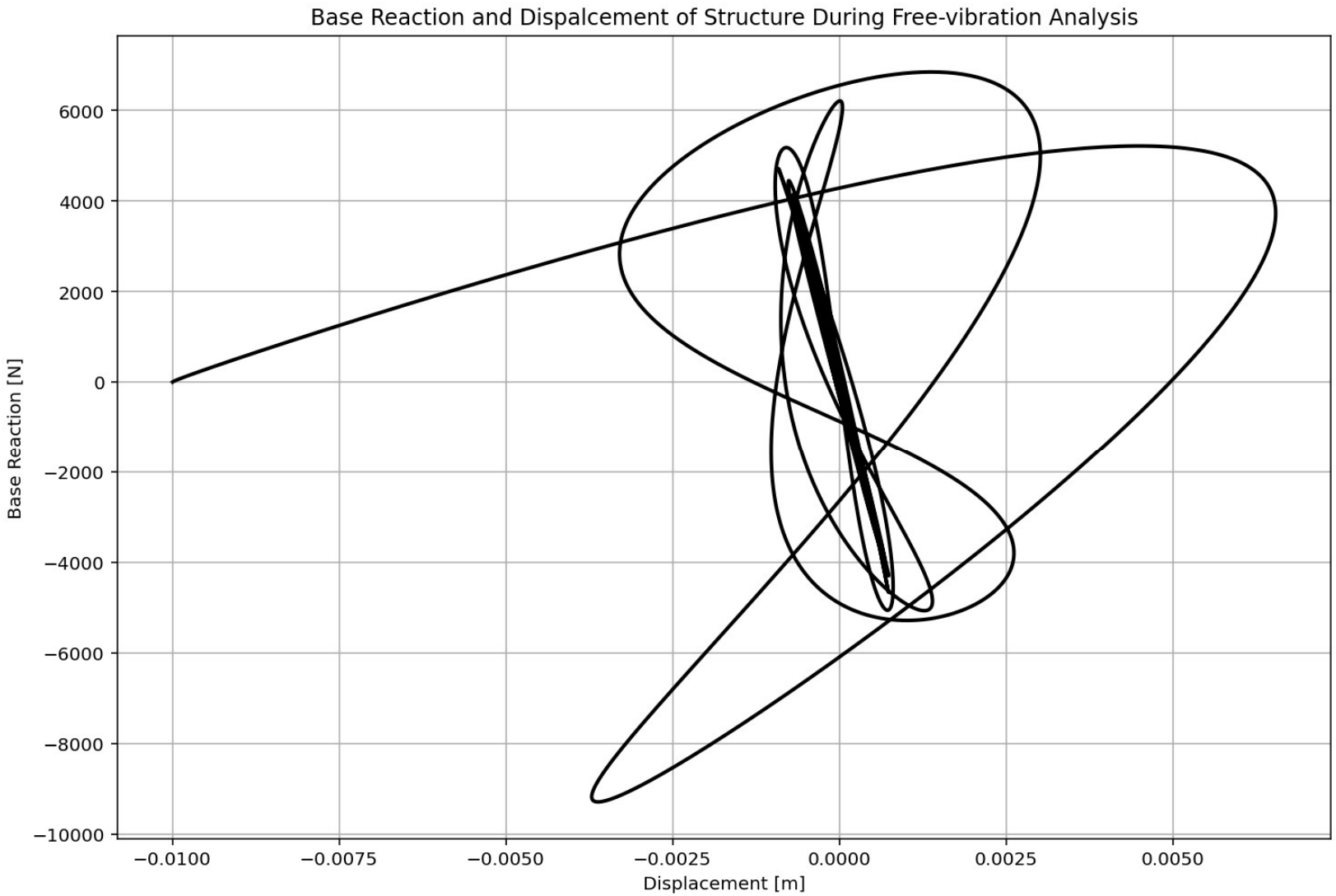
```
687
688     if ANAL_TYPE == 'FREE-VIBRATION': # DYNAMIC TIME-HISTORY ANALYSIS
689         #%% DEFINE PARAMETERS FOR FREE-VIBRATION ANALYSIS
690         u0 = -0.010                     # [m] Initial displacement
691         v0 = 0.000015                   # [m/s] Initial velocity
692         a0 = 0.000065                   # [m/s^2] Initial acceleration
693         IU = True                       # Free Vibration with Initial Di
694         IV = True                       # Free Vibration with Initial Ve
695         IA = True                       # Free Vibration with Initial Ac
696         duration = 20.0                 # [s] Analysis duration
697         dt = 0.001                      # [s] Time step
698         DR = 0.03                       # Damping Ratio
699
700         if IU == True:
701             # Define initial displacment
702             ops.setNodeDisp(center_node, 1, u0, '-commit') # INFO LINK: htt
703         if IV == True:
704             # Define initial velocity
705             ops.setNodeVel(center_node, 1, v0, '-commit')  # INFO LINK: htt
706         if IA == True:
707             # Define initial  acceleration
708             ops.setNodeAccel(center_node, 1, a0, '-commit') # INFO LINK: ht
709
710         ops.constraints('Plain')
711         ops.numberer('Plain')
712         ops.system('BandGeneral')
713         ops.test('NormDispIncr', MAX_TOLERANCE, MAX_ITERATIONS) # INFO LINK
714         #ops.integrator('CentralDifference')  # JUST FOR LINEAR ANALYSIS -
715         alpha=0.5; beta=0.25;
716         ops.integrator('Newmark', alpha, beta) # INFO LINK: https://opensee
717         #alpha=2/3;gamma=1.5-alpha; gamma=1.5-alpha;beta=(2-alpha)**2/4;
718         #ops.integrator('HHT', alpha, gamma, beta) # INFO LINK: https://ope
719         ops.algorithm('Newton')  # INFO LINK: https://openseespydoc.readthe
720         ops.analysis('Transient') # INFO LINK: https://openseespydoc.readth
```

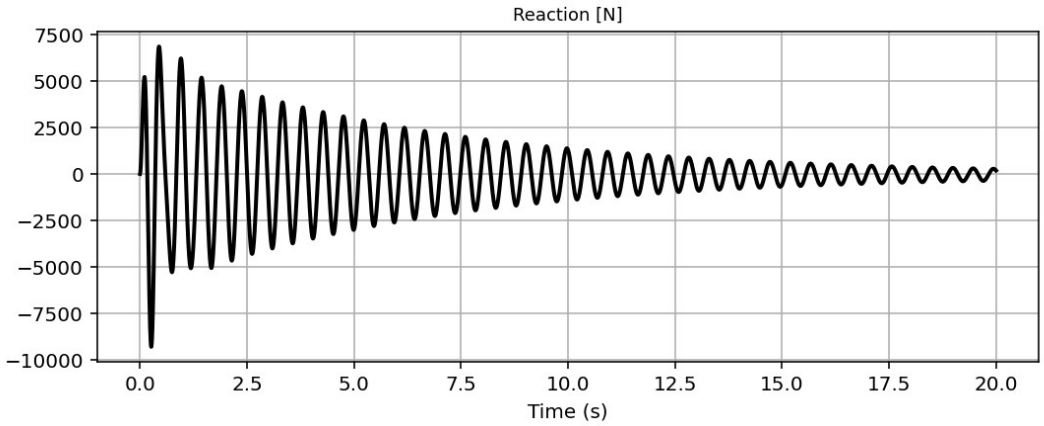


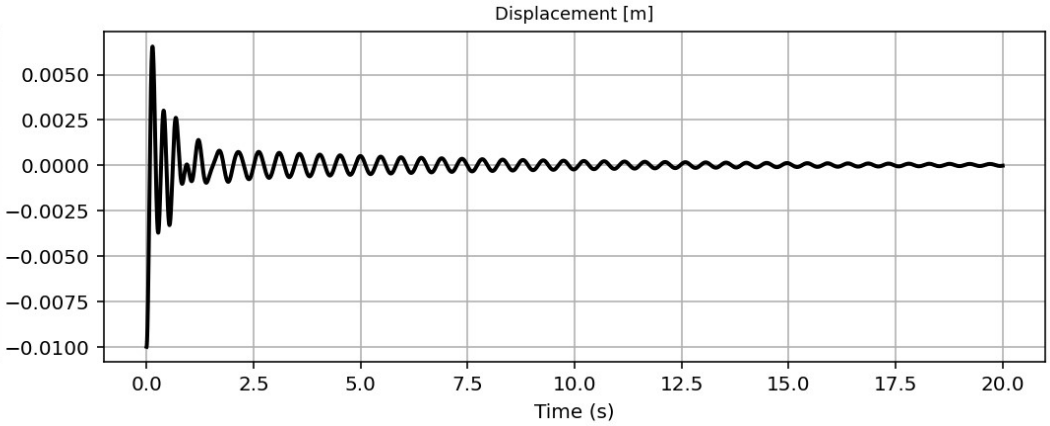
37 %

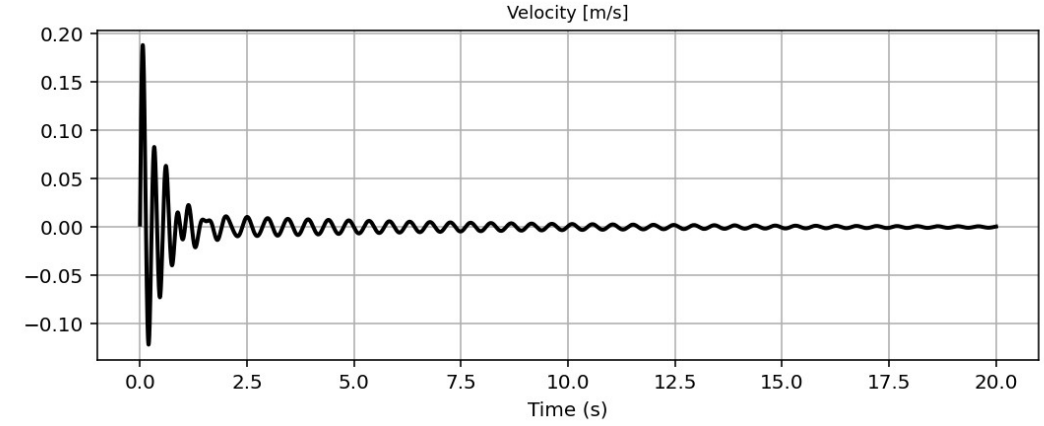
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

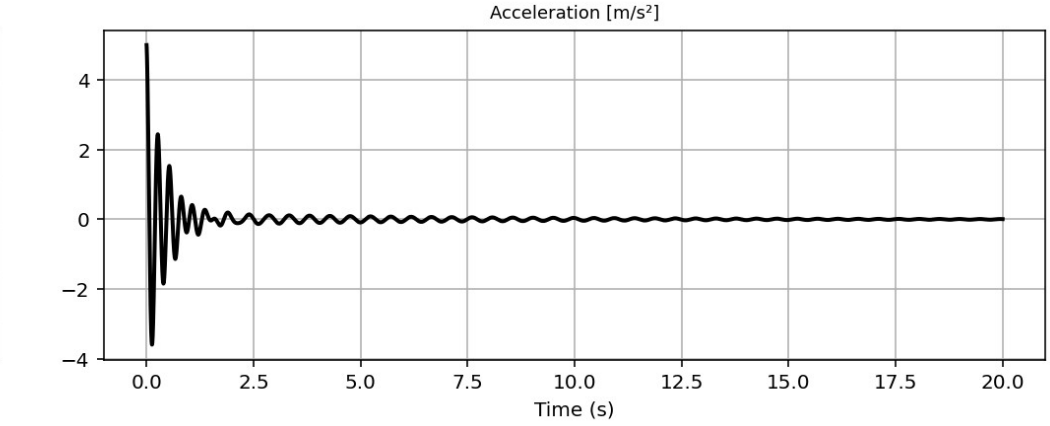
Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 362, Col 44 | UTF-8 | CRLF | RW | Mem 53%

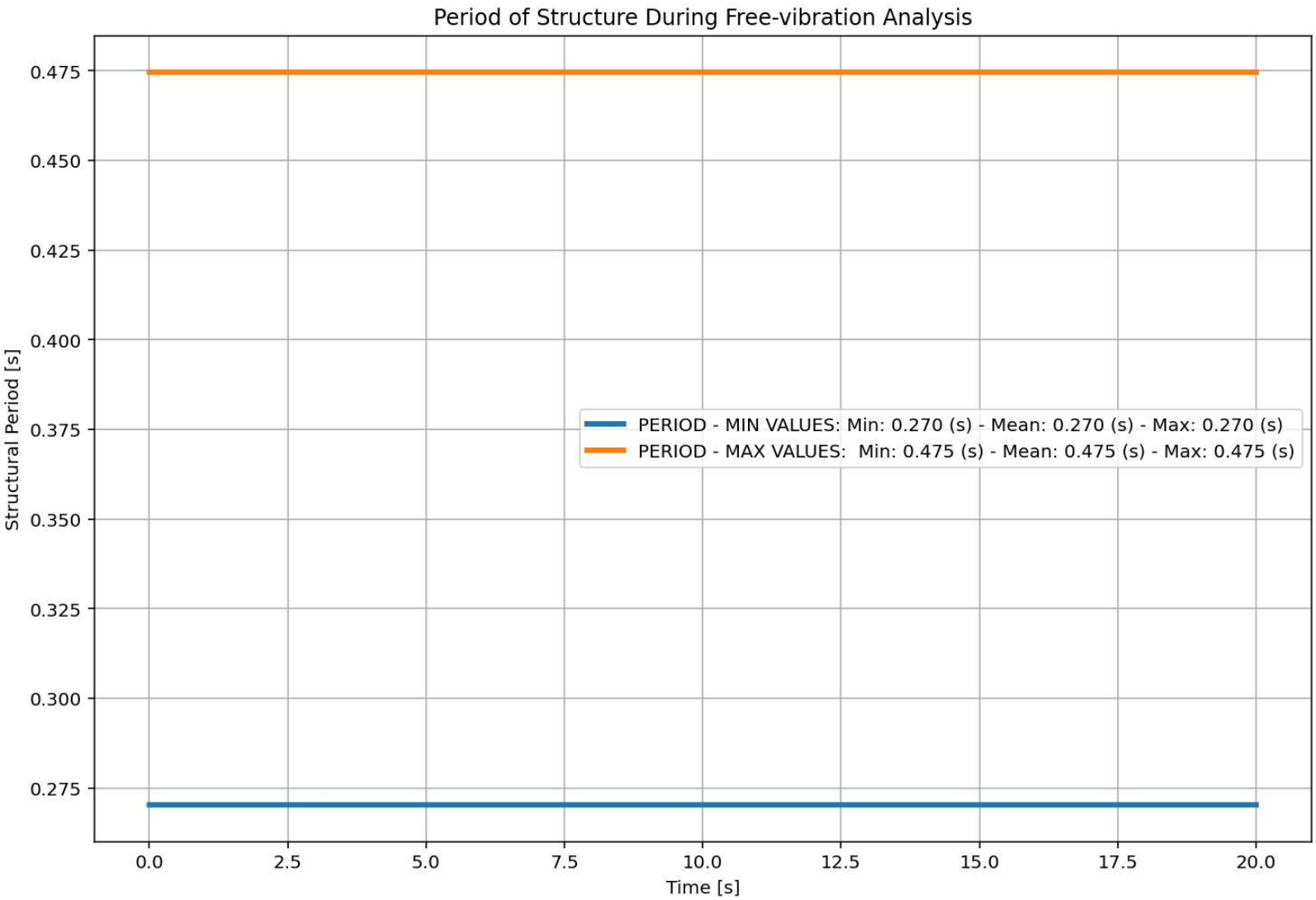

Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
6000
4000
2000
0
−2000
−4000
−6000
−8000
−10000
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
Displacement [m]

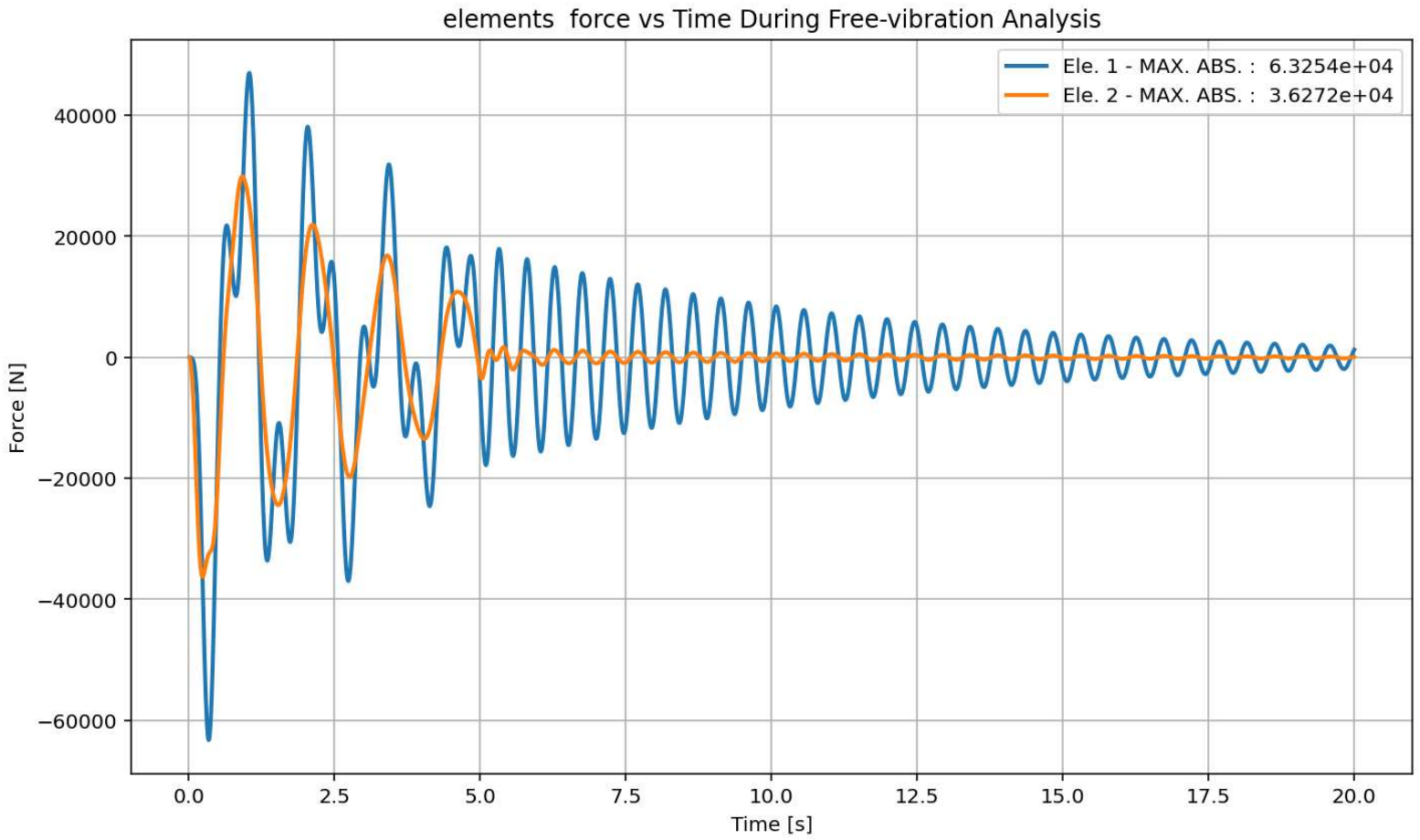

Reaction [N]
7500
5000
2500
0
−2500
−5000
−7500
−10000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

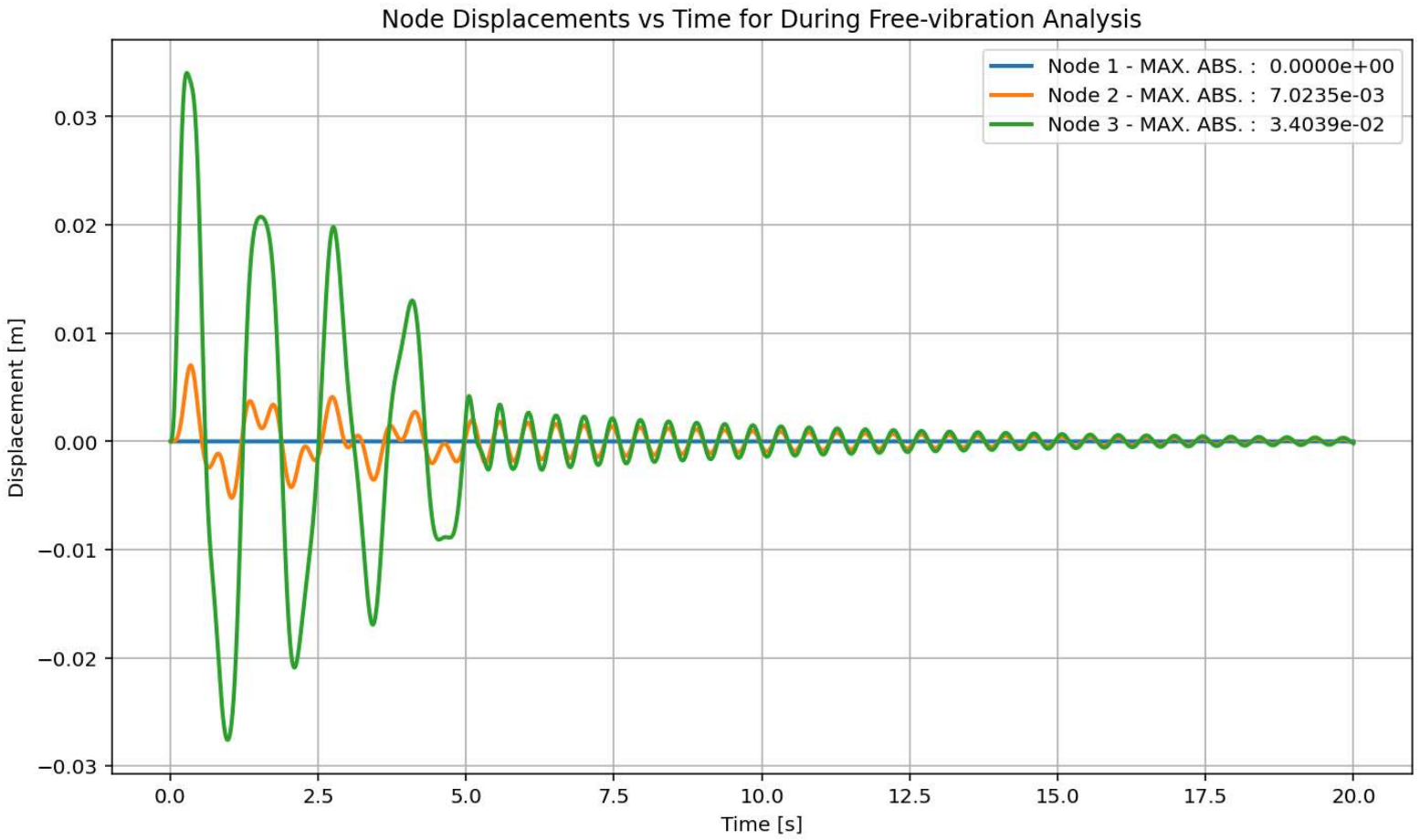

Displacement [m]
0.0050
0.0025
0.0000
−0.0025
−0.0050
−0.0075
−0.0100
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

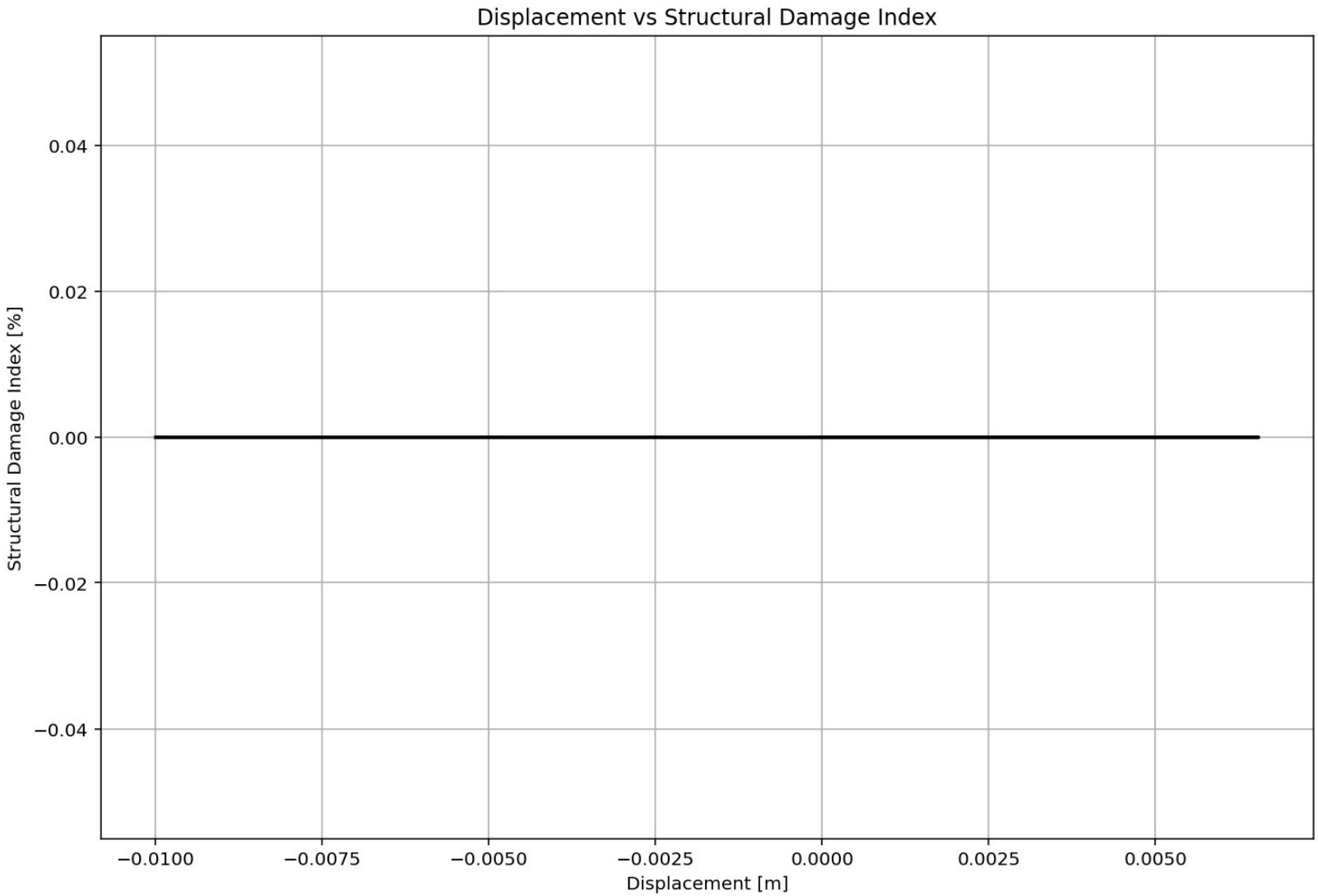

Velocity [m/s]
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

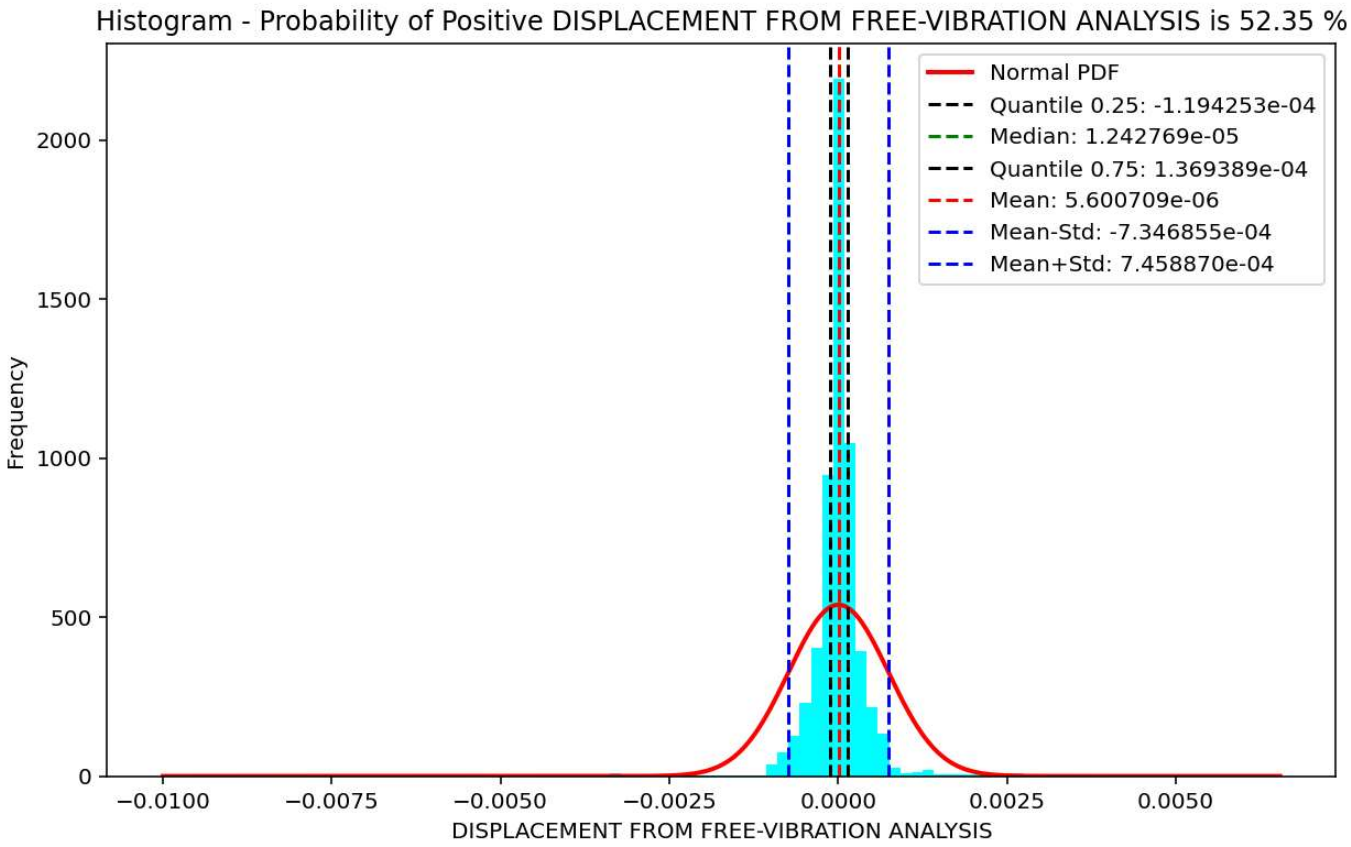

Acceleration [m/s²]
4
2
0
−2
−4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

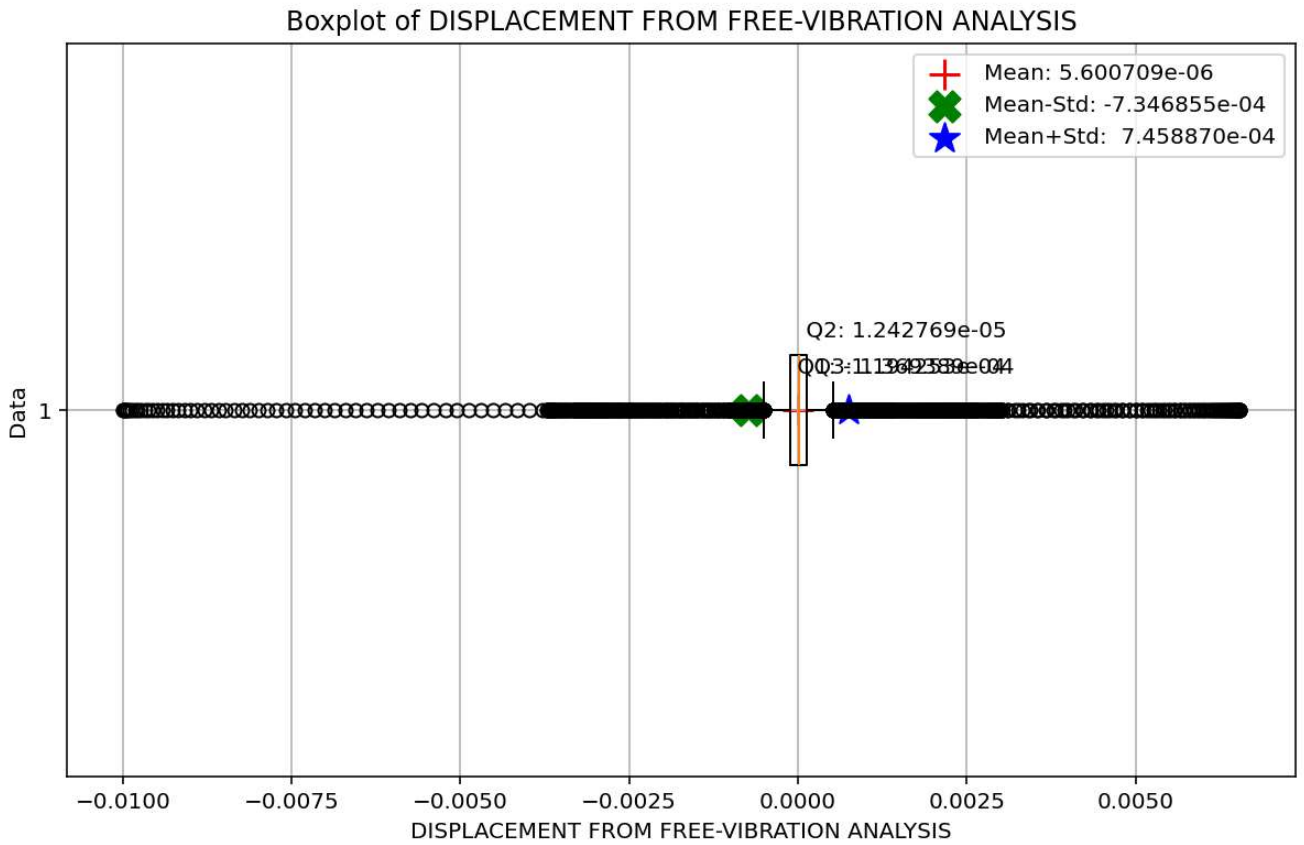

Period of Structure During Free-vibration Analysis
Structural Period [s]
0.475
0.450
0.425
0.400
0.375
0.350
0.325
0.300
0.275
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
PERIOD - MIN VALUES: Min: 0.270 (s) - Mean: 0.270 (s) - Max: 0.270 (s)
PERIOD - MAX VALUES: Min: 0.475 (s) - Mean: 0.475 (s) - Max: 0.475 (s)

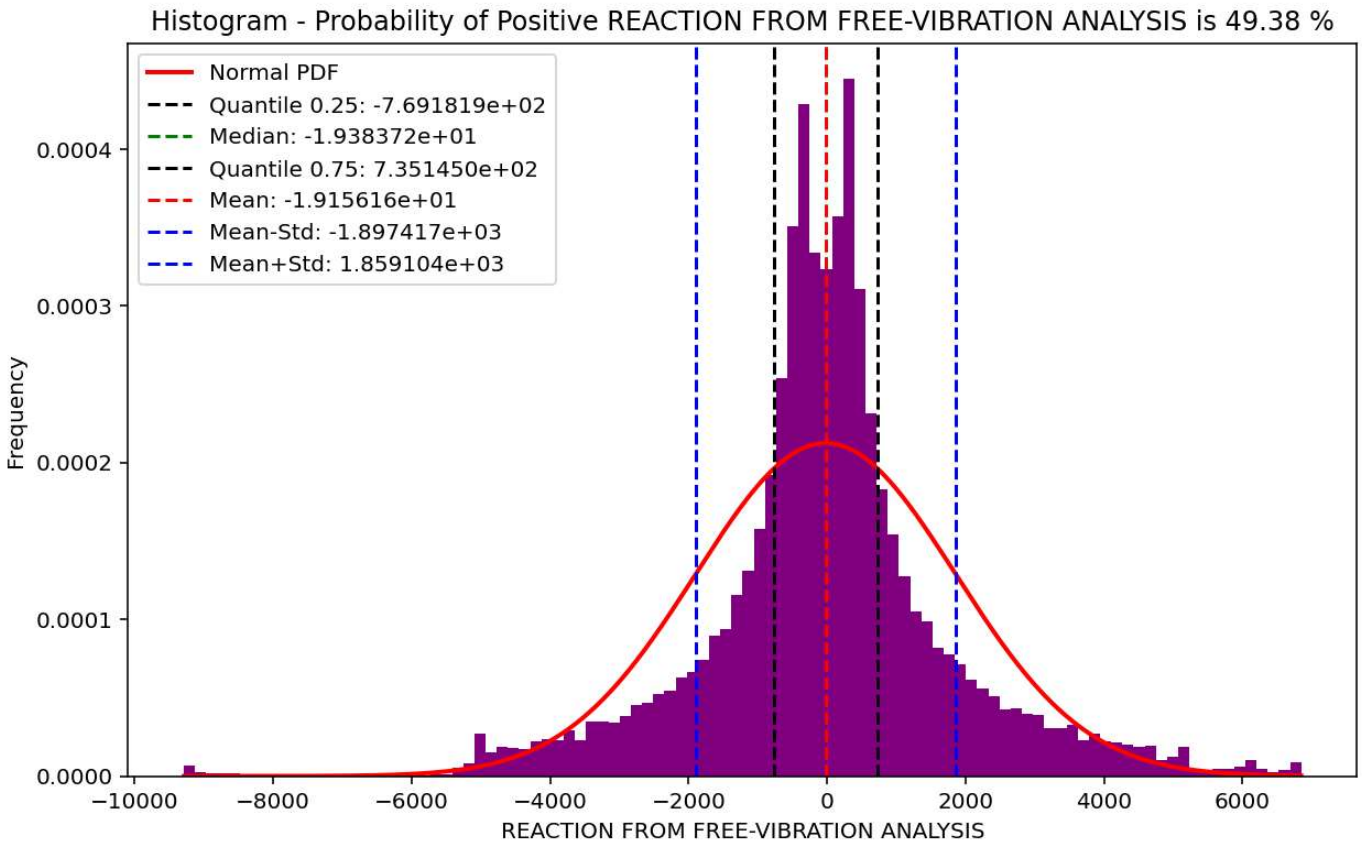

elements force vs Time During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 6.3254e+04
Ele. 2 - MAX. ABS. : 3.6272e+04
Force [N]
40000
20000
0
−20000
−40000
−60000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

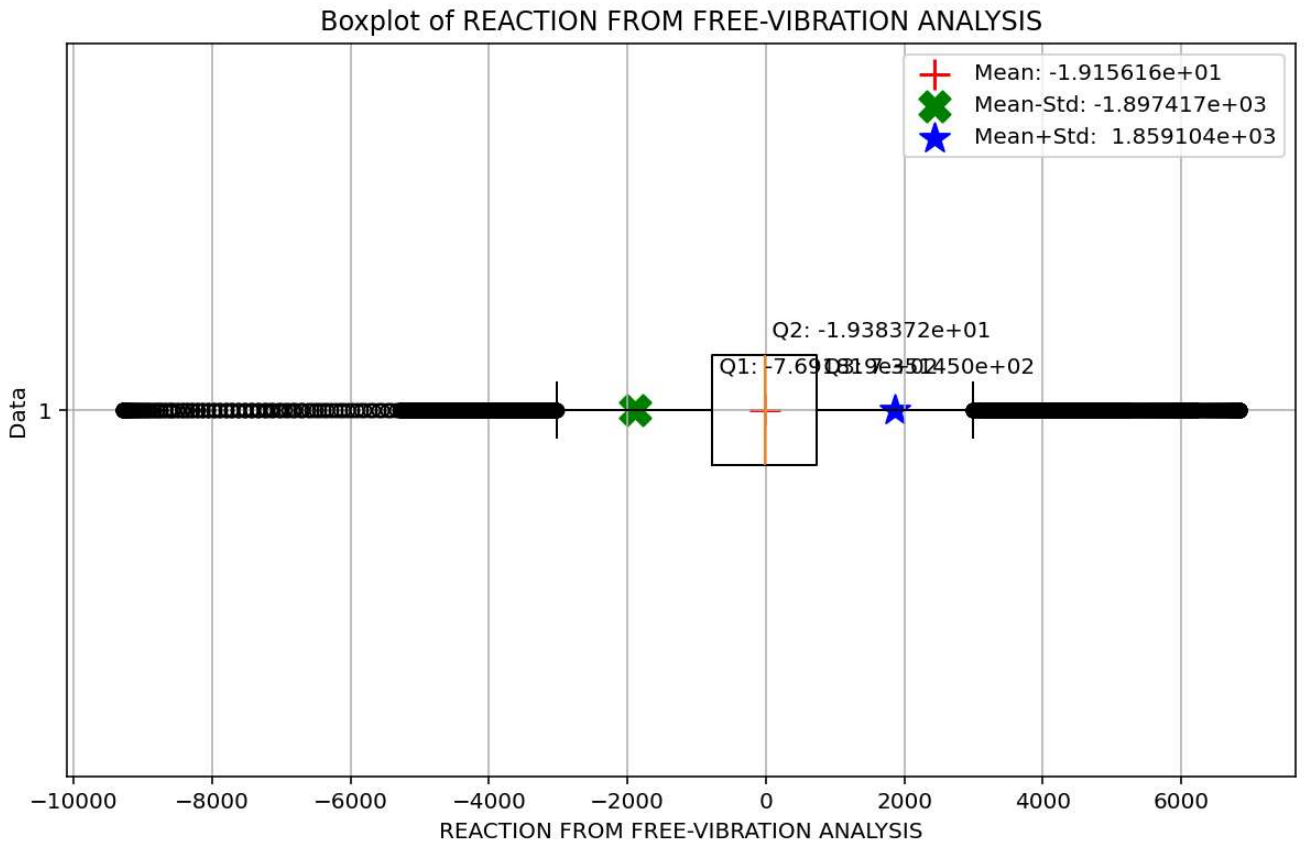

Node Displacements vs Time for During Free-vibration Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 7.0235e-03
Node 3 - MAX. ABS. : 3.4039e-02
Displacement [m]
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

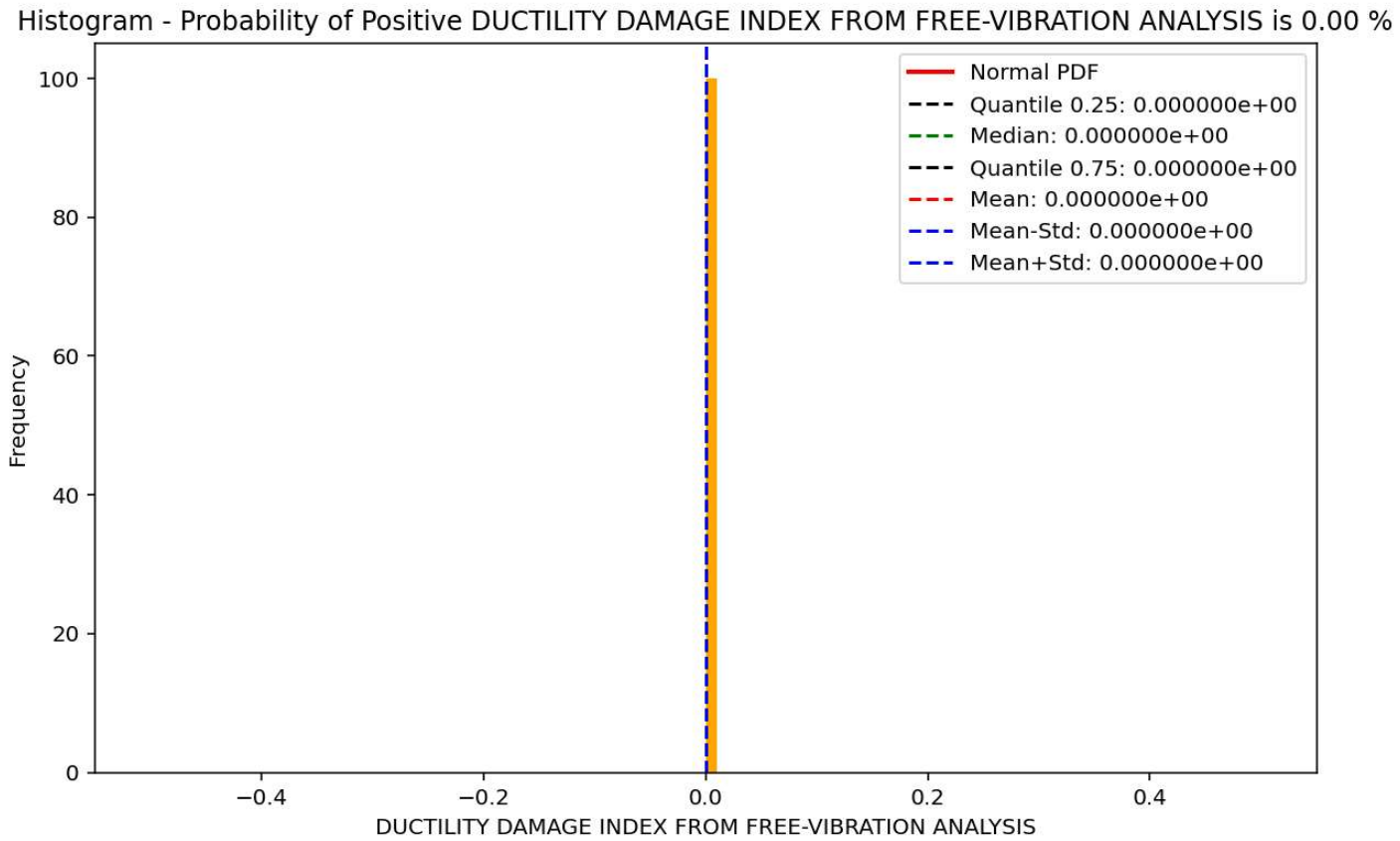

Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
Displacement [m]

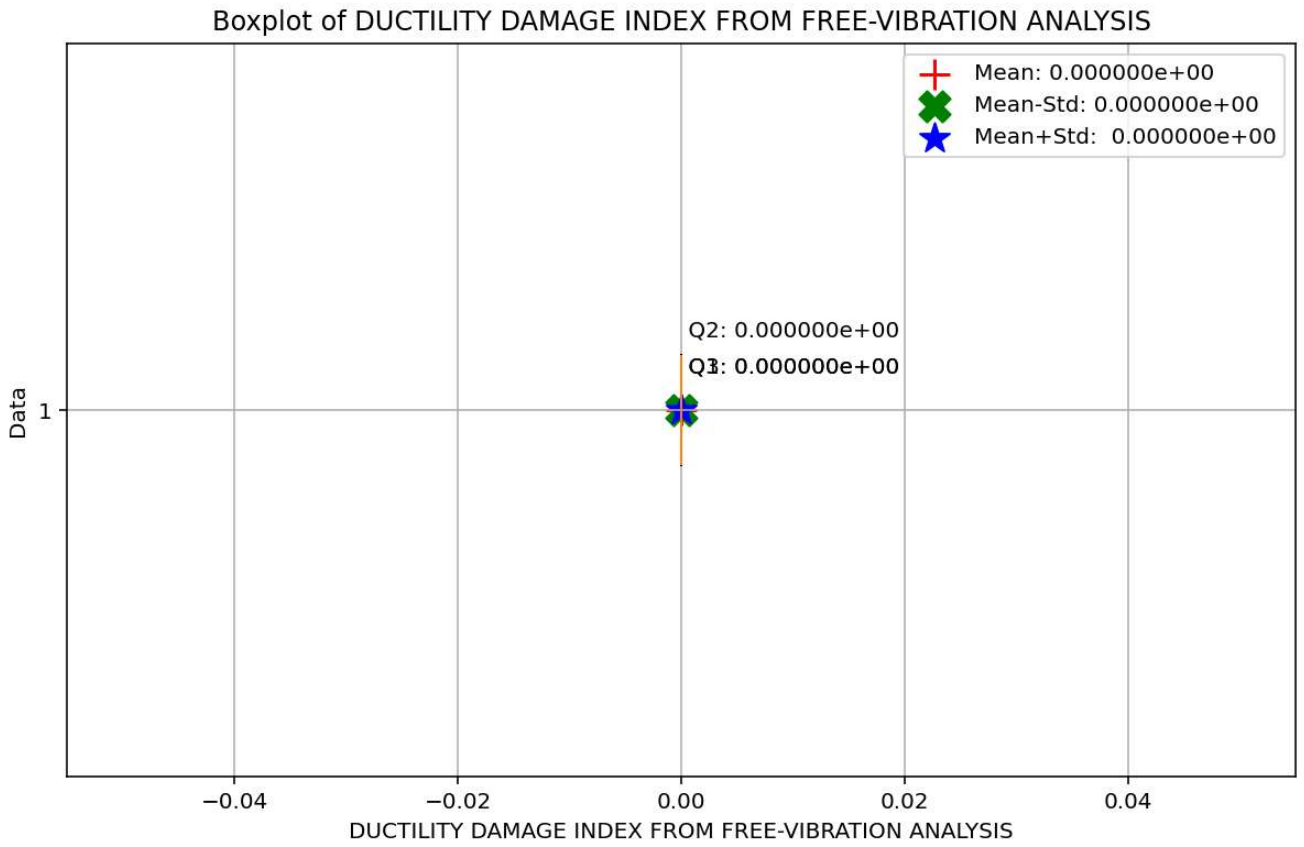

Histogram - Probability of Positive DISPLACEMENT FROM FREE-VIBRATION ANALYSIS is 52.35 %
Normal PDF
Quantile 0.25: -1.194253e-04
Median: 1.242769e-05
Quantile 0.75: 1.369389e-04
Mean: 5.600709e-06
Mean-Std: -7.346855e-04
Mean+Std: 7.458870e-04
Frequency
2000
1500
1000
500
0
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
DISPLACEMENT FROM FREE-VIBRATION ANALYSIS

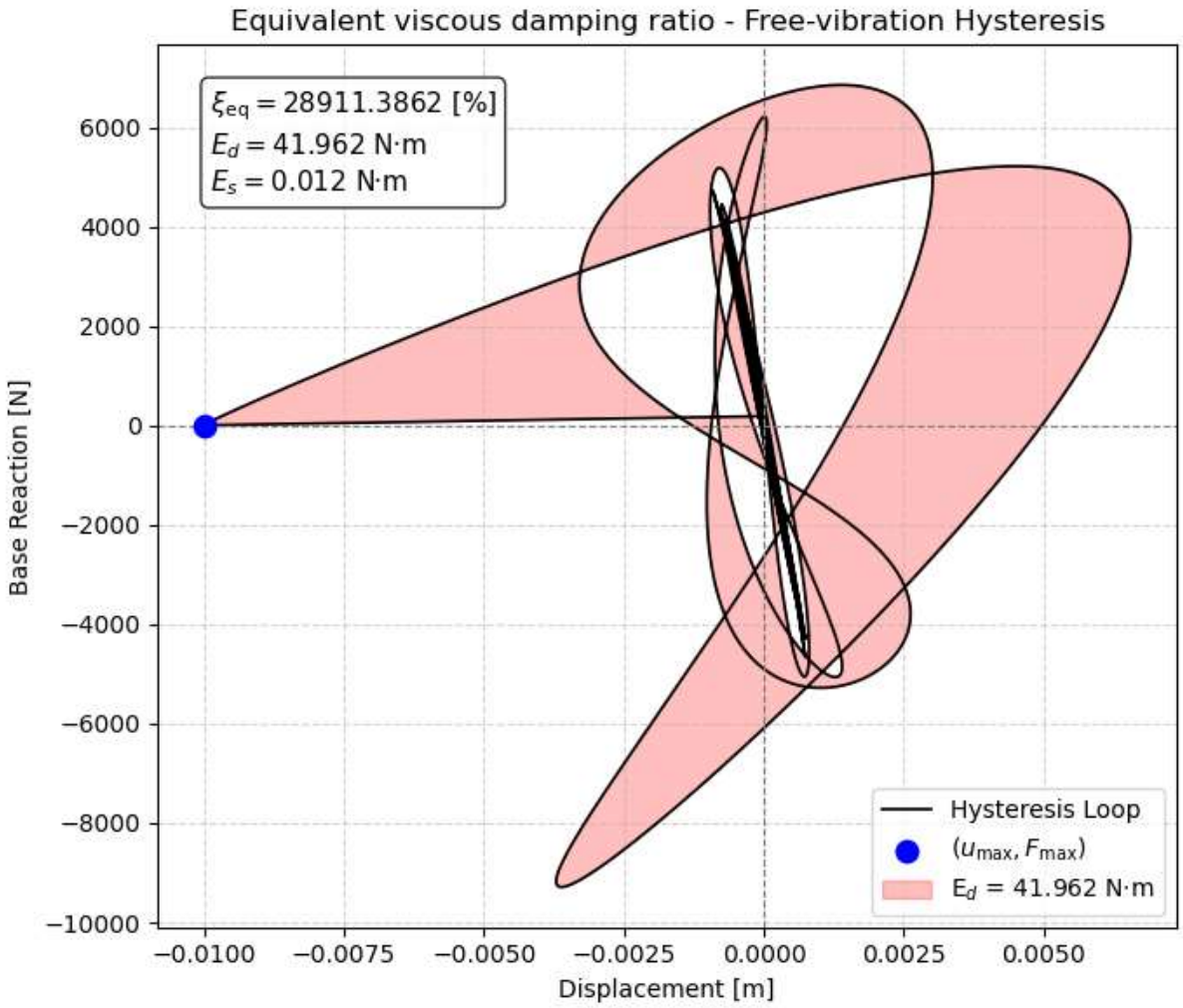

Boxplot of DISPLACEMENT FROM FREE-VIBRATION ANALYSIS
Mean: 5.600709e-06
Mean-Std: -7.346855e-04
Mean+Std: 7.458870e-04
Q2: 1.242769e-05
Data
1
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
DISPLACEMENT FROM FREE-VIBRATION ANALYSIS


Histogram - Probability of Positive REACTION FROM FREE-VIBRATION ANALYSIS is 49.38 %
Normal PDF
Quantile 0.25: -7.691819e+02
Median: -1.938372e+01
Quantile 0.75: 7.351450e+02
Mean: -1.915616e+01
Mean-Std: -1.897417e+03
Mean+Std: 1.859104e+03
Frequency
0.0000
0.0001
0.0002
0.0003
0.0004
−10000
−8000
−6000
−4000
−2000
0
2000
4000
6000
REACTION FROM FREE-VIBRATION ANALYSIS


Boxplot of REACTION FROM FREE-VIBRATION ANALYSIS
Mean: -1.915616e+01
Mean-Std: -1.897417e+03
Mean+Std: 1.859104e+03
Q2: -1.938372e+01
Q1: -7.69
Q3: 1.
e+02
Data
1
−10000
−8000
−6000
−4000
−2000
0
2000
4000
6000
REACTION FROM FREE-VIBRATION ANALYSIS


Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS is 0.00 %
Normal PDF
Quantile 0.25: 0.000000e+00
Median: 0.000000e+00
Quantile 0.75: 0.000000e+00
Mean: 0.000000e+00
Mean-Std: 0.000000e+00
Mean+Std: 0.000000e+00
Frequency
0
20
40
60
80
100
−0.4
−0.2
0.0
0.2
0.4
DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS


Boxplot of DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS
Mean: 0.000000e+00
Mean-Std: 0.000000e+00
Mean+Std: 0.000000e+00
Q2: 0.000000e+00
Q3: 0.000000e+00
Data
1
−0.04
−0.02
0.00
0.02
0.04
DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS


Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 28911.3862$ [%]
$E_d = 41.962$ N·m
$E_s = 0.012$ N·m
Base Reaction [N]
Displacement [m]
6000
4000
2000
0
−2000
−4000
−6000
−8000
−10000
−0.0100
−0.0075
−0.0050
−0.0025
0.0000
0.0025
0.0050
Hysteresis Loop
(u_{max}, F_{max})
E_d = 41.962 N·m

SEISMIC ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

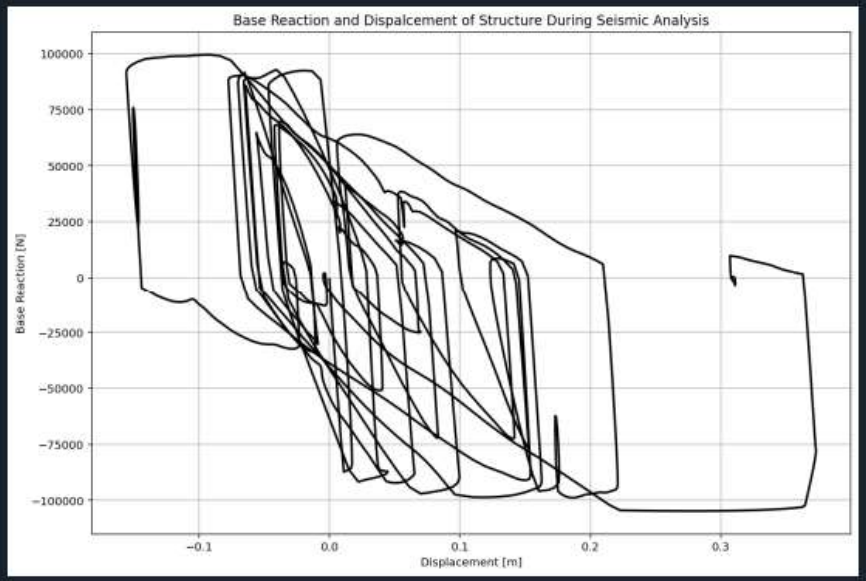
...ISSIPATION_CAPACITY_INDEX\54_ELEVATOR_STRUCTURE_MDOF_8_ANA

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..._MDOF_8_ANA\P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py

P(t)_ELEVATOR_STRUCTURE_MDOF_8_ANA.py | COMPUTE_EFFECTIVE_PROPERTIES_FUN.py

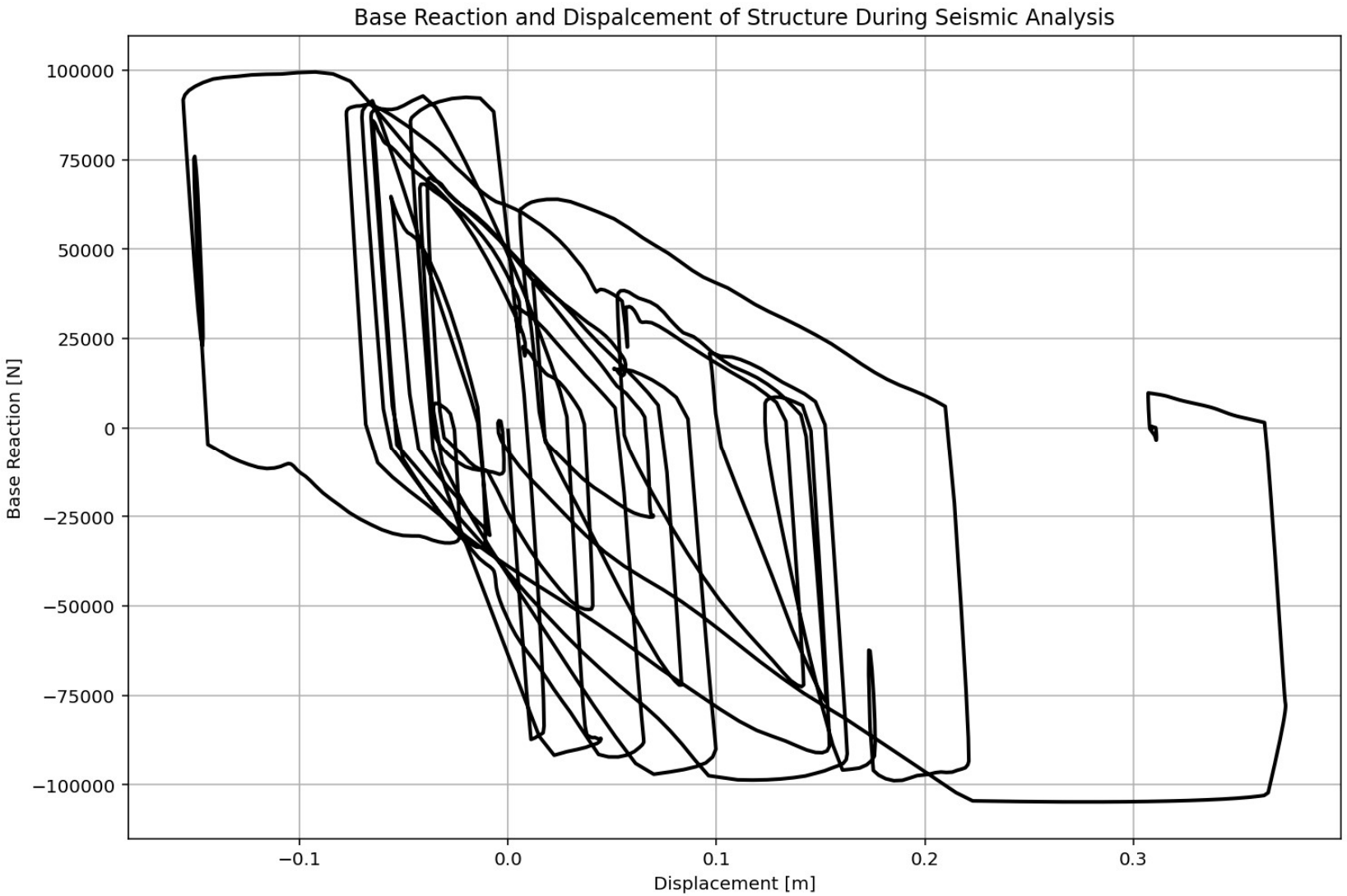
```
770     if ANAL_TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
771         #%% DEFINE PARAMETERS FOR SEISMIC ANALYSIS
772         duration = 20.0                    # [s] Analysis duration
773         dt = 0.01                          # [s] Time step
774         GMfact = 9810                      # [mm/s²]standard acceleration o
775         SSF_X = 0.05                       # Seismic Acceleration Scale Fac
776         SSF_Y = 0.05                       # Seismic Acceleration Scale Fac
777         iv0_X = 0.0005                     # [mm/s] Initial velocity applie
778         iv0_Y = 0.0005                     # [mm/s] Initial velocity applie
779         st_iv0 = 0.0                       # [s] Initial velocity applied s
780         SEI = 'X'                          # Seismic Direction
781         DR = 0.03                          # Damping ratio
782
783         # Define time series for input motion (Acceleration time history)
784         if SEI == 'X':
785             SEISMIC_TAG_01 = 100
786             ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath',
787             # Define load patterns
788             # pattern UniformExcitation $patternTag $dof -accel $tsTag <-ve
789             ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', S
790         if SEI == 'Y':
791             SEISMIC_TAG_02 = 200
792             ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath',
793             ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', S
794         if SEI == 'XY':
795             SEISMIC_TAG_01 = 100
796             ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath',
797             # Define load patterns
798             # pattern UniformExcitation $patternTag $dof -accel $tsTag <-ve
799             ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', S
800             SEISMIC_TAG_02 = 200
801             ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath',
802             ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', S
803         print('Seismic Defined Done.')
```

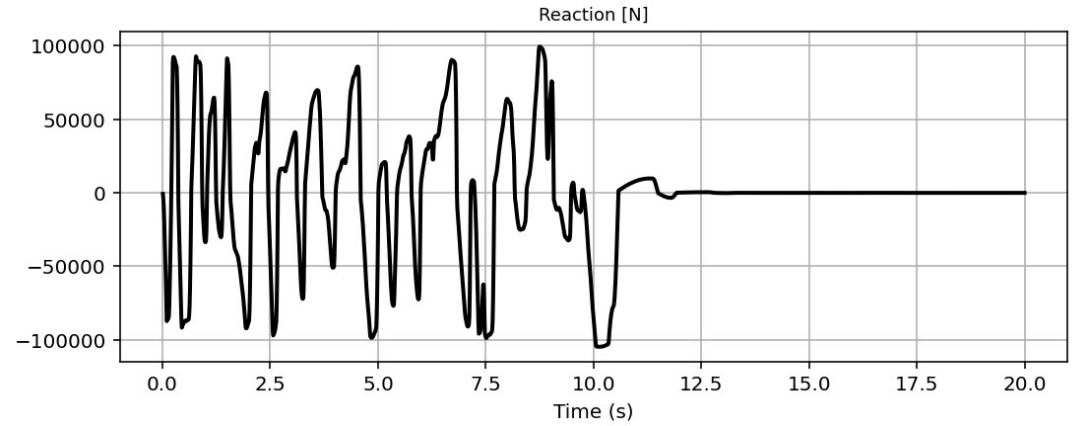
37 %

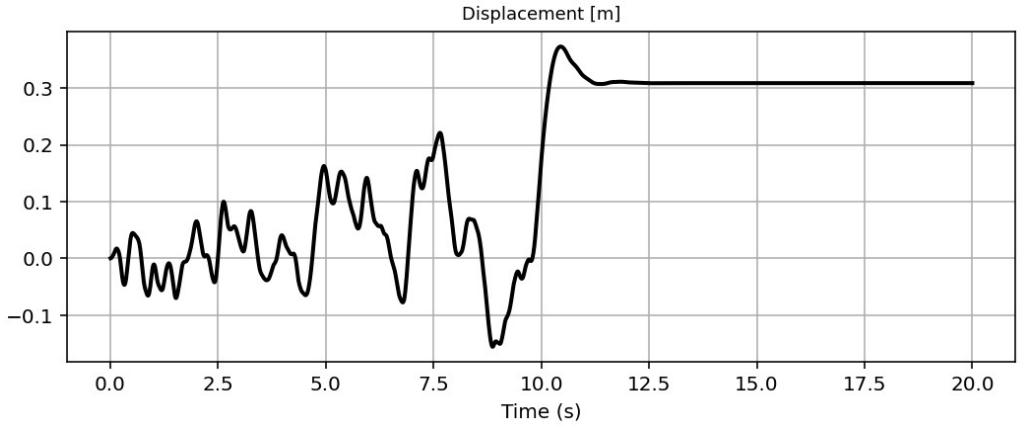


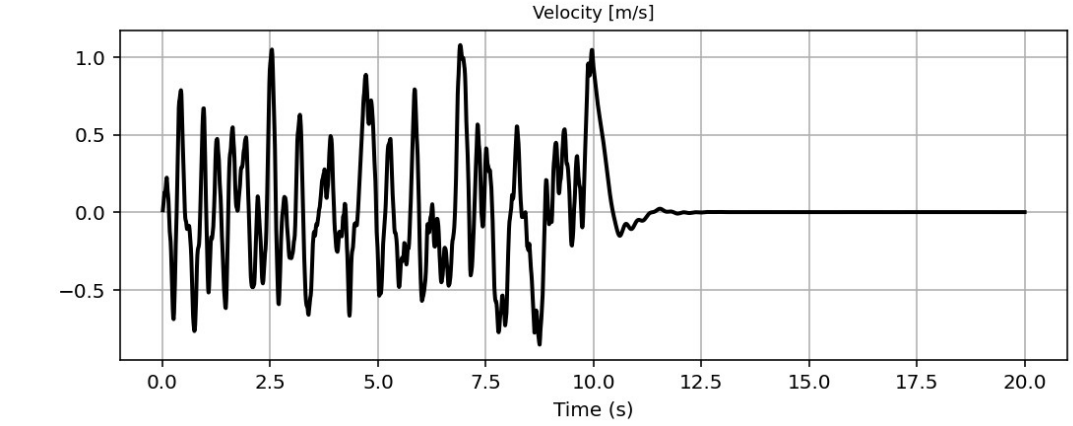
IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

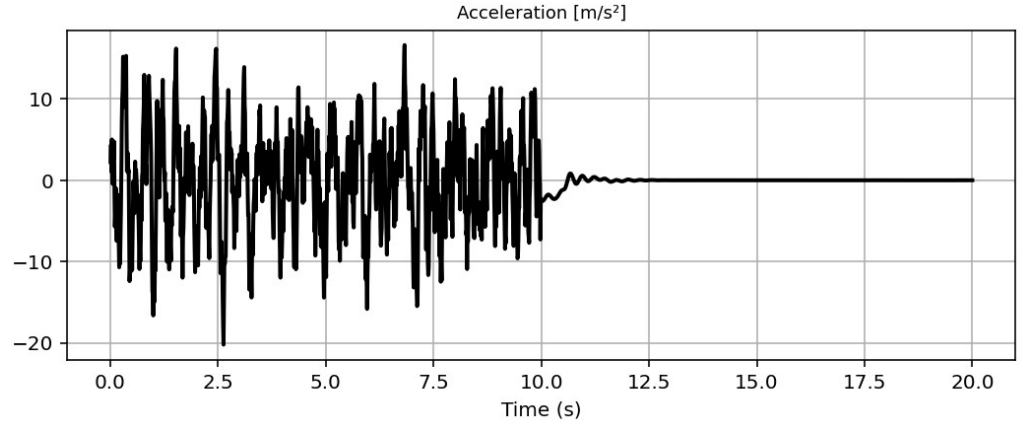
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 769, Col 5 UTF-8 CRLF RW Mem 51%

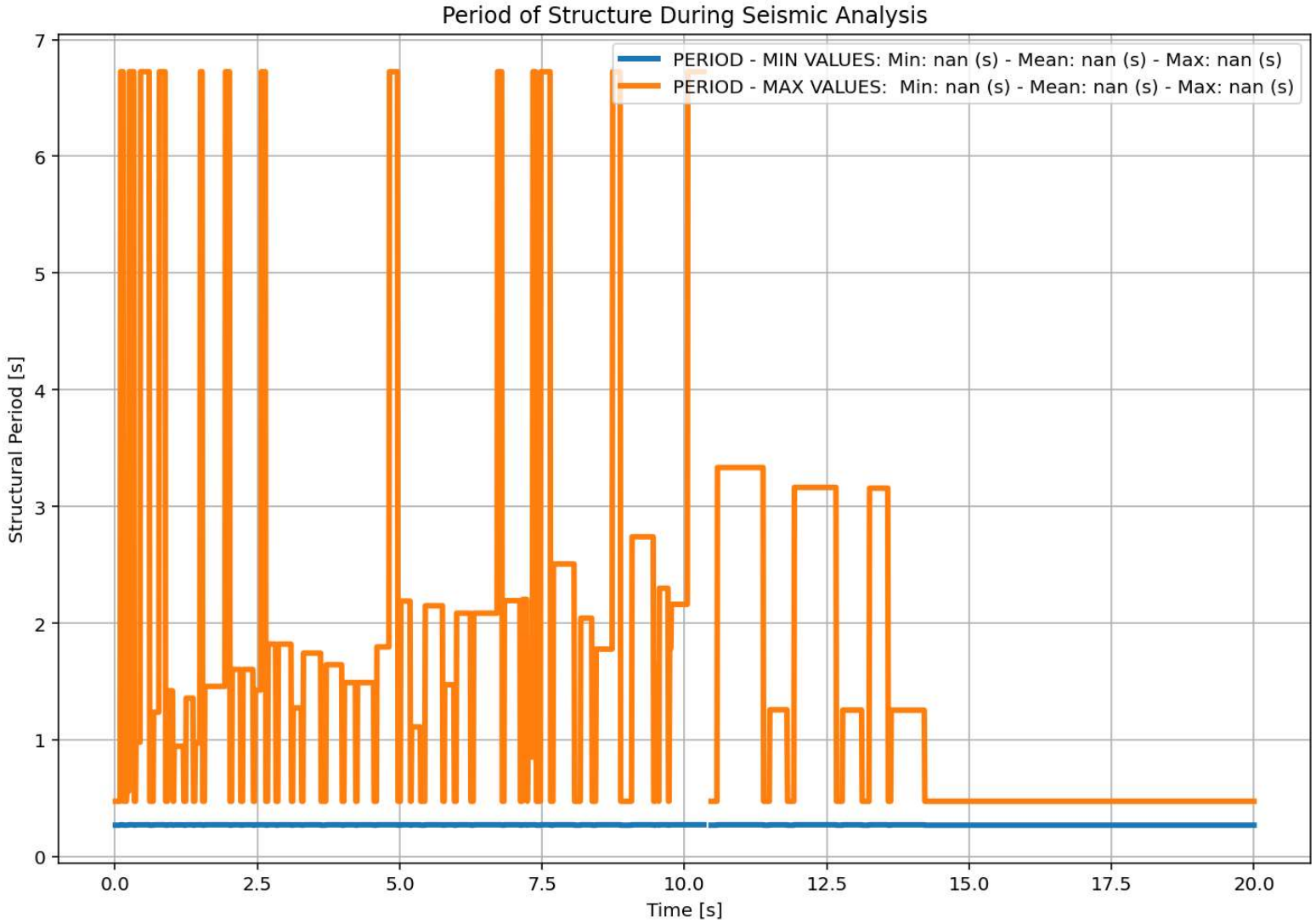

Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.1
0.0
0.1
0.2
0.3
Displacement [m]

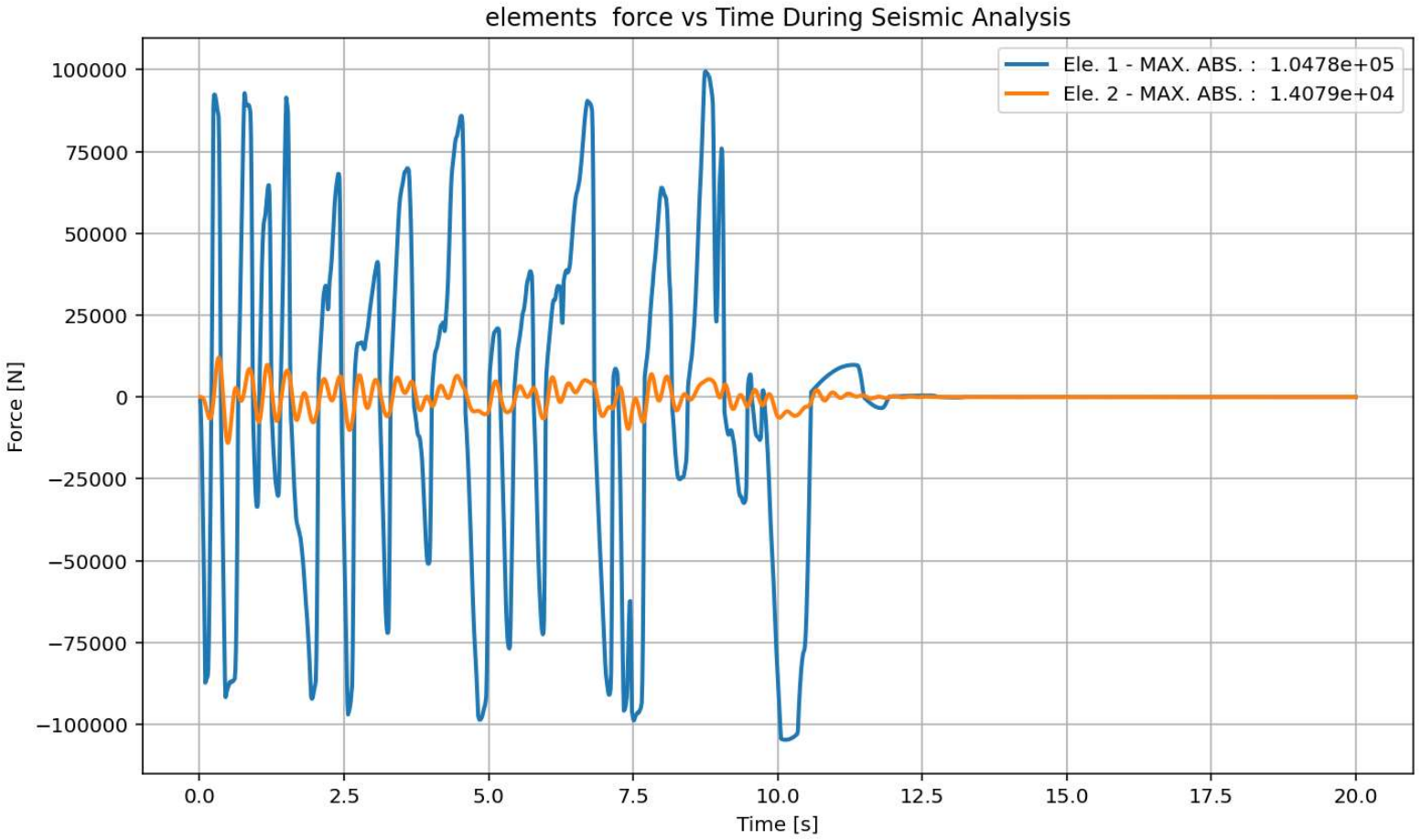

Reaction [N]
100000
50000
0
−50000
−100000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

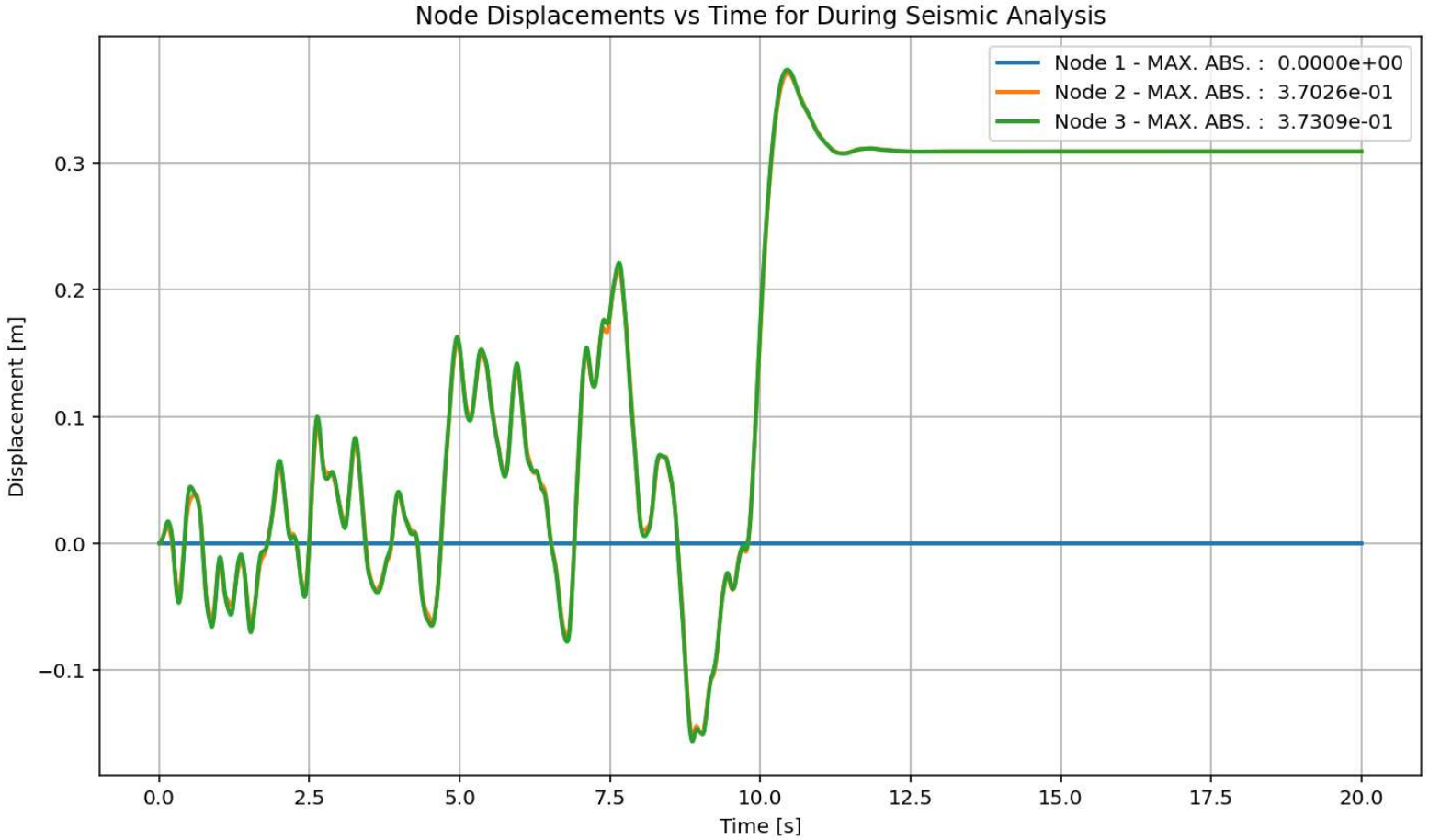

Displacement [m]
0.3
0.2
0.1
0.0
−0.1
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

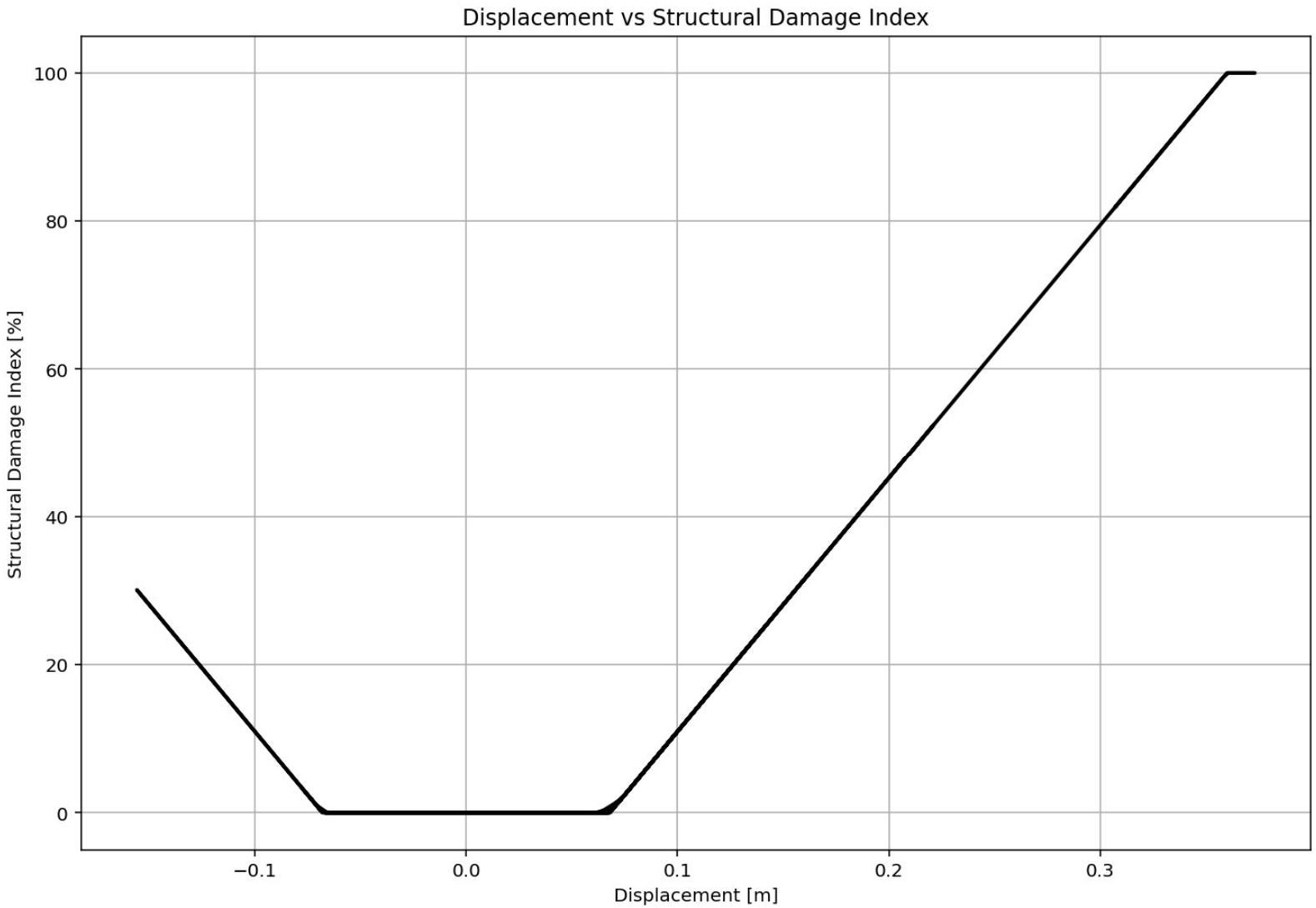

Velocity [m/s]
1.0
0.5
0.0
−0.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

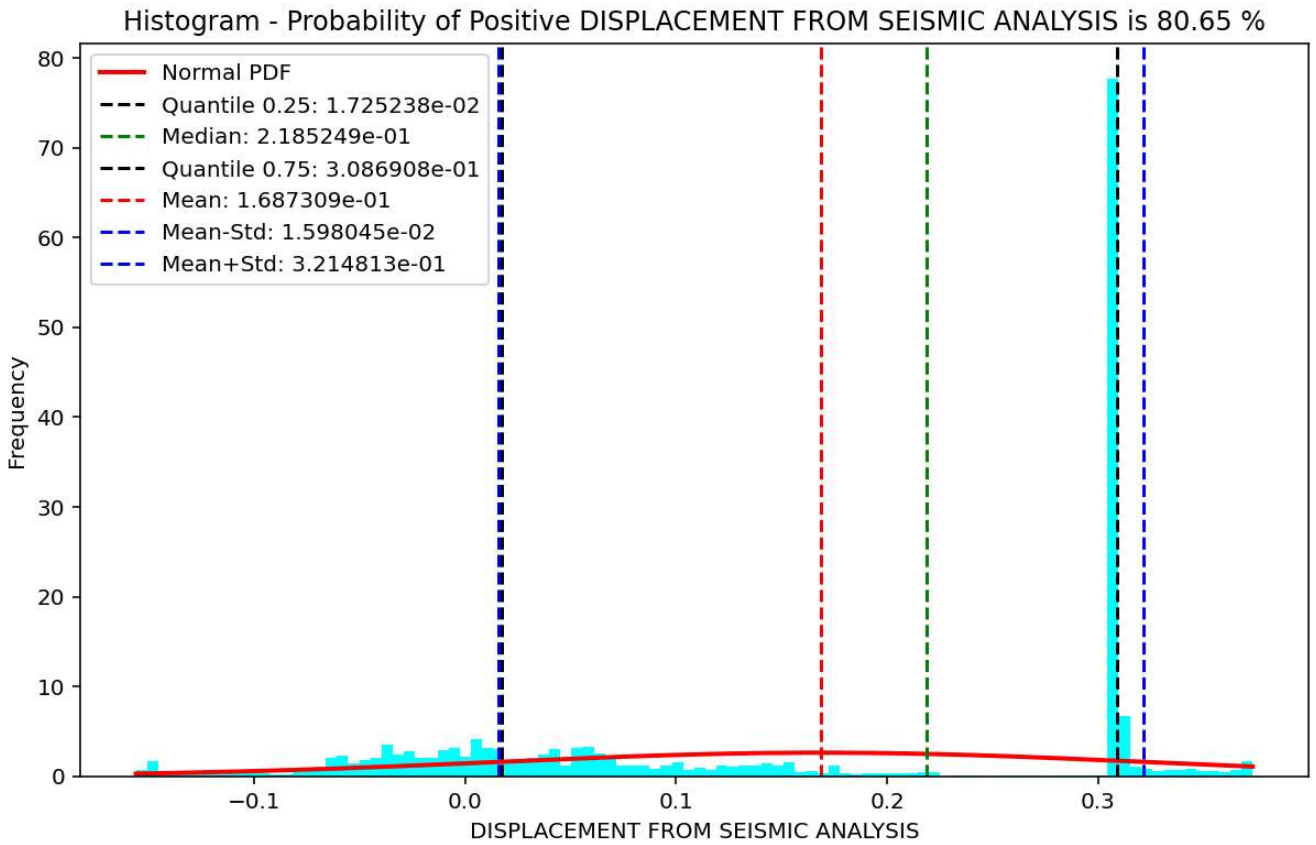

Acceleration [m/s²]
10
0
−10
−20
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)

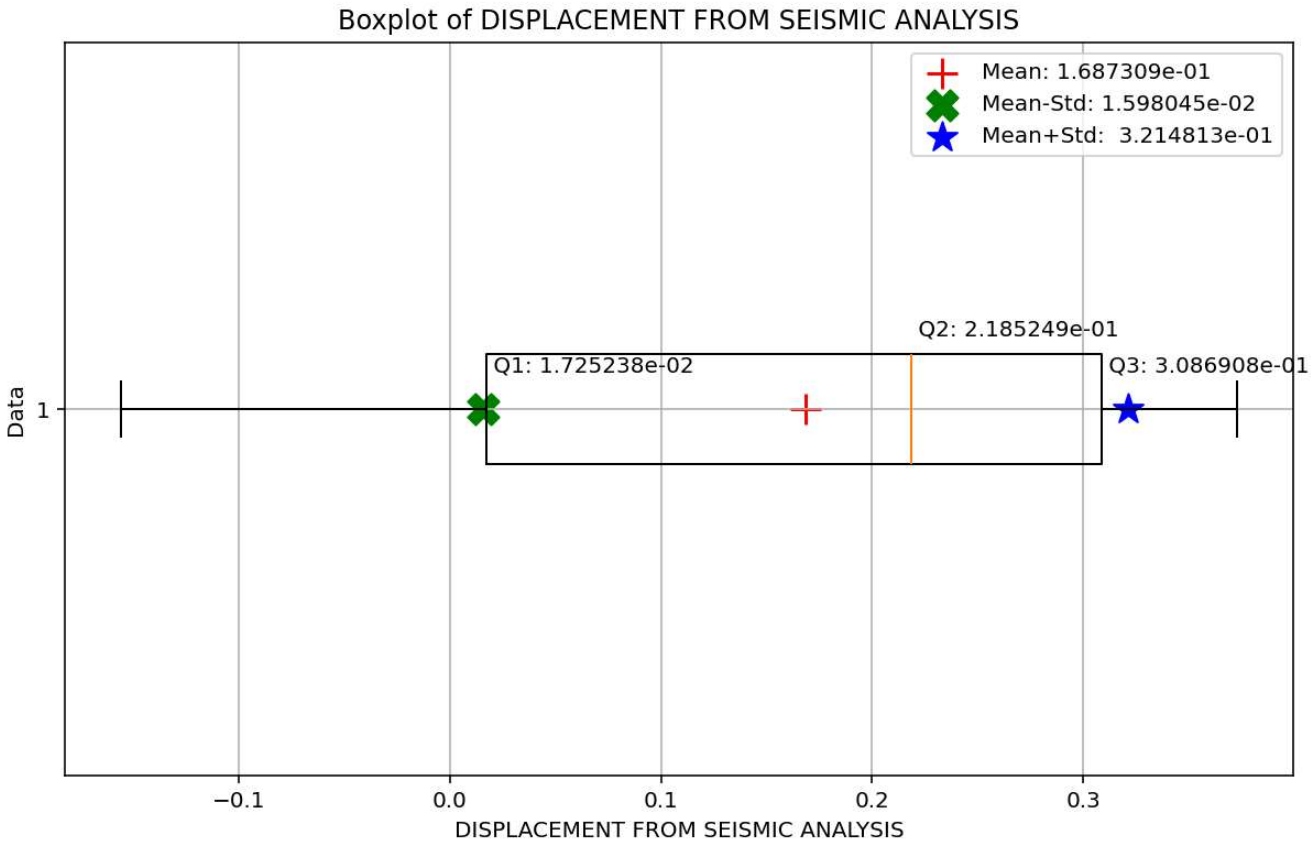

Period of Structure During Seismic Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
Time [s]
0
1
2
3
4
5
6
7
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

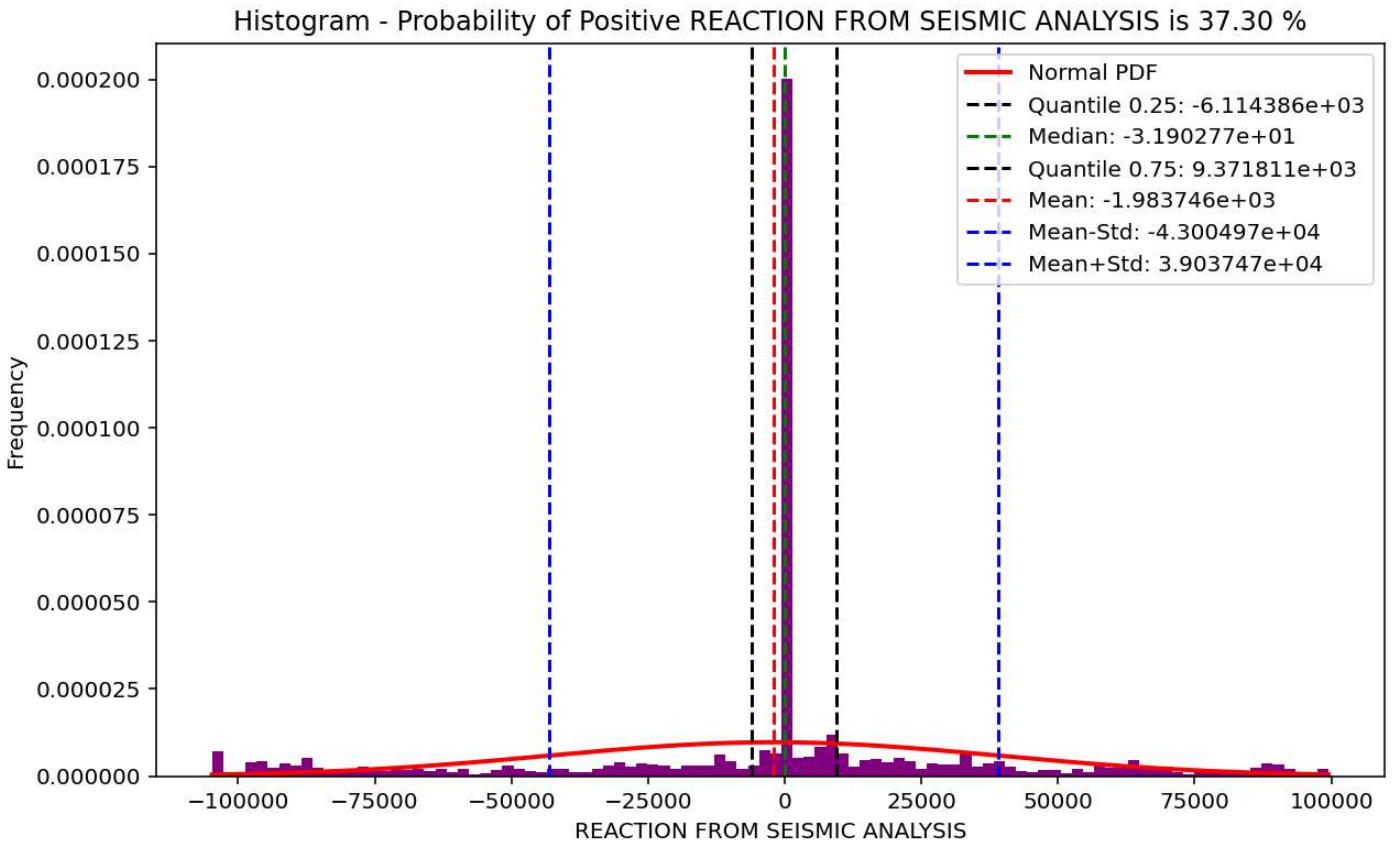

elements force vs Time During Seismic Analysis
Ele. 1 - MAX. ABS. : 1.0478e+05
Ele. 2 - MAX. ABS. : 1.4079e+04
Force [N]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

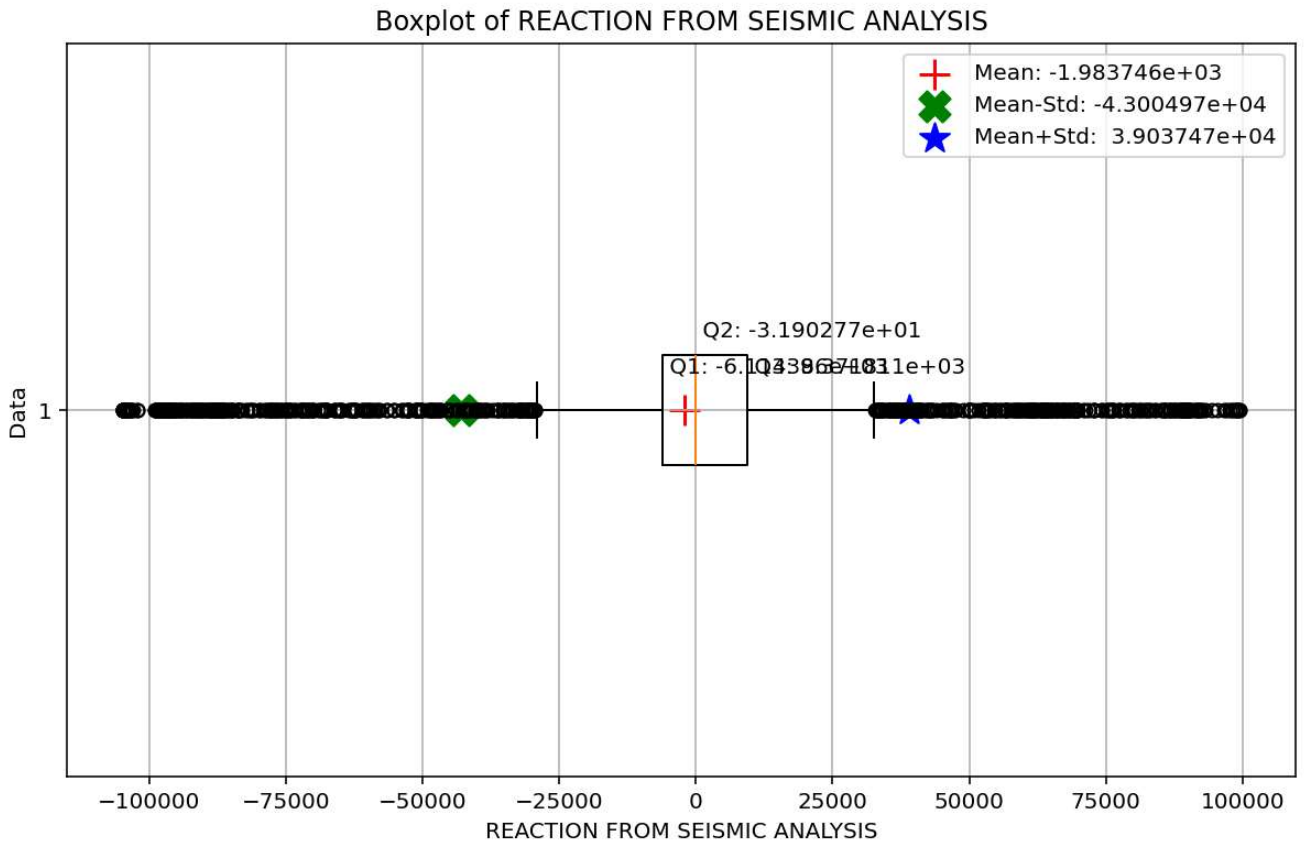

Node Displacements vs Time for During Seismic Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.7026e-01
Node 3 - MAX. ABS. : 3.7309e-01
Displacement [m]
0.3
0.2
0.1
0.0
−0.1
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

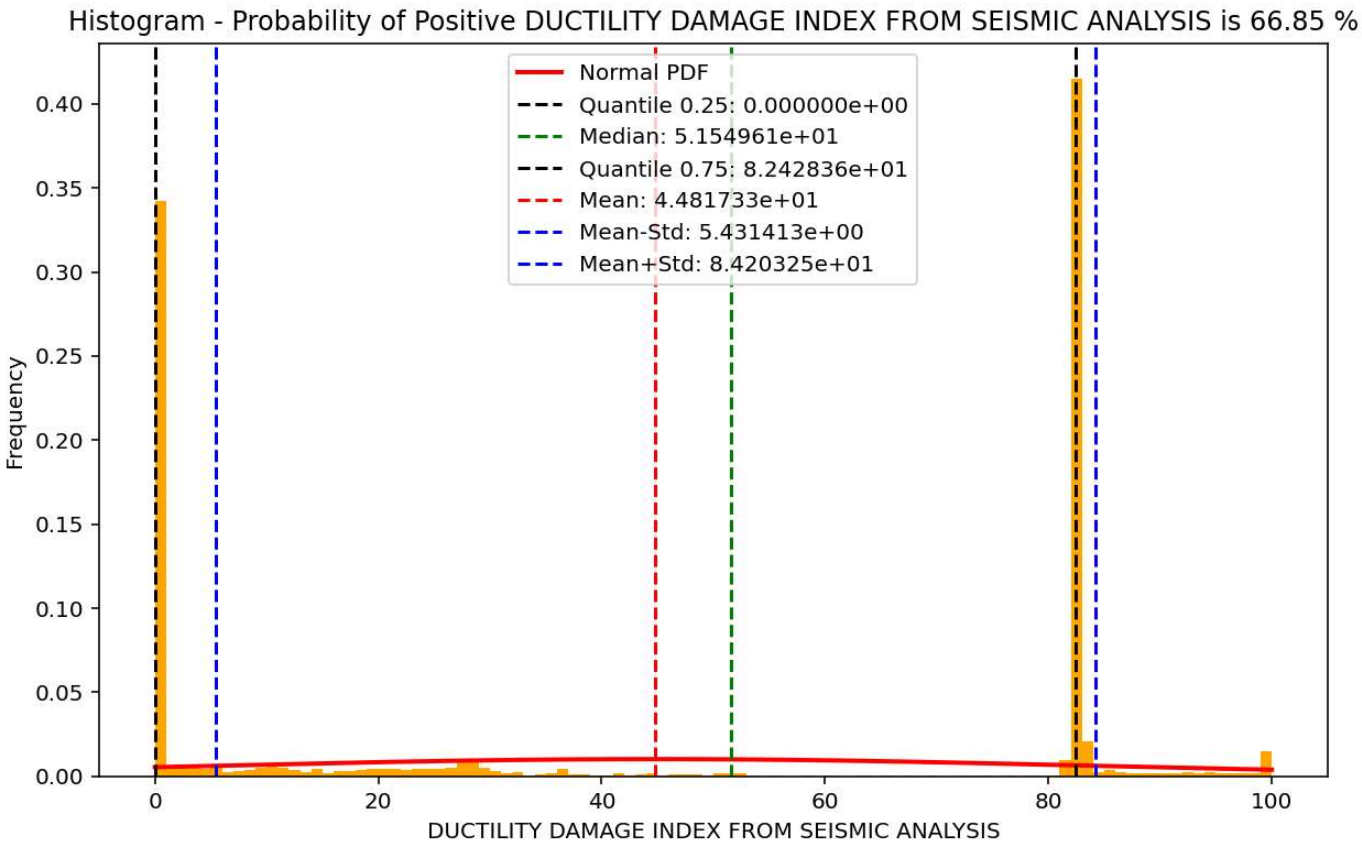

Displacement vs Structural Damage Index
Structural Damage Index [%]
100
80
60
40
20
0
−0.1
0.0
0.1
0.2
0.3
Displacement [m]

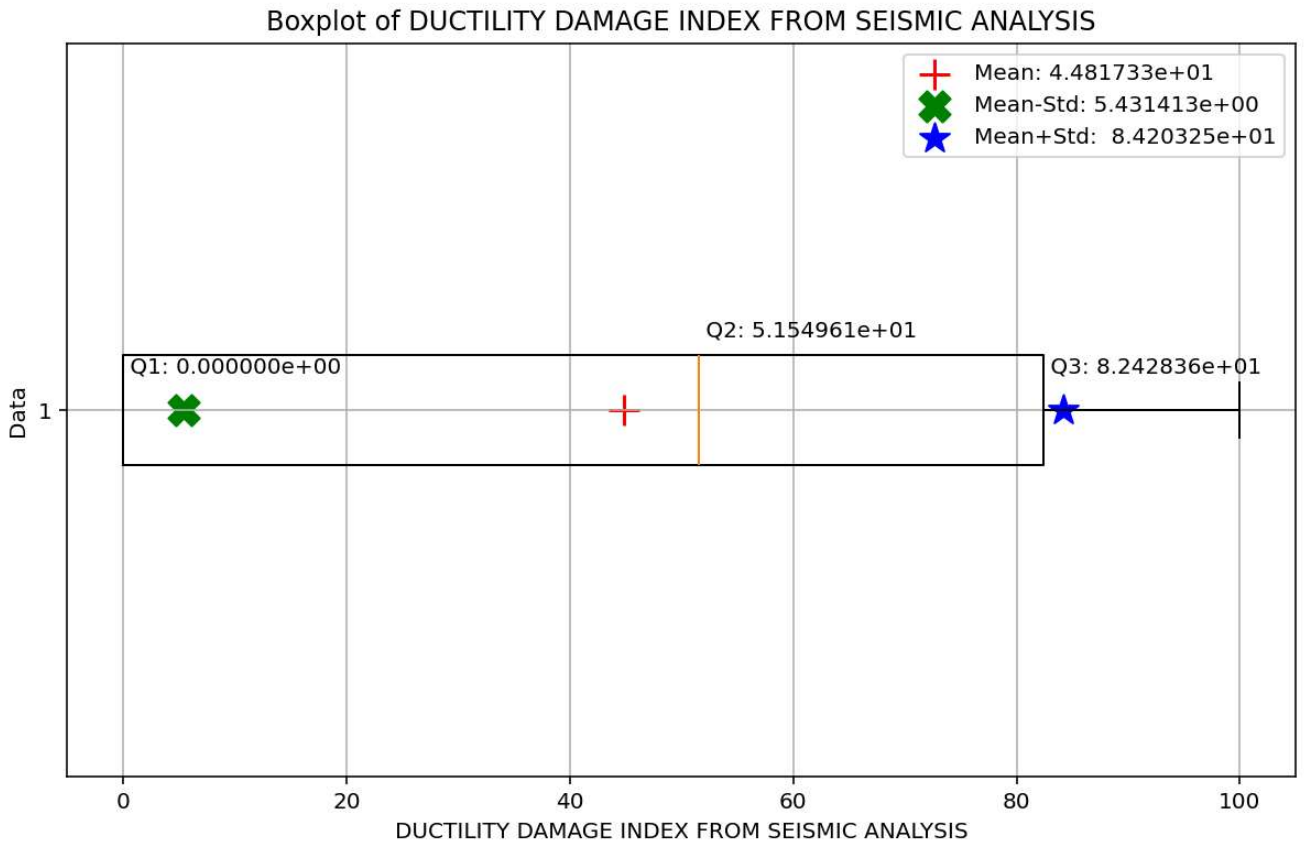

Histogram - Probability of Positive DISPLACEMENT FROM SEISMIC ANALYSIS is 80.65 %
Normal PDF
Quantile 0.25: 1.725238e-02
Median: 2.185249e-01
Quantile 0.75: 3.086908e-01
Mean: 1.687309e-01
Mean-Std: 1.598045e-02
Mean+Std: 3.214813e-01
Frequency
DISPLACEMENT FROM SEISMIC ANALYSIS
0
10
20
30
40
50
60
70
80
−0.1
0.0
0.1
0.2
0.3

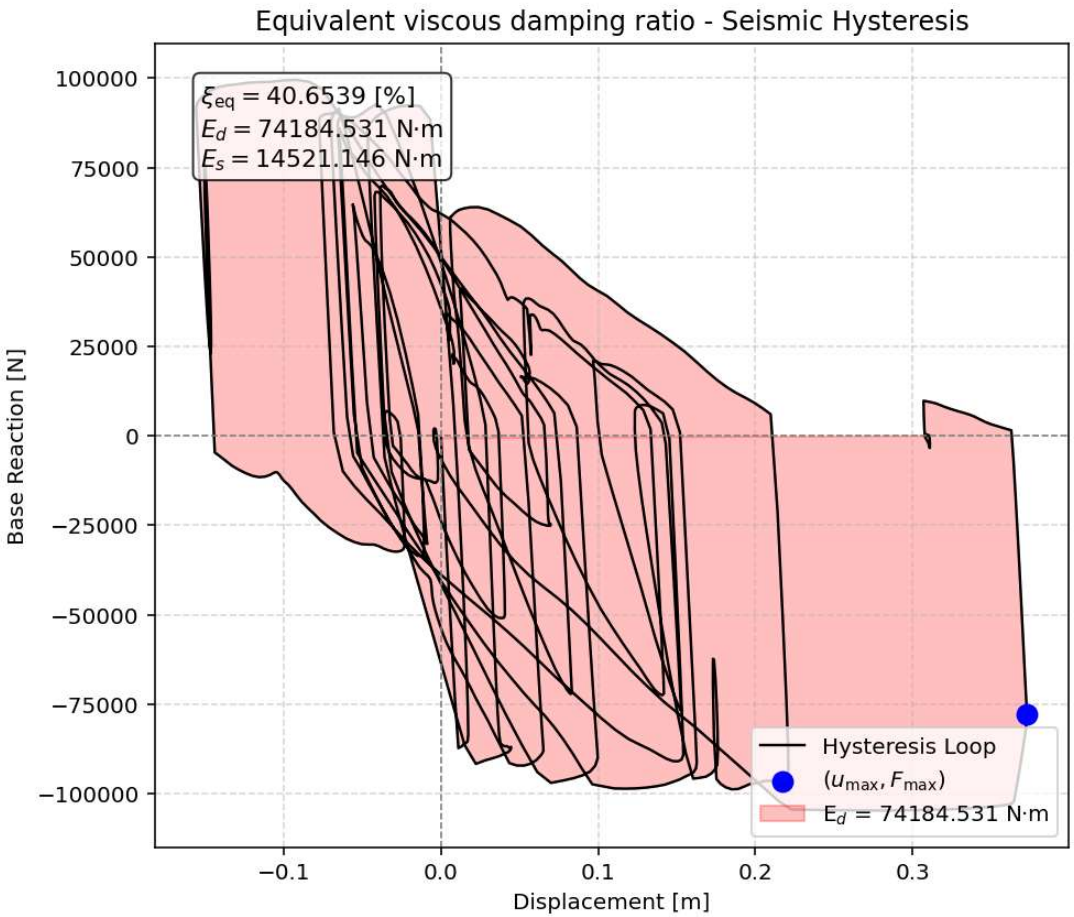

Boxplot of DISPLACEMENT FROM SEISMIC ANALYSIS
Mean: 1.687309e-01
Mean-Std: 1.598045e-02
Mean+Std: 3.214813e-01
Q2: 2.185249e-01
Q1: 1.725238e-02
Q3: 3.086908e-01
Data
1
−0.1
0.0
0.1
0.2
0.3
DISPLACEMENT FROM SEISMIC ANALYSIS

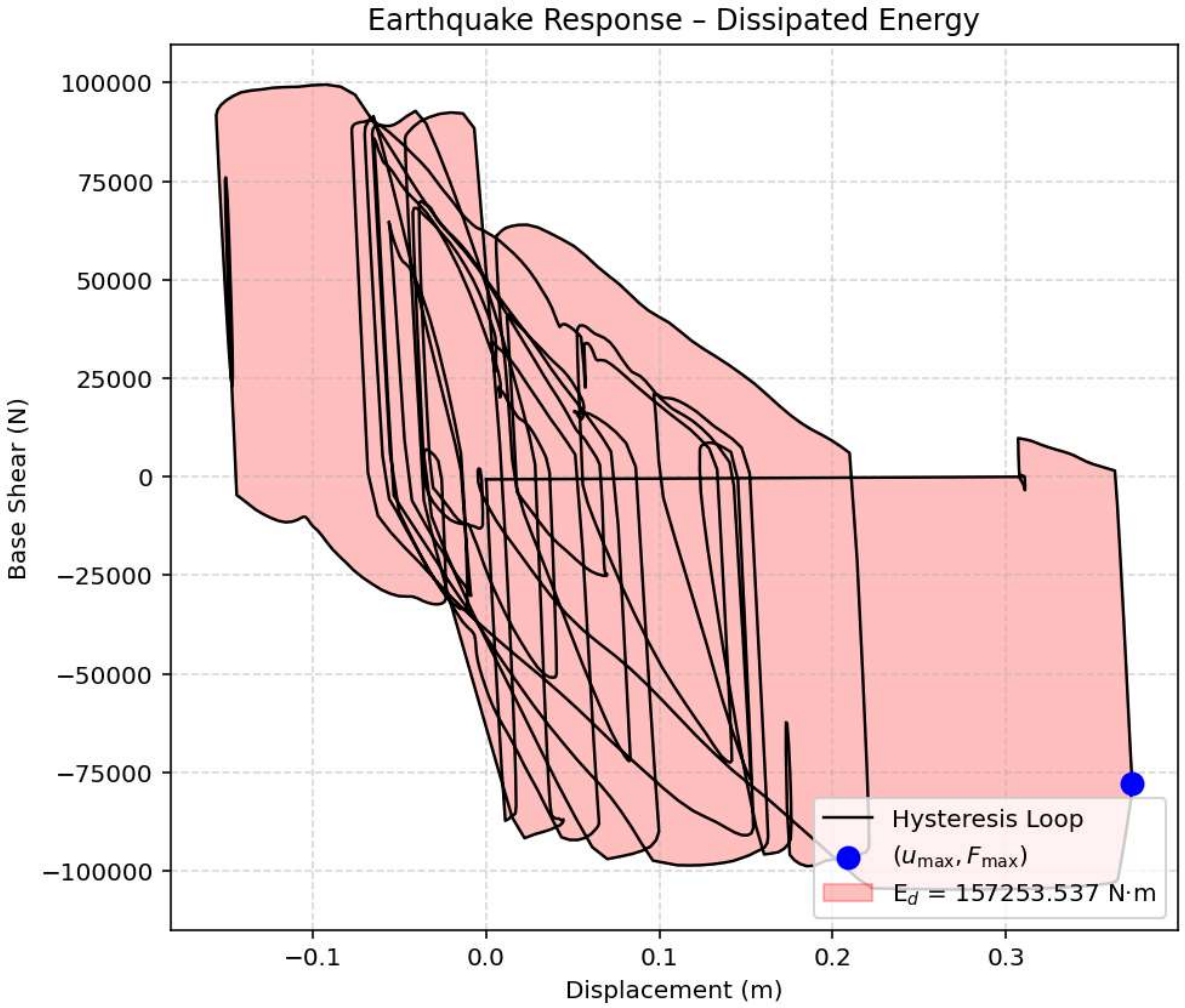

Histogram - Probability of Positive REACTION FROM SEISMIC ANALYSIS is 37.30 %
Normal PDF
Quantile 0.25: -6.114386e+03
Median: -3.190277e+01
Quantile 0.75: 9.371811e+03
Mean: -1.983746e+03
Mean-Std: -4.300497e+04
Mean+Std: 3.903747e+04
Frequency
0.000200
0.000175
0.000150
0.000125
0.000100
0.000075
0.000050
0.000025
0.000000
−100000
−75000
−50000
−25000
0
25000
50000
75000
100000
REACTION FROM SEISMIC ANALYSIS

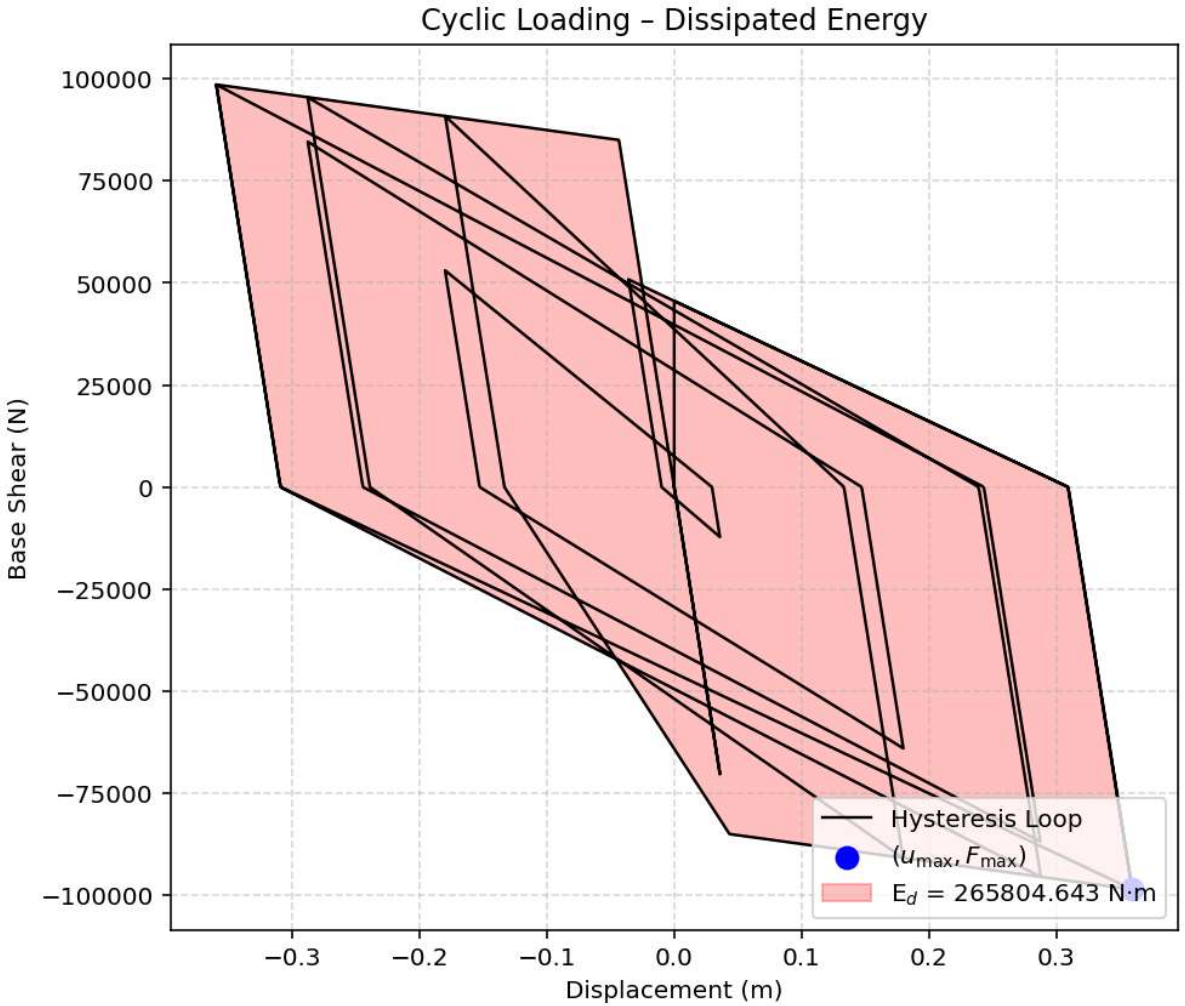

Boxplot of REACTION FROM SEISMIC ANALYSIS
Mean: -1.983746e+03
Mean-Std: -4.300497e+04
Mean+Std: 3.903747e+04
Q2: -3.190277e+01
Data
1
−100000
−75000
−50000
−25000
0
25000
50000
75000
100000
REACTION FROM SEISMIC ANALYSIS


Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS is 66.85 %
Normal PDF
Quantile 0.25: 0.000000e+00
Median: 5.154961e+01
Quantile 0.75: 8.242836e+01
Mean: 4.481733e+01
Mean-Std: 5.431413e+00
Mean+Std: 8.420325e+01
Frequency
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0
20
40
60
80
100
DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS


Boxplot of DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS
Mean: 4.481733e+01
Mean-Std: 5.431413e+00
Mean+Std: 8.420325e+01
Q2: 5.154961e+01
Q1: 0.000000e+00
Q3: 8.242836e+01
Data
1
0
20
40
60
80
100
DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS

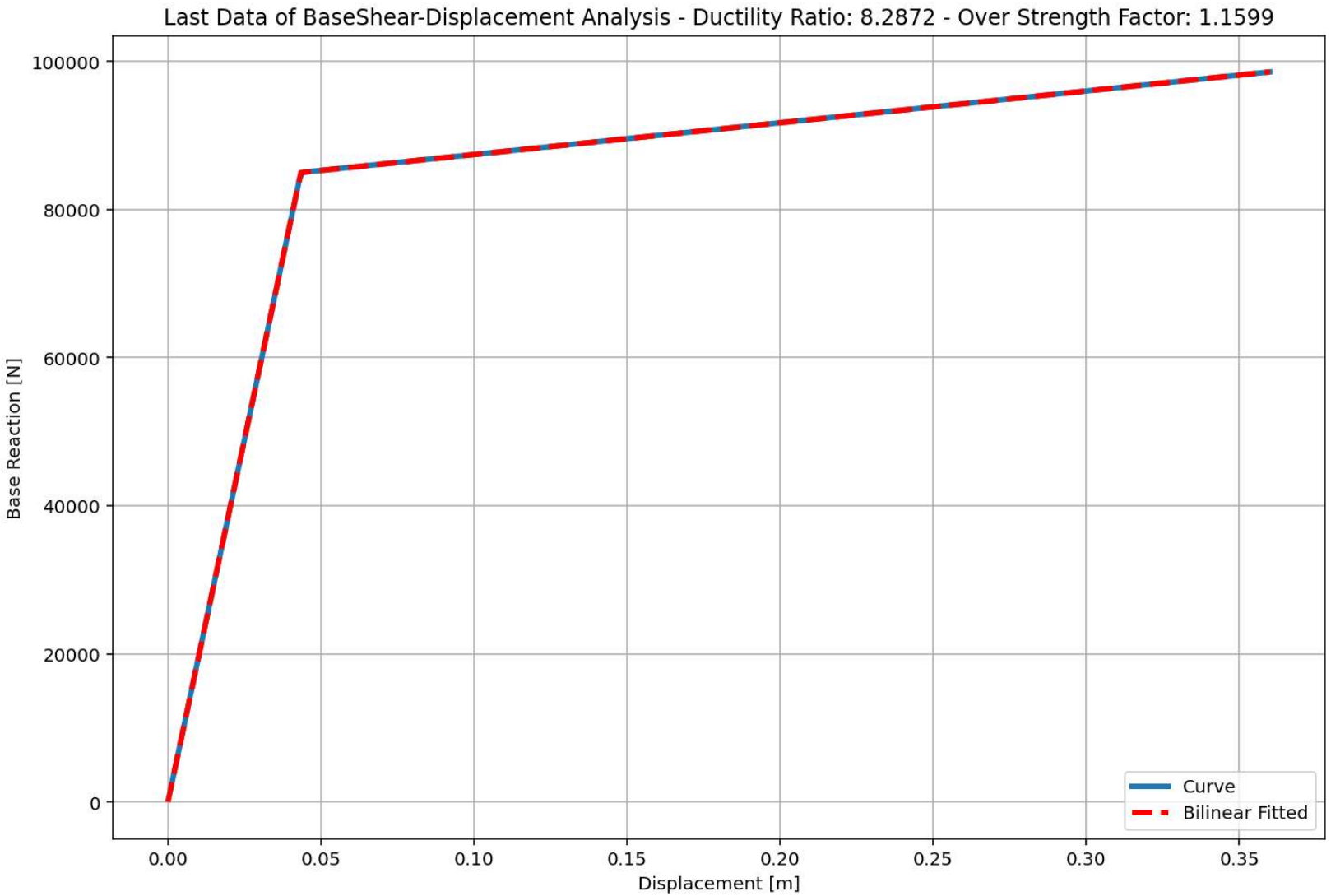

Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 40.6539$ [%]
$E_d = 74184.531$ N·m
$E_s = 14521.146$ N·m
Base Reaction [N]
Displacement [m]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
(u_{max}, F_{max})
E_d = 74184.531 N·m


Earthquake Response – Dissipated Energy
Base Shear (N)
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
(u_{max}, F_{max})
E_d = 157253.537 N·m

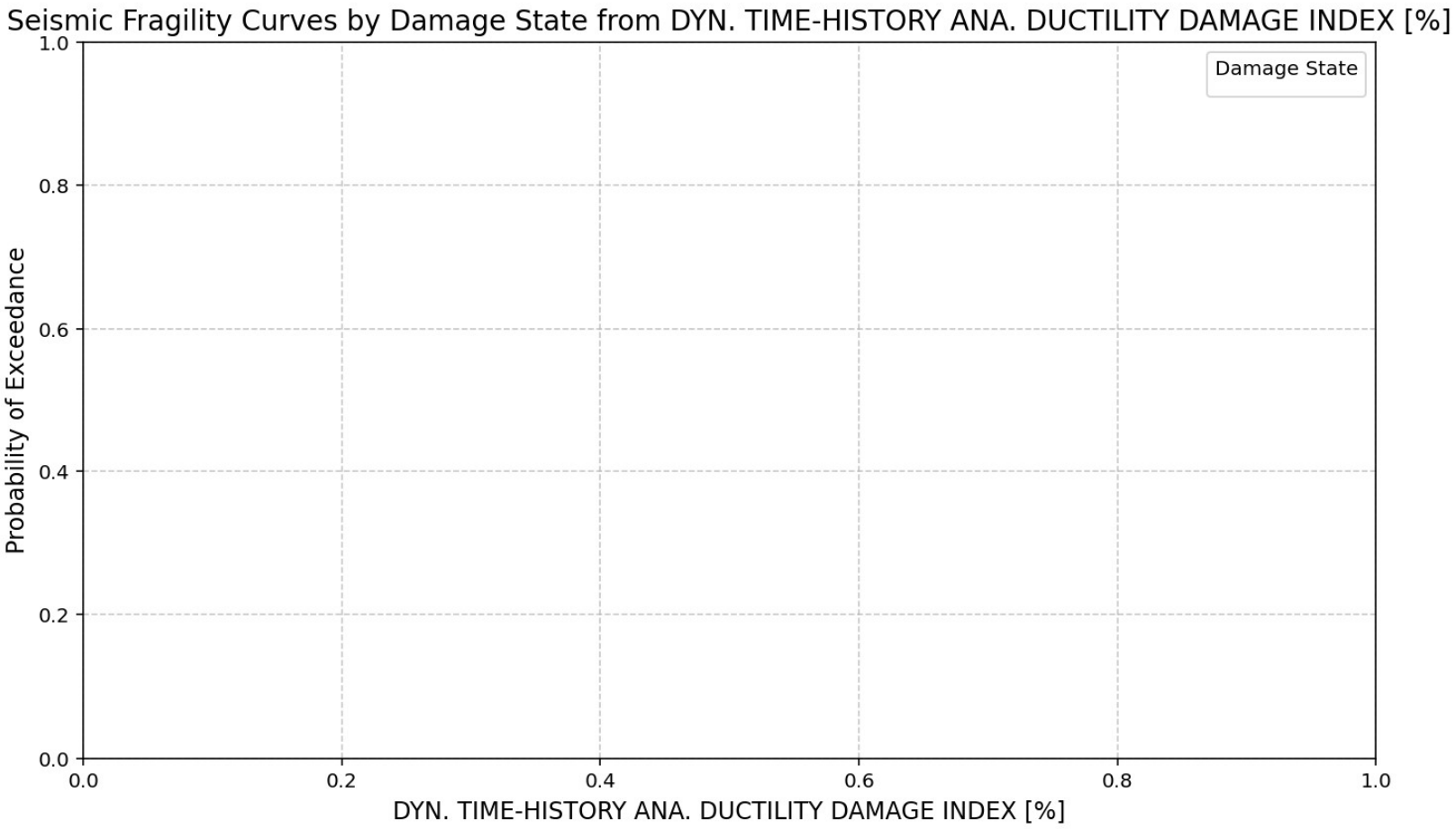

Cyclic Loading – Dissipated Energy
Base Shear (N)
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Hysteresis Loop
(u_{max}, F_{max})
E_d = 265804.643 N·m

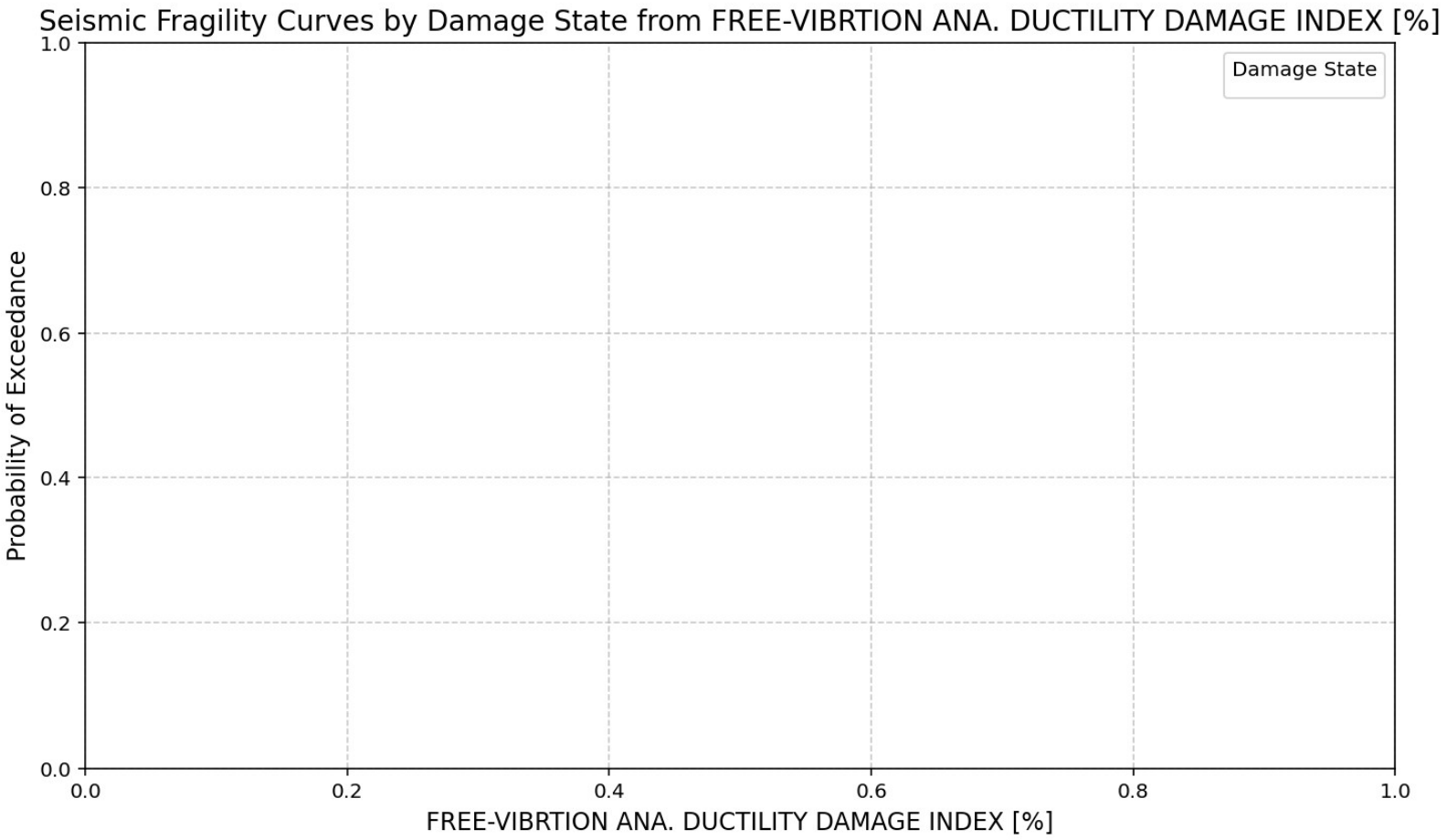
Dissipated Energy from Earthquake= 157253.54 N·m
Dissipated Energy from Cyclic Displacement= 265804.64 N·m

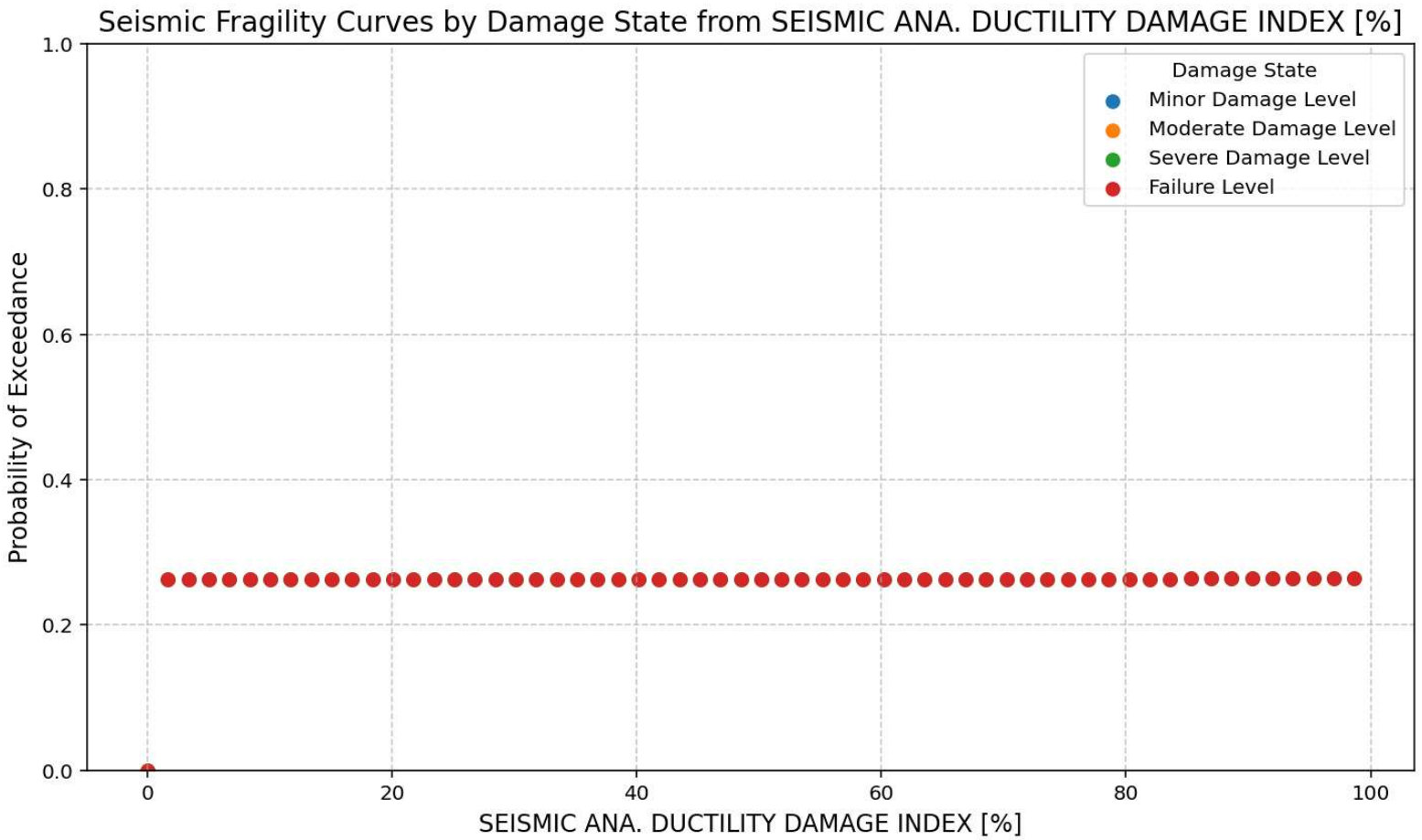
DISSIPATED ENERGY CAPACITY INDEX: 59.161 [%]
ZONE 6: SEVERE–LOW DAMAGE


Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 8.2872 - Over Strength Factor: 1.1599
Base Reaction [N]
100000
80000
60000
40000
20000
0
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Displacement [m]
Curve
Bilinear Fitted

```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
--------------------------
[[0.00000000e+00 0.00000000e+00]
 [4.34404346e-02 8.49921547e+04]
 [3.60000000e-01 9.85790074e+04]]
+==========================+
+------------------------------------------------+
 Structure Elastic Stiffness :     1956521.74
 Structure Plastic Stiffness :     273830.58
 Structure Tangent Stiffness :     42920.37
 Structure Ductility Ratio :      8.29
 Structure Over Strength Factor:   1.16
+------------------------------------------------+
Over Strength Coefficient (Ω0):     1.1599
Displacement Ductility Ratio (μ):    8.2872
Ductility Coefficient (Rμ):         8.2872
Structural Behavior Coefficient (R): 9.6120
Structural Ductility Damage Index in X Direction: 104.1352 (%)
```


Seismic Fragility Curves by Damage State from DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Probability of Exceedance
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]


Seismic Fragility Curves by Damage State from FREE-VIBRTION ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Probability of Exceedance
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
FREE-VIBRTION ANA. DUCTILITY DAMAGE INDEX [%]


Seismic Fragility Curves by Damage State from SEISMIC ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Minor Damage Level
Moderate Damage Level
Severe Damage Level
Failure Level
Probability of Exceedance
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
SEISMIC ANA. DUCTILITY DAMAGE INDEX [%]